

Solvency Capital Requirement - Standard Formula

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Application and Definitions

1.1

Unless otherwise stated, this Part applies to:

1. (1) a _UK Solvency II firm_ ; and
2. (2) in accordance with Insurance General Application 3, the _Society_ , as modified by 8.

1.2

In this Part, the following definitions shall apply:

bankruptcy remote

in relation to client assets, means that effective arrangements exist which ensure that those assets will not be available to the creditors of a CCP or of a clearing member in the event of the insolvency of that CCP or clearing member respectively, or that the assets will not be available to the clearing member to cover losses it incurred following the default of a client other than those that provided those assets.

basis risk

means the risk resulting from the situation in which the exposure covered by the risk-mitigation technique does not correspond to the risk exposure of the UK Solvency II firm.

CCP

means a CCP as defined in point (1) of Article 2 of Regulation (EU) No 648/2012 of the European Parliament and of the Council.

CCP-related transaction

means a contract or a transaction listed in of Article 301(1) of the Counterparty Credit Risk (CRR) Part between a client and a clearing member that is directly related to a contract or a transaction listed in that paragraph between that clearing member and a CCP.

clearing member

means a clearing member as defined in point (14) of Article 2 of Regulation (EU) No 648/2012.

client

means a client as defined in point (15) of Article 2 of Regulation (EU) No 648/2012 or an undertaking that has established indirect clearing arrangements with a clearing member in accordance with Article 4(3) of that Regulation.

collateral arrangement

means an arrangement under which a collateral provider does one of the following:

1. (1) transfers full ownership of the collateral to the collateral taker for the purposes of securing or otherwise covering the performance of a relevant obligation; or
2. (2) provides collateral by way of security in favour of, or to, a collateral taker, and the legal ownership of the collateral remains with the collateral provider or a custodian when the security right is established.

continuity options

means all legal or contractual policyholder rights which allow that policyholder to fully or partly establish, renew, increase, extend or resume insurance or reinsurance cover.

covered bond

means a bond that is issued by a credit institution which has its registered office in the UK or an EEA State and is subject by law to special public supervision designed to protect bondholders and in particular protection under which sums deriving from the issue of the bond must be invested in conformity with the law in assets which, during the whole period of validity of the bond, are capable of covering claims attaching to the bond and which, in the event of failure of the issuer, would be used on a priority basis for the reimbursement of the principal and payment of the accrued interest.

currency risk

means the sensitivity of the values of assets, liabilities and financial instruments to changes in the level or in the volatility of currency exchange rates.

disability-morbidity risk

means the risk of loss, or of adverse change, in the value of insurance obligations, resulting from changes in the level, trend or volatility of disability, sickness and morbidity rates.

discontinuance

means, in relation to an insurance policy, surrender, lapse without value, making a contract of insurance paid-up, automatic non-forfeiture provisions or exercising other discontinuity options or not exercising continuity options.

discontinuity options

means all legal or contractual policyholder rights which allow that policyholder to fully or partly terminate, surrender, decrease, restrict or suspend insurance cover or permit the insurance policy to lapse.

equity risk

means the sensitivity of the values of assets, liabilities and financial instruments to changes in the level or in the volatility of market prices of equities.

health underwriting risk

means

1. (1) expense risk;
2. (2) NSLT health premium and reserve risk; and
3. (3) health catastrophe risk.

interest-rate risk

means the sensitivity of the values of assets, liabilities and financial instruments to changes in the term structure of interest rates, or in the volatility of interest rates.

lapse risk

means the risk of loss, or of adverse change, in the value of insurance obligations, resulting from changes in the level or volatility of the rates of policy lapses, terminations, renewals and surrenders.

life-catastrophe risk

means the risk of loss, or of adverse change, in the value of insurance obligations, resulting from the significant uncertainty of pricing and provisioning assumptions related to extreme or irregular events.

market risk concentrations

means the additional risks to a firm stemming either from lack of diversification in the asset portfolio or from large exposure to default risk by a single issuer of securities or a group of related issuers.

pool exposure of type A

means the risk ceded by a firm to a pooling arrangement where the firm is not a party to that pooling arrangement.

pool exposure of type B

means the risk ceded by a firm to another member of a pooling arrangement, where the firm is a party to that pooling arrangement.

pool exposure of type C

means the risk ceded by a firm which is a party to a pooling arrangement to another firm which is not a member of that pooling arrangement.

pooling arrangement

means an arrangement whereby several undertakings which are UK Solvency II undertakings, third country insurance undertakings or third country reinsurance undertakings agree to share identified insurance risks in defined proportions, but the parties insured by the members of the pooling arrangement are not themselves members of the pooling arrangement.

property risk

means the sensitivity of the values of assets, liabilities and financial instruments to changes in the level or in the volatility of market prices of real estate.

resecuritisation

has the meaning given in Securitisation 1.2.

securitisation

has the meaning given in Securitisation 1.2.

securitisation position

has the meaning given in Securitisation 1.2.

senior securitisation position

means a senior securitisation position within the meaning of Article 242(6) of the CRR.

standard equity capital charge

means the standard capital requirement for equity risk calculated in accordance with 3D before any _symmetric adjustment_ is applied.

STS securitisation

means

1. (1) an STS securitisation as defined by regulation 9 of the Securitisation Regulations; 2024 (SI 2024/102);
2. (2) an overseas STS securitisation as defined by regulation 12(2) of the Securitisation Regulations 2024 (SI 2024/102); or
3. (3) a qualifying EU securitisation as defined by regulation 12(3) of the Securitisation Regulations 2024 (SI 2024/102).

symmetric adjustment

means the symmetric adjustment that may be applied to the _standard equity capital charge_ in accordance with 3D.12.

UK Solvency II undertaking

means a UK Solvency II firm or Lloyd's.

1A

General Requirements on the Use of Credit Assessments

1A.1

A firm may use an external credit assessment for the calculation of the SCR in accordance with the standard formula only where it has been issued by an external credit assessment institution or endorsed by an external credit assessment institution in accordance with Regulation (EC) No 1060/2009.

1A.2

A firm must nominate one or more external credit assessment institutions to be used for the calculation of the SCR according to the standard formula.

1A.3

A firm must use credit assessments consistently and must not use such assessments selectively.

1A.4

When using credit assessments, a firm must comply with all of the following requirements:

1. (1) where a firm decides to use the credit assessments produced by a nominated external credit assessment institution for a certain class of items, it must use those credit assessments consistently for all items belonging to that class;
2. (2) where a firm decides to use the credit assessments produced by a nominated external credit assessment institution, it must use them in a continuous and consistent way over time;
3. (3) a firm must only use nominated external credit assessment institution credit assessments that take into account all

amounts of principal and interest owed to it;

4. (4) subject to 1C.1, where only one credit assessment is available from a nominated external credit assessment institution for a rated item, a firm must use that credit assessment to determine the capital requirements for that item;

5. (5) where two credit assessments are available from nominated external credit assessment institutions and they correspond to different parameters for a rated item, a firm must use the assessment generating the higher capital requirement;

6. (6) where more than two credit assessments are available from nominated external credit assessment institutions for a rated item, a firm must use the two assessments generating the two lowest capital requirements, provided that:

1. (a) if the two lowest capital requirements are different, the firm must use the assessment generating the higher capital requirement of those two credit assessments; and

2. (b) if the two lowest capital requirements are the same, the firm must use the assessment generating that capital requirement; and

7. (7) where available, a firm must use both solicited and unsolicited credit assessments.

1A.5

Where an item is part of the larger or more complex exposures of a firm, the firm must produce its own internal credit assessment of the item and allocate it to a credit quality step, provided that where the firm's own internal credit assessment generates a lower capital requirement than the one generated by the credit assessments available from nominated external credit assessment institutions, then the firm's own internal credit assessment must not be taken into account for the purposes of this Part.

1A.6

For the purposes of 1A.5, the larger or more complex exposures of a firm must include securitisation positions as referred to in 3D21.8 and 3D21.9 and resecuritisation positions.

1B

Issuers And Issue Credit Assessment

1B.1

Where a credit assessment exists for a specific issuing program or facility to which the item constituting the exposure belongs, a firm must use that credit assessment.

1B.2

Where no directly applicable credit assessment exists for a certain item, but a credit assessment exists for a specific issuing program or facility to which the item constituting the exposure does not belong or a general credit assessment exists for the issuer, a firm must use that credit assessment in either of the following cases:

1. (1) it produces the same or higher capital requirement than would otherwise be the case and the exposure in question ranks *pari passu* or junior in all respects to the specific issuing program or facility or to senior unsecured exposures of that issuer, as relevant; or

2. (2) it produces the same or lower capital requirement than would otherwise be the case and the exposure in question ranks *pari passu* or senior in all respects to the specific issuing program or facility or to senior unsecured exposures of that issuer, as relevant.

In all other cases, a firm must treat the exposure as if there is no credit assessment by a nominated external credit assessment institution available for it.

1B.3

A firm must not use credit assessments for issuers within a corporate group as the credit assessment for another issuer within the same corporate group.

1C

Double Credit Rating For Securitisation Positions

1C.1

Notwithstanding 1A.4(4), where only one credit assessment is available from a nominated external credit assessment institution for a securitisation position, a firm must not use that credit assessment and the firm must derive the capital requirements for that item as if no credit assessment by a nominated external credit assessment institution is available.

1D

Allocation of Credit Assessments of External Credit Assessment Institutions to an Objective Scale of Credit Quality Steps

1D.1

The allocation of credit assessments of external credit assessment institutions to an objective scale of credit quality steps shall be that set out in the following table:

Credit quality step | 0 | 1 | 2 | 3 | 4 | 5 | 6

---|---|---|---|---|---|---

A.M. Best Europe Rating Services Limited

Long-term issuer credit rating scale | aaa | aa+, aa, aa- | a+, a, a- | bbb+, bbb, bbb- | bb+, bb, bb- | b+, b, b- | ccc+, ccc, ccc-, cc, c, d, e, f, s

Long-term issue rating scale

| aaa | aa+, aa, aa-

| a+, a, a-

| bbb+, bbb, bbb-

| bb+, bb, bb-

| b+, b, b-

| ccc+, ccc, ccc-, cc, c, d, s

Financial strength rating scale

| A++

| A+

| A, A-

| B++, B+

| B, B-

| C++, C+

| C, C-, D, E, F, S

Short-term issuer rating scale

| | AMB-1+

| AMB-1- | AMB-2, AMB-3

| AMB-4, d, e, f, s

| |

Short-term issue rating scale

| | AMB-1+ | AMB-1-

| AMB-2, AMB-3

| AMB-4, d, s

| |

ARC Ratings (UK) Limited

Long-term issuer rating scale

| AAA

| AA | A

| BBB | BB

| B

| CCC, CC, C, D

Long-term issue rating scale

| AAA

| AA

| A

| BBB | BB

| B

| CCC, CC, C, D

Insurance Financial Strength Rating scale

| AAA

| AA

| A

| BBB

| BB

| B

| CCC, CC, C, D

Short-term issuer rating scale

| | A-1+

| A-1

| A-2, A-3

| B, C, D

| |

Short-term issue rating scale

| | A-1+

| A-1

| A-2, A-3

| B, C, D

| |

Banque de France

Global ANACOT long-term issuer credit rating scale | | 1+

| 1, 1- | 2+, 2, 2-

| 3+, 3, 3-, 4+, 4, 4-, 5+ | 5, 5-, 6+, 6, 6- | 7, 8, P

Creditreform Rating AG

Long-term issuer rating scale

| AAA

| AA

| A

| BBB

| BB

| B

| C, SD, D

Long-term issue rating scale

| AAA

| AA

| A

| BBB

| BB

| B

| C, D

Short-term rating scale

| | L1

| L2

| L3

| NEL, D

| |

DBRS Ratings Limited

Long-term obligations rating scale

| AAA

| AA

| A

| BBB

| BB

| B

| CCC, CC, C, D

Commercial paper and short-term debt rating scale

| | R-1 H, R-1 M

| R-1 L

| R-2,

R-3

| R-4,

R-5, D

| |

Financial strength rating scale

| AAA

| AA

| A

| BBB

| BB

| B | CCC, CC, C, R

Expected loss rating scale

| AAA(el)

| AA(el)

| A(el)

| BBB(el)

| BB(el)

| B(el)

| CCC(el), CC(el), C(el)

Egan-Jones Ratings Co.

Long-term credit rating scale

| AAA

| AA

| A

| BBB

| BB

| B

| CCC, CC, C, D

Short-term credit rating scale

| | A-1+

| A-1

| A-2

| A-3, B, C, D

| |

Fitch Ratings Limited

Long-term issuer default rating scale

| AAA

| AA

| A

| BBB

| BB

| B

| CCC, CC, C, RD, D

Corporate finance obligations ? long-term rating scale

| AAA
| AA
| A
| BBB
| BB
| B
| CCC, CC, C

Long-term international Insurer Financial Strength rating scale

| AAA
| AA
| A | BBB
| BB | B | CCC, CC, C

Derivative counterparty rating scale

| AAA dcr
| AA dcr
| A dcr
| BBB dcr
| BB dcr
| B dcr
| CCC dcr, CC dcr, C dcr, RD dcr, D dcr

Short-term rating scale

| | F1+
| F1
| F2, F3
| B, C, RD, D
| |

Short-term Insurer Financial Strength rating scale

| | F1+
| F1
| F2, F3
| B, C
| |

HR Ratings de México, S.A. de C.V.

Global long-term rating scale

| HR AAA(G)
| HR AA(G)
| HR A(G)
| HR BBB(G)
| HR BB(G)
| HR B(G)

| HR C(G)/HR D(G)

Global short-term rating scale

| HR+1(G)

| HR1(G)

| HR2(G)

| HR3(G)

| HR4(G), HR5(G), HR D(G)

| |

Japan Credit Rating Agency Ltd

Long-term issuer rating scale

| AAA

| AA

| A

| BBB

| BB

| B | CCC, CC, C, LD, D

Long-term issue rating scale

| AAA

| AA

| A

| BBB

| BB | B

| CCC, CC, C, D

Short-term issuer rating scale

| | J-1+

| J-1

| J-2

| J-3, NJ, LD, D

| |

Short-term issue credit rating scale

| | J-1+

| J-1

| J-2

| J-3, NJ, D

| |

Kroll Bond Rating Agency UK Limited

Long-term credit rating scale

| AAA

| AA

| A

| BBB

| BB
 | B
 | CCC, CC, C, D

Short-term credit rating scale

| | K1+
 | K1
 | K2, K3
 | B, C, D
 | |

Insurance Financial Strength rating scale

| AAA
 | AA
 | A
 | BBB
 | BB
 | B
 | CCC, CC, C, D

Moody's Investors Service Limited

Global long-term rating scale

| Aaa
 | Aa
 | A
 | Baa
 | Ba
 | B
 | Caa, Ca, C

Global short-term rating scale

| | P-1
 | P-2
 | P-3
 | NP
 | |

Scope Ratings UK Limited

Long-term rating scale

| AAA
 | AA
 | A
 | BBB
 | BB
 | B | CCC, CC, C, D/SD

Short-term rating scale

| | S-1+

| S-1

| S-2

| S-3, S-4, D/SD

| |

S &P Global Ratings UK Limited

Long-term issuer credit rating scale

| AAA

| AA

| A | BBB

| BB

| B | CCC, CC, R, SD/D

Long-term issue credit rating scale

| AAA

| AA

| A

| BBB | BB | B** | CCC, CC, C, D

Insurer financial strength rating scale

| AAA

| AA | A | BBB

| BB | B | CCC, CC, SD/D, R

Long-term Financial Institution Resolution Counterparty Ratings

| AAA

| AA | A | BBB

| BB | B | CCC, CC, SD, D

Mid-Market Evaluation rating scale

| | | MM1

| MM2

| MM3, MM4

| MM5, MM6

| MM7, MM8, MMD

Short-term issuer credit rating scale

| | A-1+

| A-1

| A-2,

A-3

| B, C, R, SD/D

| |

Short-term issue credit rating scale

| | A-1+

| A-1
 | A-2,
 A-3
 | B, C, D
 | |
 Short-term Financial Institution Resolution Counterparty Ratings | | A-1+
 | A-1
 | A-2,
 A-3
 | B, C, SD/D
 | |
 2

Structure of the SCR Standard Formula

2.1

For a _firm_ calculating its _SCR_ on the basis of the _standard formula_ , its _SCR_ is the sum of the following items:

1. (1) the _basic SCR_ ;
2. (2) the capital requirement for _operational risk_ , as set out in 5; and
3. (3) the adjustment for the loss-absorbing capacity of _technical provisions_ and deferred taxes, as set out in 6.

[Note: Art. 103 of the _Solvency II Directive_]

2.2

Notwithstanding 2.1, a firm with a ring-fenced fund (other than a ring-fenced fund in respect of which the reconciliation reserve has been reduced by the total amount of restricted own funds items in accordance with Own Funds 3L.2) or matching adjustment portfolio must make an adjustment to the calculation of its SCR following the method set out in 9.

2.3

1. (1) A firm must calculate its SCR on the basis of each of the underlying assets of collective investment undertakings and other investments packaged as funds.
2. (2) Subject to (6), a firm must also apply the look-through approach to the following:
 1. (a) indirect exposures to market risk other than collective investment undertakings and investments packaged as funds;
 2. (b) indirect exposures to underwriting risk; and
 3. (c) indirect exposures to counterparty risk.
3. (3) Subject to 7.2, if a firm cannot apply the look-through approach to collective investment undertakings or investments packaged as funds, a firm may calculate its SCR on the basis of the target underlying asset allocation or, if the target underlying asset allocation is not available to the firm, on the basis of the last reported asset allocation, of the collective investment undertaking or fund, provided that, in either case:
 1. (a) the underlying assets are managed in accordance with that target allocation or last reported asset allocation, as applicable; and
 2. (b) exposures and risks are not expected to vary materially over a short period of time.
4. (4) For the purposes of the calculation in (3), a firm may use data groupings provided that they:

1. (a) enable all relevant sub-modules and scenarios of the standard formula to be calculated in a prudent manner; and
2. (b) do not apply to more than 20% of the total value of the firm's assets.
5. (5) For the purposes of determining the percentage of assets where data groupings are used as referred to in (4)(b), a firm must not take into account underlying assets of the collective investment undertaking, or the investments packaged as funds, backing unit-linked liabilities or index-linked liabilities for which the market risk is borne by the policyholders.
6. (6) (2) does not apply to investments in related undertakings, other than investments in respect of which all of the following requirements are met:
 1. (a) the main purpose of the related undertaking is to hold and manage assets on behalf of the participating undertaking;
 2. (b) the related undertaking supports the operations of the participating undertaking related to investment activities, following a specific and documented investment mandate; and
 3. (c) the related undertaking does not carry on any significant business other than investing for the benefit of the participating undertaking.

2A

Annexes

1

The Annexes referred to in 3A, 3C and 7 can be found here.

3

The Basic SCR

3.1

For the purposes of calculating its *_basic SCR_*, a *_firm_* must:

1. (1) calculate the capital requirements for:
 1. (a) the non-life *_underwriting risk_* module;
 2. (b) the life *_underwriting risk_* module;
 3. (c) the health underwriting risk module;
 4. (d) the *_market risk_* module; and
 5. (e) the counterparty default risk module;
2. (2) aggregate the capital requirements referred to in (1) in accordance with the following formula:

$$\text{BasicSCR} = \sqrt{\sum_{i,j} \text{Corr}_{i,j} \cdot \text{SCR}_{i,j} \cdot \text{SCR}_{i,j}} + \text{SCR}_{\text{intangibles}}$$

where:

1. (a) *?SCR_i?* and *?SCR_j?* denote the non-life *_underwriting risk_* module, the life *_underwriting risk_* module, the health underwriting risk module, the *_market risk_* module and the counterparty default risk module;
2. (b) *?i,j?* means that the sum of the different terms should cover all possible combinations of *?i?* and *?j?*;
3. (c) the factor *?Corr_{i,j}?* denotes the item set out in row *?i?* and column *?j?* of the correlation matrix in (d); and
4. (d) *_j_*

i | Market | Default | Life | Health | Non-life

---|---|---|---|---|---

Market | 1 | 0.25 | 0.25 | 0.25 | 0.25

Default | 0.25 | 1 | 0.25 | 0.25 | 0.5

Life | 0.25 | 0.25 | 1 | 0.25 | 0

Health | 0.25 | 0.25 | 0.25 | 1 | 0

Non-life | 0.25 | 0.5 | 0 | 0 | 1

; and

1. (3) include a risk module for intangible asset risk, and a firm must calculate this in accordance with the following formula:

$$\text{BasicSCR} = \sqrt{\sum_{i,j} \text{Corr}_{i,j} \cdot \text{SCR}_i \cdot \text{SCR}_j} + \text{SCR}_{\text{intangibles}}$$

where:

1. (a) in the summation, $\text{Corr}_{i,j}$, SCR_i and SCR_j are specified as set out in (2); and
2. (b) $\text{SCR}_{\text{intangibles}}$ denotes the capital requirement for intangible asset risk referred to in 3F1.

[Note: Art. 104(1) and Annex IV point (1) of the _Solvency II Directive_]

3.2

For the purposes of calculating the capital requirements in 3.1(1) for non-life _underwriting risk_, life _underwriting risk_ and health _underwriting risk_, a _firm_ must allocate its insurance and _reinsurance_ operations to the _underwriting risk_ that best reflects the technical nature of the underlying risks.

[Note: Art. 104(2) of the _Solvency II Directive_]

3.2A

For the purposes of calculating the capital requirements in 3.1(1) for non-life underwriting risk, life underwriting risk and health underwriting risk, a firm must apply:

1. (1) the non-life underwriting risk module to non-life insurance and reinsurance obligations other than health insurance obligations and health reinsurance obligations;
2. (2) the life underwriting risk module to life insurance and reinsurance obligations other than health insurance obligations and health reinsurance obligations; and
3. (3) the health underwriting risk module to health insurance obligations and health reinsurance obligations.

3.3

Each of the risk modules referred to in 3.1(1) must be calibrated using a Value-at-Risk measure, with a 99.5% confidence level over a one-year period.

3.3A

1. (1) Where the calculation of a module or sub-module of the basic SCR is based on the impact of a scenario on the basic own funds of a firm, the firm must make all of the following assumptions in that calculation:
 1. (a) the scenario does not change the amount of the risk margin included in technical provisions;
 2. (b) the scenario does not change the value of deferred tax assets and liabilities;
 3. (c) the scenario does not change the value of future discretionary benefits included in technical provisions; and
 4. (d) no management actions are taken by the firm during the scenario.
2. (2) In calculating technical provisions arising as a result of determining the impact of a scenario on its basic own funds as referred to in (1), a firm must not change the value of future discretionary benefits, and must take account of all of the

following:

1. (a) without prejudice to (1)(d), future management actions following the scenario, provided they comply with Technical Provisions ? Further Requirements 8; and
2. (b) any material adverse impact of the scenario or the future management actions referred to in (a) on the likelihood that policyholders will exercise options relating to contracts of insurance.
3. (3) A firm may use simplified methods to calculate the technical provisions arising as a result of determining the impact of a scenario as referred to in (1), provided that the simplified method does not lead to a misstatement of the SCR that could influence the decision-making or the judgement of the user of the information relating to the SCR, unless the simplified calculation produces an SCR which exceeds the SCR that results from the calculation according to the standard formula.
4. (4) In calculating the assets and liabilities arising as a result of determining the impact of a scenario as referred to in (1), a firm must take account of the impact of the scenario on the value of any relevant risk mitigation instruments held by the firm which comply with 3G2, 3G3 and 3G5 to 3G9.
5. (5) Where the scenario would result in an increase in its basic own funds, a firm must base the calculation of the module or sub-module on the assumption that the scenario has no impact on its basic own funds.

3.4

Where appropriate, diversification effects must be taken into account in the design of each risk module.

[Note: Art. 104(4) of the Solvency II Directive]

3.5

For the purposes of the basic SCR, a firm must calculate the capital requirement for the non-life underwriting risk module so that it:

1. (1) reflects the risk arising from its non-life insurance obligations, in relation to the perils covered and the processes used in the conduct of business; and
2. (2) takes account of the uncertainty in its results related to existing insurance and reinsurance obligations, as well as to new business expected to be written within the following 12 months.

[Note: Art. 105(2) of the Solvency II Directive]

3.6

For the purposes of 3.1(1)(a), the capital requirement for the non-life underwriting risk module is a combination of the capital requirements for the following sub-modules:

1. (1) the non-life premium and reserve risk sub-module covering non-life premium and reserve risk;
2. (2) the non-life catastrophe risk sub-module covering non-life catastrophe risk; and
3. (3) the non-life lapse risk sub-module covering non-life lapse risk.

[Note: Art. 105(2) of the Solvency II Directive]

3.6A

1. (1) A firm must calculate the capital requirement for non-life underwriting risk in accordance with the following formula:

$$\sqrt{\sum_{i,j} \text{CorrNL}_{\left(i,j\right)} \cdot \text{SCR}_i \cdot \text{SCR}_j}$$

where:

1. (a) the sum covers all possible combinations (i, j) of the sub-modules set out in 3.6;

2. (b) $\text{CorrNL}_{(i,j)}$ denotes the correlation coefficient for non-life underwriting risk for sub-modules i and j ; and
3. (c) SCR_i and SCR_j denote the capital requirements for risk sub-modules i and j , respectively.

(2) The correlation coefficient $\text{CorrNL}_{(i,j)}$ referred to in (1) denotes the item set out in row i and column j of the following correlation matrix:

$j \ i \mid$ Non-life premium and reserve $\mid$ Non-life catastrophe $\mid$ Non-life lapse

---|---|---|---

Non-life premium and reserve $\mid 1 \mid 0.25 \mid 0$

Non-life catastrophe $\mid 0.25 \mid 1 \mid 0$

**Non-life lapse

** $\mid 0 \mid 0 \mid 1$

3.7

For the purposes of 3.1(1)(b) a firm must calculate the capital requirement for the life underwriting risk module so as to reflect the risk arising from its life insurance obligations, in relation to the perils covered and the processes used in the conduct of business.

[Note: Art. 105(3) of the $\text{Solvency II Directive}$]

3.8

The life underwriting risk module must be calculated as:

1. (1) a combination of the capital requirements for the following sub-modules:

1. (a) mortality risk;
2. (b) longevity risk;
3. (c) disability-morbidity risk;
4. (d) life expense risk;
5. (e) revision risk;
6. (f) lapse risk; and
7. (g) life catastrophe risk;

2. (2) aggregated in accordance with the following formula:

$$\sqrt{\sum_{i,j} \{\text{Corr}\}_{i,j} \cdot \{\text{SCR}\}_i \cdot \{\text{SCR}\}_j}$$

where: SCR_i and SCR_j denote the mortality risk sub-module, the longevity risk sub-module, the disability-morbidity risk sub-module, the life expense risk sub-module, the revision risk sub-module, the lapse risk sub-module and the life catastrophe risk sub-module;

i, j means that the sum of the different terms should cover all possible combinations of i and j ; and

$\text{Corr}_{i,j}$ denotes the correlation coefficient for life underwriting risk for sub-modules i and j .

1. (3) The correlation coefficient $\text{Corr}_{i,j}$ referred to in (2) must be equal to the item set out in row i and column j of the following correlation matrix:

$j \ i \mid$

Mortality $\mid$ Longevity $\mid$ Disability $\mid$ Life expense $\mid$ Revision $\mid$ Lapse $\mid$ Life catastrophe $\mid 1 \mid -0.25 \mid 0.25 \mid 0.25 \mid 0 \mid 0 \mid 0.25$

---|---|---|---|---|---|---|---|---|---|---|---

****Longevity**

** | -0.25 | 1 | 0 | 0.25 | 0.25 | 0.25 | 0

Disability | 0.25 | 0 | 1 | 0.5 | 0 | 0 | 0.25

****Life expense**

** | 0.25 | 0.25 | 0.5 | 1 | 0.5 | 0.5 | 0.25

Revision

| 0 | 0.25 | 0 | 0.5 | 1 | 0 | 0

Lapse | 0 | 0.25 | 0 | 0.5 | 0 | 1 | 0.25

****Life catastrophe**

** | 0.25 | 0 | 0.25 | 0.25 | 0 | 0.25 | 1

[Note: Art. 105(3) and Annex IV point (3)_Solvency II Directive_]

3.9

For the purposes of 3.8:

1. (1) the mortality risk sub-module covers mortality risk;
2. (2) the longevity risk sub-module covers longevity risk;
3. (3) the disability-morbidity risk sub-module covers disability-morbidity risk;
4. (4) the life expense risk sub-module covers life expense risk;
5. (5) the revision risk sub-module covers revision risk;
6. (6) the lapse risk sub-module covers lapse risk; and
7. (7) the life-catastrophe risk sub-module covers life-catastrophe risk.

[Note: Art. 105(3) of the Solvency II Directive)]

3.10

For the purposes of 3.1(1)(c):

1. (1) a _firm_ must calculate the capital requirement for the health _underwriting risk_ module to reflect the risk arising from its underwriting of health insurance obligations, whether it is pursued on a similar technical basis to that of life insurance or not, following from both the perils covered and the processes used in the conduct of business; and

2. (2) the health _underwriting risk_ module must cover at least health _underwriting risk_.

1. (a) [deleted]

2. (b) [deleted]

3. (c) [deleted]

[Note: Art. 105(4) of the _Solvency II Directive_]

3.10A

1. (1) The health underwriting risk module must consist of all of the following sub-modules:

1. (a) the NSLT health insurance underwriting risk sub-module;
2. (b) the SLT health insurance underwriting risk sub-module; and
3. (c) the health catastrophe risk sub-module.

2. (2) A firm must calculate the capital requirement for health underwriting risk in accordance with the following formula:

$$\sqrt{\sum_{i,j} \{CorrH\}_{\left(i,j\right)} \cdot \{SCR\}_i \cdot \{SCR\}_j}$$

where:

1. (a) the sum covers all possible combinations $_i, _j$ of the sub-modules set out in (1);
 2. (b) $_CorrH(i, j)$ denotes the correlation coefficient for health underwriting risk for sub-modules $_i$ and $_j$; and
 3. (c) $_SCR _i$ and $_SCR _j$ denote the capital requirements for risk sub-modules $_i$ and $_j$, respectively.
1. (3) The correlation coefficient $_CorrH(i, j)$ referred to in (2) denotes the item set out in row $_i$ and column $_j$ of the following correlation matrix:

$_j$			
$_i$	NSLT health underwriting	SLT health underwriting	Health catastrophe
---	---	---	---
NSLT health underwriting	1	0.5	0.25
SLT health underwriting	0.5	1	0.25
Health catastrophe	0.25	0.25	1

3.10B

A firm must apply:

1. (1) the NSLT health underwriting risk sub-module to health insurance obligations and health reinsurance obligations included in lines of business 1, 2, 3, 13, 14, 15 and 25;
2. (2) the SLT health underwriting risk sub-module to health insurance obligations and health reinsurance obligations included in lines of business 29, 33 and 35; and
3. (3) the health catastrophe risk sub-module to health insurance obligations and health reinsurance obligations.

3.11

For the purposes of 3.1(1)(d):

1. (1) a $_firm$ must calculate the capital requirement for the $_market \ risk$ module so that it:
 1. (a) reflects the risk arising from the level or volatility of market prices of financial instruments which have an impact upon the value of the assets and liabilities of the $_firm$;
 2. (b) properly reflects the structural mismatch between assets and liabilities, in particular with respect to the duration of assets and liabilities; and
2. (2) the capital requirement for the $_market \ risk$ module is a combination of the capital requirements for at least the following sub-modules:
 1. (a) an interest-rate risk sub-module covering interest-rate risk;
 2. (b) an equity risk sub-module covering equity risk;
 3. (c) a property risk sub-module covering property risk;
 4. (d) a spread risk sub-module covering spread risk;
 5. (e) a currency risk sub-module covering currency risk; and
 6. (f) a market risk concentrations sub-module covering market risk concentrations.

[Note: Art. 105(5) of the $_Solvency \ II \ Directive$]

3.11A

1. (1) A firm must calculate the capital requirement for market risk in accordance with the following formula:

$$\sqrt{\sum_{i,j} \{Corr\}_{\left(i,j\right)} \cdot \{SCR\}_i \cdot \{SCR\}_j}$$

where:

1. (a) the sum covers all possible combinations i, j of sub-modules set out in 3.11(2);
2. (b) $\text{Corr}_{(i,j)}$ denotes the correlation coefficient for market risk for sub-modules i and j ; and
3. (c) SCR_i and SCR_j denote the capital requirements for risk sub-modules i and j , respectively.

(2) The correlation coefficient $\text{Corr}(i,j)$ referred to in (1) must be equal to the item set out in row i and column j of the following correlation matrix:

j							
i	Interest rate	Equity	Property	Spread	Concentration	Currency	
	---	---	---	---	---	---	---
Interest rate	1	A	A	A	0	0.25	
Equity	A	1	0.75	0.75	0	0.25	
Property	A	0.75	1	0.5	0	0.25	
Spread	A	0.75	0.5	1	0	0.25	
Concentration	0	0	0	0	1	0	
Currency	0.25	0.25	0.25	0.25	0	1	

1. (3) The coefficient A in the table in (2) must be equal to 0 where the capital requirement for interest-rate risk set out in 3D4 is the capital requirement referred to in 3D4.1(1). In all other cases, the coefficient A must be equal to 0.5.

3.12

For the purposes of 3.1(1)(e), the counterparty default risk module:

1. (1) must reflect possible losses due to unexpected default, or deterioration in the credit standing, of the counterparties and debtors of the firm over the following 12 months ;
2. (2) must cover risk-mitigating contracts, such as reinsurance arrangements, securitisations and derivatives , and receivables from intermediaries, as well as any other credit exposures which are not covered in the spread risk sub-module;
3. (3) must take appropriate account of collateral or other security held by, or for the account of, the firm and the associated risks;
4. (4) for each counterparty, must take account of the overall counterparty risk exposure of the firm to that counterparty, irrespective of the legal form of the counterparty's contractual obligations to the firm .

[Note: Art. 105(6) of the $\text{Solvency II Directive}$]

3.13

A firm must calculate the capital requirement for counterparty default risk in accordance with the following formula:

$$\sqrt{\{\text{SCR}\}_{\text{def}} = \sqrt{\{\text{SCR}\}_{\text{left}(\text{def},1\text{right})}^2 + 1.5 \cdot \{\text{SCR}\}_{\text{left}(\text{def},1\text{right})} \cdot \{\text{SCR}\}_{\text{left}(\text{def},2\text{right})} + \{\text{SCR}\}_{\text{left}(\text{def},2\text{right})}^2}}$$

where:

1. (a) $\text{SCR}(\text{def},1)$ denotes the capital requirement for counterparty default risk on type 1 exposures as set out in 3.14; and
2. (b) $\text{SCR}(\text{def},2)$ denotes the capital requirement for counterparty default risk on type 2 exposures as set out in 3.15.

3.14

A firm must treat exposures in relation to the following as type 1 exposures:

1. (1) risk-mitigation contracts including reinsurance arrangements, special purpose vehicles and insurance securitisations;
2. (2) cash at bank as referred to in Schedule 3 to the Large and Medium-Sized Companies and Groups (Accounts and Reports) Regulations 2008/410 as amended from time to time;
3. (3) deposits with ceding undertakings where the number of single name exposures does not exceed 15;
4. (4) commitments received by the firm which have been called up but are unpaid, where the number of single name exposures does not exceed 15, including called up but unpaid ordinary share capital and preference shares, called up but unpaid legally binding commitments to subscribe and pay for subordinated liabilities, called up but unpaid initial funds, members' contributions or the equivalent basic own fund item for mutual and mutual-type undertakings, called up but unpaid guarantees, called up but unpaid letters of credit, called up but unpaid claims which mutual or mutual-type associations may have against their members by way of a call for supplementary contributions;
5. (5) legally binding commitments which the firm has provided or arranged and which may create payment obligations depending on the credit standing or default of a counterparty including guarantees, letters of credit, and letters of comfort; and
6. (6) derivatives other than credit derivatives covered in the spread risk sub-module.

3.15

A firm must treat all credit exposures which are not covered in the spread risk sub-module and which are not type 1 exposures as type 2 exposures, including the following:

1. (1) receivables from intermediaries;
2. (2) policyholder debtors;
3. (3) mortgage loans which meet the requirements in 3E3.2 to 3E3.13;
4. (4) deposits with ceding undertakings where the number of single name exposures exceeds 15; and
5. (5) commitments received by the firm which have been called up but are unpaid as referred to in 3.14(4), where the number of single name exposures exceeds 15.

3.16

A firm may, at its discretion, consider all exposures referred to in 3.15(4) and (5) as type 1 exposures, regardless of the number of single name exposures.

3.17

Where a letter of credit, a guarantee or an equivalent risk-mitigation technique has been provided to fully secure an exposure and this risk mitigation technique complies with the requirements of 3G2, 3G3 and 3G5 to 3G9, then the firm may treat the provider of that letter of credit, guarantee or equivalent risk mitigation technique as the counterparty on the secured exposure for the purposes of assessing the number of single name exposures.

3.18

A firm must not include the following credit risks in the counterparty default risk module:

1. (1) the credit risk transferred by a credit derivative;
2. (2) the credit risk on debt issuance by special purpose vehicles;
3. (3) the underwriting risk of credit and suretyship insurance or reinsurance as referred to in lines of business 9, 21 and 28;
4. (4) the credit risk on mortgage loans which do not meet the requirements in 3E3.2 to 3E3.9; and
5. (5) the credit risk on assets posted as collateral to a CCP or a clearing member that are bankruptcy remote.

3.19

A firm must treat investment guarantees on contracts of insurance provided to policyholders by a third party and for which the firm would be liable should the third party default as derivatives in the counterparty default risk module.

3A

Non-life Underwriting Risk Module

3A1 Non-Life Premium And Reserve Risk Sub-Module

1\.

A firm must calculate the capital requirement for non-life premium and reserve risk in accordance with the following formula:

$$\sqrt{\{SCR\}_{nl\ prem\ res}} = 3 \cdot \sigma_{nl} \cdot V_{nl}$$

where:

1. (1) σ_{nl} denotes the standard deviation for non-life premium and reserve risk determined in accordance with 3A4; and
2. (2) V_{nl} denotes the volume measure for non-life premium and reserve risk determined in accordance with 3A2.

Export Article as

3A2 Volume Measure For Non-Life Premium And Reserve Risk

1\.

A firm must calculate the volume measure for non-life premium and reserve risk as equal to the sum of the volume measures for premium and reserve risk of the segments set out in 3A3.

2\.

For all segments set out in 3A3 a firm must calculate the volume measure of a particular segment s in accordance with the following formula:

$$V_s = (V_{(prem,s)} + V_{(res,s)}) \cdot (0.75 + 0.25 \cdot \{DIV\}_s)$$

where:

1. (1) $V_{(prem,s)}$ denotes the volume measure for premium risk of segment s ;
2. (2) $V_{(res,s)}$ denotes the volume measure for reserve risk of segment s ; and
3. (3) DIV_s denotes the factor for geographical diversification of segment s .

3\.

For all segments set out in 3A3, a firm must calculate the volume measure for premium risk of a particular segment s in accordance with the following formula:

$$V_{(prem,s)} = \max[P_s; P_{(last,s)}] + \{FP\}_{(existing,s)} + \{FP\}_{(future,s)}$$

where:

1. (1) $_P s_$ denotes an estimate of the premiums to be earned by the firm for segment $_s_$ during the following 12 months;
 2. (2) $_P(\text{last}, s)_$ denotes the premiums earned by the firm for segment s during the last 12 months;
 3. (3) $_FP(\text{existing}, s)_$ denotes the expected present value of premiums to be earned by the firm for segment $_s_$ after the following 12 months for existing contracts of insurance; and
 4. (4) $_FP(\text{future}, s)_$ denotes the following amount with respect to contracts of insurance where the initial recognition date falls in the following 12 months:
 1. (a) for all such contracts of insurance with an initial term of one year or less, the expected present value of premiums to be earned by the firm for segment s , but excluding the premiums to be earned during the 12 months after the initial recognition date; and
 2. (b) for all such contracts of insurance with an initial term of more than one year, the amount equal to 30% of the expected present value of premiums to be earned by the firm for segment $_s_$ after the following 12 months.
- 4\.

For all segments set out in 3A3, a firm may, as an alternative to the calculation set out in 3A2.3, choose to calculate the volume measure for premium risk of a particular segment $_s_$ in accordance with the following formula:

$$\backslash[V_{\{(prem,s)\}} = P_{s\} + \{FP\}_{\{(existing,s)\}} + \{FP\}_{\{(future,s)\}} \backslash]$$

provided that all of the following requirements are met:

1. (1) the governing body of the firm has decided that its earned premiums for segment $_s_$ during the following 12 months will not exceed $_P s_$;
2. (2) the firm has established effective control mechanisms to ensure that the limits on earned premiums referred to in (1) will be met; and
3. (3) the firm has informed the PRA in writing about the decision referred to in (1) and the reasons for it.

For the purposes of this calculation, the terms $_P s_$, $_FP(\text{existing}, s)_$ and $_FP(\text{future}, s)_$ must be calculated in accordance with 3A2.3(1), (3) and (4).

5\.

For the purposes of the calculations set out in 3A2.3 and 3A2.4, premiums must be net, after deduction of premiums for reinsurance contracts, except for premiums for the following types of reinsurance contracts which must not be deducted:

1. (1) premiums in relation to non-insurance events or settled insurance claims that are not accounted for in the cash-flows referred to in Technical Provisions ? Further Requirements 23.3; and
2. (2) premiums for reinsurance contracts that do not comply with 3G2, 3G3, 3G5 and 3G7.

6\.

For all segments set out in 3A3, a firm must calculate the volume measure for reserve risk of a particular segment as equal to the best estimate of the provisions for claims outstanding for the segment, after deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, provided that:

1. (1) the reinsurance contracts or special purpose vehicles comply with 3G2, 3G3, 3G5 and 3G7; and
2. (2) the volume measure must not be a negative amount.

7\.

For all segments set out in 3A3, the default factor for geographical diversification of a particular segment must be either 1 or calculated in accordance with 3A5.

Export Article as

3A3 Segmentation Of Non-Life Insurance And Reinsurance Obligations And Standard Deviations For The Non-Life Premium And Reserve Risk Sub-Module

| Segment | Lines of business that the segment consists of | Standard deviation for gross premium risk of the segment | Standard deviation for reserve risk of the segment

---|---|---|---|---

1 | Motor vehicle liability insurance and proportional reinsurance | 4 and 16 | 10% | 9%

2 | Other motor insurance and proportional reinsurance

| 5 and 17

| 8% | 8%

3 | Marine, aviation and transport insurance and proportional reinsurance

| 6 and 18 | 15% | 11%

4 | Fire and other damage to property insurance and proportional reinsurance

| 7 and 19

| 8% | 10%

5 | General liability insurance and proportional reinsurance

| 8 and 20

| 14% | 11%

6 | Credit and suretyship insurance and proportional reinsurance

| 9 and 21

| 19% | 17.2%

7 | Legal expenses insurance and proportional reinsurance

| 10 and 22

| 8.3% | 5.5%

8 | Assistance and its proportional reinsurance

| 11 and 23 | 6.4% | 22%

9 | Miscellaneous financial loss insurance and proportional reinsurance

| 12 and 24 | 13% | 20%

10 | Non-proportional casualty reinsurance

| 26 | 17% | 20%

11 | Non-proportional marine, aviation and transport reinsurance

| 27 | 17% | 20%

12 | Non-proportional property reinsurance

| 28 | 17% | 20%

Export Article as

3A4 Standard Deviation For Non-Life Premium And Reserve Risk

1

A firm must calculate the standard deviation for non-life premium and reserve risk in accordance with the following formula:

$$\sigma_{nl} = \frac{1}{V_{nl}} \sqrt{\sum_{s,t} \text{CorrS}_{(s,t)} \sigma_s \cdot V_s \cdot \sigma_t \cdot V_t}$$

where:

1. (1) V_{nl} denotes the volume measure for non-life premium and reserve risk;
2. (2) the sum covers all possible combinations (s, t) of the segments set out in 3A3;
3. (3) $\text{CorrS}_{(s, t)}$ denotes the correlation coefficient for non-life premium and reserve risk for segment s and segment t set out in Annex IV;
4. (4) σ_s and σ_t denote standard deviations for non-life premium and reserve risk of segments s and t respectively; and
5. (5) V_s and V_t denote volume measures for premium and reserve risk of segments s and t , referred to in 3A2, respectively.

2\.

For all segments set out in 3A3, a firm must calculate the standard deviation for non-life premium and reserve risk of a particular segment s in accordance with the following formula:

$$\sigma_s = \frac{\sqrt{\sigma_{\text{prem}}^2 \cdot V_{\text{prem}}^2 + \sigma_{\text{res}}^2 \cdot V_{\text{res}}^2}}{V_{\text{prem}} + V_{\text{res}}}$$

where:

1. (1) $\sigma_{\text{prem}}(s)$ denotes the standard deviation for non-life premium risk of segment s determined in accordance with 3A4.3;
2. (2) $\sigma_{\text{res}}(s)$ denotes the standard deviation for non-life reserve risk of segment s as set out in 3A3;
3. (3) $V_{\text{prem}}(s)$ denotes the volume measure for premium risk of segment s referred to in 3A2; and
4. (4) $V_{\text{res}}(s)$ denotes the volume measure for reserve risk of segment s referred to in 3A2.

3\.

For all segments set out in 3A3, a firm must calculate the standard deviation for non-life premium risk of a particular segment as equal to the product of the standard deviation for non-life gross premium risk of the segment set out in 3A3 and the adjustment factor for non-proportional reinsurance.

4\.

For segments 1, 4 and 5 set out in 3A3 the adjustment factor for non-proportional reinsurance must be equal to 80%. For all other segments set out in 3A3 the adjustment factor for non-proportional reinsurance must be equal to 100%.

Export Article as

3A5 Factor For Geographical Diversification Of Premium And Reserve Risk

1\.

Subject to 3A5.5, 3A5.6 and 3A5.7, for all segments set out in 3A3 and 3C4, a firm must calculate the factor for geographical diversification of a particular segment s referred to in 3A2 and 3C3 in accordance with the following formula:

$$\sqrt{\frac{\sum_r \{V_{\text{prem},r,s} + V_{\text{res},r,s}\}^2}{\left(\sum_r \{V_{\text{prem},r,s} + V_{\text{res},r,s}\}\right)^2}}$$

where:

1. (1) each of the sums cover all the geographical regions set out in 3A5.8;
2. (2) $V_{\text{prem},r,s}$ denotes the volume measure for premium risk of the segment s and the region r ; and
3. (3) $V_{\text{res},r,s}$ denotes volume measure for reserve risk of the segment s and the region r .

2\.

For all segments set out in 3A3 and 3C4 and all geographical regions set out in 3A5.8, a firm must calculate the volume measure for premium risk of a particular segment s and a particular region r in the same way as the volume measure for non-life or NSLT health premium risk of the segment s as referred to in 3A2 and 3C3, but taking into account only insurance and reinsurance obligations where the underlying risk is situated in the region r .

3\.

For all segments set out in 3A3 and 3C4 and all geographical regions set out in 3A5.8 a firm must calculate the volume measure for reserve risk of a particular segment s and a particular region r in the same way as the volume measure for non-life or NSLT health reserve risk of the segment s as referred to in 3A2 and 3C3, but taking into account only insurance and reinsurance obligations where the underlying risk is situated in the region r .

4\.

For the purpose of the calculations set out in 3A5.2 and 3A5.3, the following criteria apply:

1. (1) In the case of non-life insurance, the region in which a risk is situated is,
 1. (a) if the insurance relates to a building or to a building and its contents (so far as the contents are covered by the same policy), to the region in which the building is situated;
 2. (b) if the insurance relates to a vehicle of any type, to the region of registration;
 3. (c) in the case of policies of a duration of four months or less covering travel or holiday risks (whatever the class concerned), to the region in which the policyholder took out the policy; or
 4. (d) in a case not covered by (a) to (c):
 1. (i) if the policyholder is an individual, to the region in which the individual has their habitual residence at the date when the contract of insurance is entered into; and
 2. (ii) otherwise, to the region in which the establishment of the policyholder to which the policy relates is situated and that date; and
2. (2) In the case of life insurance, the region of the commitment, in relation to a commitment entered into at any date, is
 1. (a) if the policyholder is an individual, the region in which the individual had their habitual residence at that date; or
 2. (b) if the policyholder is not an individual, the region in which the establishment of the policyholder to which the commitment relates was situated at that date;
3. where for these purposes ?commitment? means a commitment represented by contracts of insurance of a prescribed class.

5\.

Notwithstanding 3A5.1, the factor for geographical diversification must be equal to 1 for segments 6, 10, 11 and 12 set out in 3A3 and for segment 4 set out in 3C4.

Notwithstanding 3A5.1, the factor for geographical diversification for a segment set out in 3A3 must be equal to 1 if a firm uses an undertaking specific parameter for the standard deviation for non-life premium risk or non-life reserve risk of the segment to calculate the non-life premium and reserve risk sub-module.

7\.

Notwithstanding 3A5.1, the factor for geographical diversification for a segment set out in 3C4 must be equal to 1 if a firm uses an undertaking specific parameter for the standard deviation for NSLT health premium risk or NSLT health reserve risk of the segment to calculate the NSLT health premium and reserve risk sub-module.

8\.

Regions for the calculation of the factor for geographical diversification.

| Region | Territories that the region consists of

---|---|---

1 | Northern Europe | Denmark (except Greenland), Estonia, Finland, Guernsey, Iceland, Ireland, Isle of Man, Jersey, Latvia, Lithuania, Norway, Sweden, United Kingdom (except Anguilla, Bermuda, British Virgin Islands, Cayman Islands, Falkland Islands, Gibraltar, Montserrat, Pitcairn Islands, Saint Helena, Turks and Caicos Islands)

2 | Western Europe | Austria, Belgium, France (except French Guiana, French Polynesia, Guadeloupe, Martinique, Mayotte, New Caledonia, Réunion, Saint Barthélemy, Saint Martin, Saint Pierre and Miquelon, Wallis and Futuna), Germany, Liechtenstein, Luxembourg, Monaco, Netherlands (except Aruba, Bonaire, Curaçao, Saba, Sint Eustatius, Sint Maarten), Switzerland

3 | Eastern Europe

| Belarus, Bulgaria, Czech Republic, Hungary, Moldova, Poland, Romania, Russia, Slovakia, Ukraine

4 | Southern Europe

| Albania, Andorra, Bosnia and Herzegovina, Croatia, Cyprus, the former Yugoslav Republic of Macedonia, Gibraltar, Greece, Italy, Malta, Montenegro, Portugal, San Marino, Serbia, Slovenia, Spain, Vatican City State

5 | Central and Western Asia

| Armenia, Azerbaijan, Bahrain, Georgia, Iraq, Israel, Jordan, Kazakhstan, Kuwait, Kyrgyzstan, Lebanon, Oman, Qatar, Saudi Arabia, Syria, Tajikistan, Turkey, Turkmenistan, United Arab Emirates, Uzbekistan, Yemen

6 | Eastern Asia

| China, Japan, Mongolia, North Korea, South Korea, Taiwan

7 | South and South-Eastern Asia

| Afghanistan, Bangladesh, Bhutan, Brunei, Burma/Myanmar, Cambodia, India, Indonesia, Iran, Laos, Malaysia, Maldives, Nepal, Pakistan, Philippines, Singapore, Sri Lanka, Thailand, East Timor, Vietnam

8 | Oceania

| American Samoa, Australia, Cook Islands, Fiji, French Polynesia, Guam, Kiribati, Marshall Islands, Micronesia, Nauru, New Caledonia, New Zealand, Niue, Northern Mariana Islands, Palau, Papua New Guinea, Pitcairn Islands, Samoa, Solomon Islands, Tonga, Tuvalu, Vanuatu, Wallis and Futuna

9 | Northern Africa

| Algeria, Benin, Burkina Faso, Cameroon, Cape Verde, Central African Republic, Chad, Côte d'Ivoire, Egypt, Gambia, Ghana, Guinea, Guinea-Bissau, Liberia, Libya, Mali, Mauritania, Morocco, Niger, Nigeria, Saint Helena, Senegal, Sierra Leone, South Sudan, Sudan, Togo, Tunisia

10 | Southern Africa

| Angola, Botswana, Burundi, Comoros, Democratic Republic of the Congo, Djibouti, Equatorial Guinea, Eritrea, Ethiopia, Gabon, Kenya, Lesotho, Madagascar, Malawi, Mauritius, Mayotte, Mozambique, Namibia, Congo, Réunion, Rwanda, São Tomé and Príncipe, Seychelles, Somalia, South Africa, Swaziland, Uganda, Tanzania, Zambia, Zimbabwe

11 | Northern America excluding the United States of America

| Bermuda, Canada, Greenland, Saint Pierre and Miquelon

12 | Caribbean and Central America

| Anguilla, Antigua & Barbuda, Aruba, Bahamas, Barbados, Belize, Bonaire, British Virgin Islands, Cayman Islands, Costa Rica, Cuba, Curaçao, Dominica, Dominican Republic, El Salvador, Grenada, Guadeloupe, Guatemala, Haiti, Honduras, Jamaica, Martinique, Mexico, Montserrat, Nicaragua, Panama, Puerto Rico, Saint Barthélemy, Saba, Saint Kitts and Nevis, Saint Lucia, Saint Martin, Saint Vincent and the Grenadines, Sint Eustatius, Sint Maarten, Trinidad and Tobago, Turks and Caicos Islands, US Virgin Islands

13 | Eastern South America

| Brazil, Falkland Islands, French Guiana, Guyana, Paraguay, Suriname, Uruguay

14 | Northern, southern and western South America

| Argentina, Bolivia, Chile, Colombia, Ecuador, Peru, Venezuela

15 | North-east United States of America

| Connecticut, Delaware, District of Columbia, Maine, Maryland, Massachusetts, New Hampshire, New Jersey, New York, Pennsylvania, Rhode Island, Vermont

16 | South-east United States of America | Alabama, Arkansas, Florida, Georgia (US), Kentucky, Louisiana, Mississippi, North Carolina, South Carolina, Tennessee, Virginia, West Virginia

17 | Mid-west United States of America

| Illinois, Indiana, Iowa, Kansas, Michigan, Minnesota, Missouri, Nebraska, North Dakota, Ohio, Oklahoma, South Dakota, Wisconsin

18 | Western United States of America

| Alaska, Arizona, California, Colorado, Hawaii, Idaho, Montana, Nevada, New Mexico, Oregon, Texas, Utah, Washington, Wyoming

Export Article as

3A6 Non-Life Lapse Risk Sub-Module

1\.

A firm must calculate the capital requirement for the non-life lapse risk sub-module as equal to the loss in its basic own funds resulting from a combination of the following instantaneous events:

1. (1) the discontinuance of 40% of the insurance policies for which discontinuance would result in an increase in technical provisions without the risk margin; and
2. (2) where reinsurance contracts cover contracts of insurance or reinsurance contracts that will be written in the future, the decrease of 40% of the number of those future contracts of insurance or reinsurance contracts used in the calculation of technical provisions.

2\.

A firm must apply the events referred to in 3A6.1 uniformly to all relevant contracts of insurance and reinsurance

contracts and, in respect of any such reinsurance contracts, the firm must apply the event referred to in 3A6.1(1) to the underlying contracts of insurance.

3\.

For the purposes of determining the loss in its basic own funds under the event referred to in 3A6.1(1), the firm must base the calculation on the type of discontinuance that most negatively affects its basic own funds on a per policy basis.

Export Article as

3A7 Non-Life Catastrophe Risk Sub-Module

1\.

The non-life catastrophe risk sub-module must consist of all of the following sub-modules:

1. (1) the natural catastrophe risk sub-module;
2. (2) the sub-module for catastrophe risk of non-proportional property reinsurance;
3. (3) the man-made catastrophe risk sub-module; and
4. (4) the sub-module for other non-life catastrophe risk.

2\.

A firm must calculate the capital requirement for the non-life catastrophe risk sub-module in accordance with the following formula:

$$\sqrt{\left\{\text{SCR}_{\text{natCAT}}\right\}^2+\left\{\text{SCR}_{\text{npproperty}}\right\}^2}+\left\{\text{SCR}_{\text{mmCAT}}\right\}^2+\left\{\text{SCR}_{\text{CATother}}\right\}^2$$

where:

1. (1) $\text{SCR}_{\text{natCAT}}$ denotes the capital requirement for natural catastrophe risk;
2. (2) $\text{SCR}_{\text{npproperty}}$ denotes the capital requirement for the catastrophe risk of non-proportional property reinsurance;
3. (3) $\text{SCR}_{\text{mmCAT}}$ denotes the capital requirement for man-made catastrophe risk; and
4. (4) $\text{SCR}_{\text{CATother}}$ denotes the capital requirement for other non-life catastrophe risk.

Export Article as

3A8 Natural Catastrophe Risk Sub-Module

1\.

The natural catastrophe risk sub-module must consist of all of the following sub-modules:

1. (1) the windstorm risk sub-module;
2. (2) the earthquake risk sub-module;
3. (3) the flood risk sub-module;
4. (4) the hail risk sub-module; and
5. (5) the subsidence risk sub-module.

2\.

A firm must calculate the capital requirement for natural catastrophe risk in accordance with the following formula:

$$\sqrt{\sum_i \{SCR\}_i^2}$$

where:

1. (1) the sum includes all possible combinations of the sub-modules i set out in 3A8.1; and
2. (2) SCR_i denotes the capital requirement for sub-module i .

Export Article as

3A9 Windstorm Risk Sub-Module

1\.

A firm must calculate the capital requirement for windstorm risk in accordance with the following formula:

$$\sqrt{\left(\sum_{(r,s)} \{CorrWS\}_{(r,s)} \cdot \{SCR\}_{windstorm,r} \cdot \{SCR\}_{windstorm,s}\right) + \{SCR\}_{windstorm,other}^2}$$

where:

1. (1) the sum includes all possible combinations (r, s) of the regions set out in Annex V;
2. (2) $CorrWS(r, s)$ denotes the correlation coefficient for windstorm risk for region r and region s as set out in Annex V;
3. (3) $SCR(windstorm, r)$ and $SCR(windstorm, s)$ denote the capital requirements for windstorm risk in region r and s respectively; and
4. (4) $SCR(windstorm, other)$ denotes the capital requirement for windstorm risk in regions other than those set out in 3A10.

2\.

For all regions set out in Annex V, a firm must calculate the capital requirement for windstorm risk in a particular region r as the higher of the following two capital requirements:

1. (1) the capital requirement for windstorm risk in region r according to scenario A as set out in 3A9.3; and
2. (2) the capital requirement for windstorm risk in region r according to scenario B as set out in 3A9.4.

3\.

For all regions set out in Annex V, a firm must calculate the capital requirement for windstorm risk in a particular region r according to scenario A as equal to the loss in its basic own funds that would result from the following sequence of events:

1. (1) an instantaneous loss of an amount that, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, is equal to 80% of the specified windstorm loss in region r ; and
2. (2) a loss of an amount that, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, is equal to 40% of the specified windstorm loss in region r .

4\.

For all regions set out in Annex V, a firm must calculate the capital requirement for windstorm risk in a particular region r according to scenario B as equal to the loss in its basic own funds that would result from the following sequence of events:

1. (1) an instantaneous loss of an amount that, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, is equal to 100% of the specified windstorm loss in region r ; and
2. (2) a loss of an amount that, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, is equal to 20% of the specified windstorm loss in region r .

5\.

For all regions set out in Annex V, a firm must calculate the specified windstorm loss in a particular region r in accordance with the following formula:

$$L_{\text{left(windstorm}, r, \text{right})} = \sqrt{\sum_{(i, j)} \{ \text{Corr}_{\text{left(windstorm}, r, i, j, \text{right})} \} \cdot \{ \text{WSI}_{\text{left(windstorm}, r, i, \text{right})} \} \cdot \{ \text{WSI}_{\text{left(windstorm}, r, j, \text{right})} \} }$$

where:

1. (1) the sum includes all possible combinations of risk zones (i, j) of region r set out in Annex IX;
2. (2) $\text{Corr}_{\text{left(windstorm}, r, i, j, \text{right})}$ denotes the correlation coefficient for windstorm risk in risk zones i and j of region r set out in Annex XXII; and
3. (3) $\text{WSI}_{\text{left(windstorm}, r, i, \text{right})}$ and $\text{WSI}_{\text{left(windstorm}, r, j, \text{right})}$ denote the weighted sums insured for windstorm risk in risk zones i and j of region r set out in Annex IX.

6\.

For all regions set out in Annex V and all risk zones of those regions set out in Annex IX, a firm must calculate the weighted sum insured for windstorm risk in a particular windstorm zone i of a particular region r in accordance with the following formula:

$$\{ \text{WSI}_{\text{left(windstorm}, r, i, \text{right})} \} = Q_{\text{left(windstorm}, r, \text{right})} \cdot W_{\text{left(windstorm}, r, i, \text{right})} \cdot \{ \text{SI}_{\text{left(windstorm}, r, i, \text{right})} \}$$

where:

1. (1) $W_{\text{left(windstorm}, r, i, \text{right})}$ denotes the risk weight for windstorm risk in risk zone i of region r set out in Annex X;
2. (2) $\text{SI}_{\text{left(windstorm}, r, i, \text{right})}$ denotes the sum insured for windstorm risk in windstorm zone i of region r ; and
3. (3) $Q_{\text{left(windstorm}, r, \text{right})}$ denotes the windstorm risk factor for region r as set out in Annex V.

7\.

Where the amount determined for a particular risk zone in accordance with 3A9.6 exceeds an amount (referred to for these purposes as 'the lower amount') equal to the sum of the potential losses without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, that the firm could suffer for windstorm risk in that risk zone, taking into account the terms and conditions of its specific policies, including any contractual payment limits, the firm may, as an alternative calculation, determine the weighted sum insured for windstorm risk in that risk zone as the lower amount.

8\.

For all regions set out in Annex V and all risk zones of those regions set out in Annex IX, a firm must calculate the sum insured for windstorm risk in a particular windstorm zone i of a particular region r in accordance with the following formula:

$$\{ \text{SI}_{\text{left(windstorm}, r, i, \text{right})} \} = \{ \text{SI}_{\text{left(property}, r, i, \text{right})} \} + \{ \text{SI}_{\text{left(onshore-property}, r, i, \text{right})} \}$$

where:

1. (1) $\text{_SI}(\text{property}, r, i)$ denotes the sum insured by the firm for lines of business 7 and 19 in relation to contracts of insurance that cover windstorm risk and where the risk is situated in risk zone _i of region _r ; and
2. (2) $\text{_SI}(\text{onshore-property}, r, i)$ denotes the sum insured by the firm for lines of business 6 and 18 in relation to contracts of insurance that cover onshore property damage by windstorm and where the risk is situated in risk zone _i of region _r .

9\.

A firm must calculate the capital requirement for windstorm risk in regions other than those set out in 3A10 as equal to the loss in its basic own funds that would result from an instantaneous loss in relation to each contract of insurance that covers any of the following insurance or reinsurance obligations:

1. (1) obligations of lines of business 7 or 19 that cover windstorm risk and where the risk is not situated in one of the regions set out in 3A10; and
2. (2) obligations of lines of business 6 or 18 in relation to onshore property damage by windstorm and where the risk is not situated in one of the regions set out in 3A10.

10\.

A firm must calculate the amount of the instantaneous loss, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, referred to in 3A9.9 in accordance with the following formula:

$$\text{_L}_{\{\text{left}(\text{windstorm}, \text{other}\right)\}} = 1.75 \cdot \text{left}(0.5 \cdot \text{_DIV}_{\{\text{windstorm}\}} + 0.5 \cdot \text{right}) \cdot \text{_P}_{\{\text{windstorm}\}}$$

where:

1. (1) $\text{_DIV}_{\text{windstorm}}$ is calculated in accordance with 3A5, but based on the premiums in respect of the obligations referred to in 3A9.9 and restricted to the regions 5 to 18 set out in 3A5.8; and
2. (2) $\text{_P}_{\text{windstorm}}$ is an estimate of the premiums to be earned by the firm for each contract of insurance that covers the obligations referred to in 3A9.9 during the following 12 months provided that, for this purpose premiums must be gross, without deduction of premiums for reinsurance contracts.

Export Article as

3A10 List Of Regions For Which Natural Catastrophe Risk Is Not Calculated Based On Premiums

1\.

The regions for which natural catastrophe risk is not calculated based on premiums are:

1. (1) Member States of the European Union;
2. (2) Principality of Andorra;
3. (3) Republic of Iceland;
4. (4) Principality of Lichtenstein;
5. (5) Principality of Monaco;
6. (6) Kingdom of Norway;
7. (7) Republic of San Marino;
8. (8) Swiss Confederation;
9. (9) Vatican City State; and
10. (10) The United Kingdom.

Export Article as

3A11 Earthquake Risk Sub-Module

1\.

A firm must calculate the capital requirement for earthquake risk in accordance with the following formula:

$$\sqrt{\left\{\sum_{r,s}\text{CorrEQ}(r,s)\cdot\text{SCR}(\text{earthquake},r)\cdot\text{SCR}(\text{earthquake},s)\right\}+\left\{\text{SCR}^2(\text{earthquake},\text{other})\right\}}$$

where:

- (1) the sum includes all possible combinations (r, s) of the regions set out in Annex VI;
- (2) $\text{CorrEQ}(r, s)$ denotes the correlation coefficient for earthquake risk for region r and region s as set out in Annex VI;
- (3) $\text{SCR}(\text{earthquake}, r)$ and $\text{SCR}(\text{earthquake}, s)$ denote the capital requirements for earthquake risk in region r and s respectively; and
- (4) $\text{SCR}(\text{earthquake}, \text{other})$ denotes the capital requirement for earthquake risk in regions other than those set out in 3A10.

2\.

For all regions set out in Annex VI, a firm must calculate the capital requirement for earthquake risk in a particular region r as equal to the loss in its basic own funds that would result from an instantaneous loss of an amount that, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, is calculated in accordance with the following formula:

$$\sqrt{\sum_{i,j}\text{Corr}(\text{earthquake}, r, i, j)\cdot\text{WSI}(\text{earthquake}, r, i)\cdot\text{WSI}(\text{earthquake}, r, j)}$$

where:

- (1) the sum includes all possible combinations of risk zones (i, j) of region r set out in Annex IX;
- (2) $\text{Corr}(\text{earthquake}, r, i, j)$ denotes the correlation coefficient for earthquake risk in risk zones i and j of region r set out in Annex XXIII; and
- (3) $\text{WSI}(\text{earthquake}, r, i)$ and $\text{WSI}(\text{earthquake}, r, j)$ denote the weighted sums insured for earthquake risk in risk zones i and j of region r set out in Annex IX.

3\.

For all regions set out in Annex VI and all risk zones of those regions set out in Annex IX, a firm must calculate the weighted sum insured for earthquake risk in a particular earthquake zone i of a particular region r in accordance with the following formula:

$$\text{WSI}(\text{earthquake}, r, i) = Q(\text{earthquake}, r) \cdot W(\text{earthquake}, r, i) \cdot \text{SI}(\text{earthquake}, r, i)$$

where:

- (1) $W(\text{earthquake}, r, i)$ denotes the risk weight for earthquake risk in risk zone i of region r set out in Annex X;
- (2) $\text{SI}(\text{earthquake}, r, i)$ denotes the sum insured for earthquake risk in earthquake zone i of region r ; and
- (3) $Q(\text{earthquake}, r)$ denotes the earthquake risk factor for region r as set out in Annex VI.

4\.

Where the amount determined for a particular risk zone in accordance with 3A11.3 exceeds an amount (referred to for these purposes as 'the lower amount') equal to the sum of the potential losses, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, that the firm could suffer for earthquake risk in that risk zone, taking into account the terms and conditions of its specific policies, including any contractual payment limits, the firm may, as an alternative calculation, determine the weighted sum insured for earthquake risk in that risk zone as the lower amount.

5\.

For all regions set out in Annex VI and all risk zones of those regions set out in Annex IX, a firm must calculate the sum insured for earthquake risk in a particular earthquake zone i of a particular region r in accordance with the following formula:

$$\text{SI}_{(i)} = \text{SI}_{(i)}^{\text{property}} + \text{SI}_{(i)}^{\text{onshore-property}}$$

where:

1. (1) $\text{SI}_{(i)}^{\text{property}}$ denotes the sum insured of the firm for lines of business 7 and 19 in relation to contracts of insurance that cover earthquake risk and where the risk is situated in risk zone i of region r ; and
2. (2) $\text{SI}_{(i)}^{\text{onshore-property}}$ denotes the sum insured of the firm for lines of business 6 and 18 in relation to contracts of insurance that cover onshore property damage by earthquake and where the risk is situated in risk zone i of region r .

6\.

A firm must calculate the capital requirement for earthquake risk in regions other than those set out in 3A10 as equal to the loss in its basic own funds that would result from an instantaneous loss in relation to each contract of insurance that covers one or both of the following insurance or reinsurance obligations:

1. (1) obligations of lines of business 7 or 19 that cover earthquake risk, where the risk is not situated in one of the regions set out in 3A10; and
2. (2) obligations of lines of business 6 or 18 in relation to onshore property damage by earthquake, where the risk is not situated in one of the regions set out in 3A10.

7\.

A firm must calculate the amount of the instantaneous loss, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, referred to in 3A11.6, in accordance with the following formula:

$$L = 1.2 \times \left(0.5 \times \text{DIV}_{(i)} + 0.5 \times P_{(i)} \right) \times P_{(i)}$$

where:

1. (1) $\text{DIV}_{(i)}$ is calculated in accordance with 3A5, but based on the premiums in respect of the obligations referred to in 3A11.6(1) and 3A11.6(2) and restricted to the regions 5 to 18 set out in 3A5; and
2. (2) $P_{(i)}$ is an estimate of the premiums to be earned by the firm for each contract of insurance that covers the obligations referred to in 3A11.6(1) and 3A11.6(2) during the following 12 months provided that, for this purpose premiums must be gross, without deduction of premiums for reinsurance contracts.

Export Article as

3A12 Flood Risk Sub-module

1\.

A firm must calculate the capital requirement for flood risk in accordance with the following formula:

$$\sqrt{\left\{\sum_{r,s} \left\{ \text{CorrFL}_{r,s} \cdot \text{SCR}_{r,\text{flood}} \cdot \text{SCR}_{s,\text{flood}} \right\} \right\}} + \text{SCR}^2_{\text{flood, other}}$$

where:

1. (1) the sum includes all possible combinations (r, s) of the regions set out in Annex VII;
2. (2) $\text{CorrFL}_{r,s}$ denotes the correlation coefficient for flood risk for region r and region s as set out in Annex VII;
3. (3) $\text{SCR}_{r,\text{flood}}$ and $\text{SCR}_{s,\text{flood}}$ denote the capital requirements for flood risk in region r and s respectively; and
4. (4) $\text{SCR}_{\text{flood, other}}$ denotes the capital requirement for flood risk in regions other than those set out in 3A10.

2\.

For all regions set out in Annex VII, the capital requirement for flood risk in a particular region r must be the higher of the following capital requirements:

1. (1) the capital requirement for flood risk in region r according to scenario A as set out in 3A12.3; and
2. (2) the capital requirement for flood risk in region r according to scenario B as set out in 3A12.4.

3\.

For all regions set out in Annex VII, a firm must calculate the capital requirement for flood risk in a particular region r according to scenario A as equal to the loss in its basic own funds that would result from the following sequence of events:

1. (1) an instantaneous loss of an amount that, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, is equal to 65% of the specified flood loss in region r ; and
2. (2) a loss of an amount that, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, is equal to 45% of the specified flood loss in region r .

4\.

For all regions set out in Annex VII, a firm must calculate the capital requirement for flood risk in a particular region r according to scenario B as equal to the loss in its basic own funds that would result from the following sequence of events:

1. (1) an instantaneous loss of an amount that, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, is equal to 100% of the specified flood loss in region r ; and
2. (2) a loss of an amount that, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, is equal to 10% of the specified flood loss in region r .

5\.

For all regions set out in Annex VII, a firm must calculate the specified flood loss in a particular region r in accordance with the following formula:

$$L_{r,\text{flood}} = \sqrt{\sum_{i,j} \left\{ \text{Corr}_{r,i,j} \cdot \text{WSI}_{r,i,\text{flood}} \cdot \text{WSI}_{r,j,\text{flood}} \right\}}$$

where:

1. (1) the sum includes all possible combinations of risk zones $_i, _j$ of region $_r$ set out in Annex IX;
2. (2) $_Corr(flood, r, i, j)$ denotes the correlation coefficient for flood risk in flood zones $_i$ and $_j$ of region $_r$ set out in Annex XXIV; and
3. (3) $_WSI(flood, r, i)$ and $_WSI(flood, r, j)$ denote the weighted sums insured for flood risk in risk zones $_i$ and $_j$ of region $_r$ set out in Annex IX.

6\.

For all regions set out in Annex VII and all risk zones of those regions set out in Annex IX, a firm must calculate the weighted sum insured for flood risk in a particular flood zone $_i$ of a particular region $_r$ in accordance with the following formula:

$$_WSI_{\left(flood, r, i\right)} = Q_{\left(flood, r\right)} \cdot W_{\left(flood, r, i\right)} \cdot SI_{\left(flood, r, i\right)}$$

where:

1. (1) $_W(flood, r, i)$ denotes the risk weight for flood risk in risk zone $_i$ of region $_r$ set out in Annex X;
2. (2) $_SI(flood, r, i)$ denotes the sum insured for flood risk in flood zone $_i$ of region $_r$; and
3. (3) $_Q(flood, r)$ denotes the flood risk factor for region $_r$ as set out in Annex VII.

7\.

Where the amount determined for a particular risk zone in accordance with 3A12.6 exceeds an amount (referred to for these purposes as 'the lower amount') equal to the sum of the potential losses, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, that the firm could suffer for flood risk in that risk zone, taking into account the terms and conditions of its specific policies, including any contractual payment limits, the firm may, as an alternative calculation, determine the weighted sum insured for flood risk in that risk zone as the lower amount.

8\.

For all regions set out in Annex VII and all risk zones of those regions set out in Annex IX, a firm must calculate the sum insured for flood risk for a particular risk zone $_i$ of a particular region $_r$ in accordance with the following formula:

$$SI_{\left(flood, r, i\right)} = SI_{\left(property, r, i\right)} + SI_{\left(onshore-property, r, i\right)} + 1.5 \cdot SI_{\left(motor, r, i\right)}$$

where:

1. (1) $_SI(property, r, i)$ denotes the sum insured by the firm for lines of business 7 and 19 in relation to contracts of insurance that cover flood risk, where the risk is situated in risk zone $_i$ of region $_r$;
2. (2) $_SI(onshore-property, r, i)$ denotes the sum insured by the firm for lines of business 6 and 18 in relation to contracts of insurance that cover onshore property damage by flood and where the risk is situated in risk zone $_i$ of region $_r$; and
3. (3) $_SI(motor, r, i)$ denotes the sum insured by the firm for lines of business 5 and 17 in relation to contracts of insurance that cover flood risk, where the risk is situated in risk zone $_i$ of region $_r$.

9\.

A firm must calculate the capital requirement for flood risk in regions other than those set out in 3A10, as equal to the loss in its basic own funds that would result from an instantaneous loss in relation to each contract of insurance that covers any of the following insurance or reinsurance obligations:

1. (1) obligations of lines of business 7 or 19 that cover flood risk, where the risk is not situated in one of the regions set out in 3A10;
2. (2) obligations of lines of business 6 or 18 in relation to onshore property damage by flood, where the risk is not situated in one of the regions set out in 3A10; and
3. (3) obligations of lines of business 5 or 17 that cover flood risk, where the risk is not situated in one of the regions set out in 3A10.

10\.

A firm must calculate the amount of the instantaneous loss, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, referred to in 3A12.9, in accordance with the following formula:

$$\mathbb{L}_{\{\text{flood, other}\}} = 1.1 \cdot \left(0.5 \cdot \text{DIV}_{\{\text{flood}\}} + 0.5 \right) \cdot P_{\{\text{flood}\}}$$

where:

1. (1) $\text{DIV}_{\{\text{flood}\}}$ is calculated in accordance with 3A5, but based on the premiums in respect of the obligations referred to in 3A12.9(1), (2) and (3) and restricted to the regions 5 to 18 set out in 3A5.8; and
2. (2) $P_{\{\text{flood}\}}$ is an estimate of the premiums to be earned by the firm for each contract of insurance that covers the obligations referred to in 3A12.9(1), (2) and (3) during the following 12 months provided that, for this purpose, premiums must be gross, without deduction of premiums for reinsurance contracts.

Export Article as

3A13 Hail Risk Sub-Module

1\.

A firm must calculate the capital requirement for hail risk in accordance with the following formula:

$$\mathbb{SCR}_{\{\text{hail}\}} = \sqrt{\left(\sum_{\{\text{r, s}\}} \{\text{CorrHL}\}_{\{\text{r, s}\}} \cdot \text{SCR}_{\{\text{hail, r}\}} \cdot \text{SCR}_{\{\text{hail, s}\}} \right) + \{\text{SCR}\}^2_{\{\text{hail, other}\}}}$$

where:

1. (1) the sum includes all possible combinations $\{\text{r, s}\}$ of the regions set out in Annex VIII;
2. (2) $\text{CorrHL}_{\{\text{r, s}\}}$ denotes the correlation coefficient for hail risk for region $\{\text{r}\}$ and region $\{\text{s}\}$ as set out in Annex VIII;
3. (3) $\text{SCR}_{\{\text{hail, r}\}}$ and $\text{SCR}_{\{\text{hail, s}\}}$ denote the capital requirements for hail risk in regions $\{\text{r}\}$ and $\{\text{s}\}$ respectively; and
4. (4) $\text{SCR}_{\{\text{hail, other}\}}$ denotes the capital requirement for hail risk in regions other than those set out in 3A10.

2\.

For all regions set out in Annex VIII, a firm must calculate the capital requirement for hail risk in a particular region $\{\text{r}\}$ as the higher of the following capital requirements:

1. (1) the capital requirement for hail risk in region $\{\text{r}\}$ according to scenario A as set out in 3A13.3; and
2. (2) the capital requirement for hail risk in region $\{\text{r}\}$ according to scenario B as set out in 3A13.4.

3\.

For all regions set out in Annex VIII, a firm must calculate the capital requirement for hail risk in a particular region $\{\text{r}\}$

according to scenario A as equal to the loss in its basic own funds that would result from the following sequence of events:

1. (1) an instantaneous loss of an amount that, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, is equal to 70% of the specified hail loss in region r ; and
2. (2) a loss of an amount that, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, is equal to 50% of the specified hail loss in region r .

4\.

For all regions set out in Annex VIII, a firm must calculate the capital requirement for hail risk in a particular region r according to scenario B as equal to the loss in its basic own funds that would result from the following sequence of events:

1. (1) an instantaneous loss of an amount that, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, is equal to 100% of the specified hail loss in region r ; and
2. (2) a loss of an amount that, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, is equal to 20% of the specified hail loss in region r .

5\.

For all regions set out in Annex VIII, a firm must calculate the specified hail loss in a particular region r in accordance with the following formula:

$$L_{\left(hail, r\right)} = \sqrt{\sum_{\left(i, j\right)} \left\{ Corr_{\left(hail, r, i, j\right)} \cdot WSI_{\left(hail, r, i\right)} \cdot WSI_{\left(hail, r, j\right)} \right\}}$$

where:

1. (1) the sum includes all possible combinations of risk zones (i, j) of region r set out in Annex IX;
2. (2) $Corr(hail, r, i, j)$ denotes the correlation coefficient for hail risk in risk zones i and j of region r set out in Annex XXV; and
3. (3) $WSI(hail, r, i)$ and $WSI(hail, r, j)$ denote the weighted sums insured for hail risk in risk zones i and j of region r set out in Annex IX.

6\.

For all regions set out in Annex VIII and all risk zones of those regions set out in Annex IX, a firm must calculate the weighted sum insured for hail risk in a particular hail zone i of a particular region r in accordance with the following formula:

$$WSI_{\left(hail, r, i\right)} = Q_{\left(hail, r\right)} \cdot W_{\left(hail, r, i\right)} \cdot SI_{\left(hail, r, i\right)}$$

where:

1. (1) $W(hail, r, i)$ denotes the risk weight for hail risk in risk zone i of region r set out in Annex X;
2. (2) $SI(hail, r, i)$ denotes the sum insured for hail risk in hail zone i of region r ; and
3. (3) $Q(hail, r)$ denotes the hail risk factor for region r as set out in Annex VIII.

7\.

Where the amount determined for a particular risk zone in accordance with 3A13.6 exceeds an amount (referred to for these purposes as 'the lower amount') equal to the sum of the potential losses, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, that the firm could suffer for hail risk in that risk

zone taking into account the terms and conditions of its specific policies, including any contractual payment limits, the firm may, as an alternative calculation, determine the weighted sum insured for hail risk in that risk zone as the lower amount.

8\.

For all regions set out in Annex VIII and all hail zones, a firm must calculate the sum insured for hail risk in a particular hail zone i of a particular region r in accordance with the following formula:

$$\text{SI}(i, r) = \text{SI}_{\text{property}}(i, r) + \text{SI}_{\text{onshore-property}}(i, r) + 5 \cdot \text{SI}_{\text{motor}}(i, r)$$

where:

1. (1) $\text{SI}_{\text{property}}(i, r)$ denotes the sum insured by the firm for lines of business 7 and 19 in relation to contracts of insurance that cover hail risk, where the risk is situated in risk zone i of region r ;
2. (2) $\text{SI}_{\text{onshore-property}}(i, r)$ denotes the sum insured by the firm for lines of business 6 and 18 in relation to contracts of insurance that cover onshore property damage by hail, where the risk is situated in risk zone i of region r ; and
3. (3) $\text{SI}_{\text{motor}}(i, r)$ denotes the sum insured by the firm for insurance or reinsurance obligations for lines of business 5 and 17 in relation to contracts of insurance that cover hail risk, where the risk is situated in risk zone i of region r .

9\.

A firm must calculate the capital requirement for hail risk in regions other than those set out in 3A10, as equal to the loss in its basic own funds that would result from an instantaneous loss in relation to each contract of insurance that covers one or more of the following insurance or reinsurance obligations:

1. (1) obligations of lines of business 7 or 19 that cover hail risk, where the risk is not situated in one of the regions set out in 3A10;
2. (2) obligations of lines of business 6 or 18 in relation to onshore property damage by hail, where the risk is not situated in one of the regions set out in 3A10; and
3. (3) obligations of lines of business 5 or 17 that cover hail risk, where the risk is not situated in one of the regions set out in 3A10.

10\.

A firm must calculate the amount of the instantaneous loss, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, referred to in 3A13.9, in accordance with the following formula:

$$L(i, \text{other}) = 0.3 \cdot \left(0.5 \cdot \text{DIV}(i) + 0.5 \right) \cdot P(i)$$

where:

1. (1) $\text{DIV}(i)$ is calculated in accordance with 3A5, but based on the premiums in respect of the obligations referred to in 3A13.9(1), (2) and (3) and restricted to the regions 5 to 18 set out in 3A5; and
2. (2) $P(i)$ is an estimate of the premiums to be earned by the firm for each contract of insurance that covers the obligations referred to in 3A13.9(1), (2) and (3) during the following 12 months provided that, for this purpose premiums must be gross, without deduction of premiums for reinsurance contracts.

Export Article as

3A14 Subsidence Risk Sub-Module

1\.

A firm must calculate the capital requirement for subsidence risk as equal to the loss in its basic own funds that would result from an instantaneous loss of an amount that, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, is calculated in accordance with the following formula:

$$\sqrt{L_{\left\{ \left(\text{subsidence} \right) \right\}} = \sqrt{\sum_{\left\{ \left(i, j \right) \right\}} \left\{ \text{Corr} \right\}_{\left\{ \left(\text{subsidence}, i, j \right) \right\}} \cdot \left\{ \text{WSI} \right\}_{\left\{ \left(\text{subsidence}, i \right) \right\}} \cdot \left\{ \text{WSI} \right\}_{\left\{ \left(\text{subsidence}, j \right) \right\}}} }$$

where:

1. (1) the sum includes all possible combinations of risk zones $_i, j_$ of France set out in Annex IX;
2. (2) $_Corr(\text{subsidence}, i, j)_$ denotes the correlation coefficient for subsidence risk in risk zones $_i_$ and $_j_$ set out in Annex XXVI; and
3. (3) $_WSI(\text{subsidence}, i)_$ and $_WSI(\text{subsidence}, j)_$ denote the weighted sums insured for subsidence risk in risk zones $_i_$ and $_j_$ of France set out in Annex IX.

2\.

For all subsidence zones, a firm must calculate the weighted sum insured for subsidence risk in a particular risk zone $_i_$ of France set out in Annex IX in accordance with the following formula:

$$\left\{ \left\{ \text{WSI} \right\}_{\left\{ \left(\text{subsidence}, i \right) \right\}} \right\} = 0.0005 \cdot W_{\left\{ \left(\text{subsidence}, i \right) \right\}} \cdot \left\{ \left\{ \text{SI} \right\}_{\left\{ \left(\text{subsidence}, i \right) \right\}} \right\}$$

where:

1. (1) $_W(\text{subsidence}, i)_$ denotes the risk weight for subsidence risk in risk zone $_i_$ set out in Annex X; and
2. (2) $_SI(\text{subsidence}, i)_$ denotes the sum insured of the firm for lines of business 7 and 19 in relation to contracts of insurance that cover subsidence risk of residential buildings in subsidence zone $_i_$.

3\.

Where the amount determined for a particular risk zone in accordance with 3A14.2 exceeds an amount (referred to for these purposes as 'the lower amount') equal to the sum of the potential losses, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, that the firm could suffer for subsidence risk in that risk zone, taking into account the terms and conditions of its specific policies, including any contractual payment limits, the firm may, as an alternative calculation, determine the weighted sum insured for subsidence risk in that risk zone as the lower amount.

Export Article as

3A15 Interpretation Of Catastrophe Scenarios

1\.

For the purposes of 3A9.3 and 3A9.4, 3A12.3 and 3A12.4 and 3A13.3 and 3A13.4, a firm must base the calculation of the capital requirement on the following assumptions:

1. (1) the two consecutive events referred to in those rules are independent; and
2. (2) the firm does not enter into new insurance risk-mitigation techniques between the occurrence of the two events.

2\.

Notwithstanding 3.3A(1)(d), where current reinsurance contracts allow for reinstatements:

1. (1) a firm must take into account future management actions in relation to the reinstatements between the occurrence of the first and the second event; and
2. (2) the assumptions about future management actions must be realistic, objective and verifiable.

Export Article as

3A16 Sub-Module For Catastrophe Risk Of Non-Proportional Property Reinsurance

1\.

A firm must calculate the capital requirement for catastrophe risk of non-proportional property reinsurance as equal to the loss in its basic own funds that would result from an instantaneous loss in relation to each reinsurance contract that covers reinsurance obligations of line of business 28 other than non-proportional reinsurance obligations relating to insurance obligations included in lines of business 9 and 21.

2\.

A firm must calculate the amount of the instantaneous loss, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, referred to in 3A16.1 in accordance with the following formula:

$$\llbracket L_{\{npproperty\}} = 2.5 \cdot \left(0.5 \cdot \{DIV\}_{\{npproperty\}} + 0.5 \right) \cdot P_{\{npproperty\}} \rrbracket$$

where:

1. (1) $\{DIV\}_{npproperty}$ is calculated in accordance with 3A5, but based on the premiums earned by the firm in line of business 28, other than non-proportional reinsurance obligations relating to insurance obligations included in lines of business 9 and 21;
2. (2) $\{P\}_{npproperty}$ is an estimate of the premiums to be earned by the firm during the following 12 months for each reinsurance contract that covers the reinsurance obligations of line of business 28 other than non-proportional reinsurance obligations relating to insurance obligations included in lines of business 9 and 21 provided that for this purpose premiums must be gross, without deduction of premiums for reinsurance contracts.

Export Article as

3A17 Man-Made Catastrophe Risk Sub-Module

1\.

The man-made catastrophe risk sub-module must consist of all of the following sub-modules:

1. (1) the motor vehicle liability risk sub-module;
2. (2) the marine risk sub-module;
3. (3) the aviation risk sub-module;
4. (4) the fire risk sub-module;
5. (5) the liability risk sub-module; and
6. (6) the credit and suretyship risk sub-module.

2\.

A firm must calculate the capital requirement for the man-made catastrophe risk in accordance with the following formula:

$$\sqrt{\sum_i \{SCR\}_i^2}$$

where:

1. (1) the sum includes all sub-modules set out in 3A17.1; and
2. (2) $\{SCR\}_i$ denotes the capital requirements for sub-module i .

Export Article as

3A18 Motor Vehicle Liability Risk Sub-Module

1\.

A firm must calculate the capital requirement for motor vehicle liability risk as equal to the loss in its basic own funds that would result from an instantaneous loss that, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, is calculated in accordance with the following formula in GBP:

$$L_{\{motor\}} = \max\left(5,300,000; 44,000 \cdot \sqrt{N_a + 0.05 \cdot N_b + 0.95 \cdot \min(N_b, 20,000)}\right)$$

where:

1. (1) N_a is the number of vehicles insured by the firm in lines of business 4 and 16 with a deemed policy limit above GBP 21,200,000; and
2. (2) N_b is the number of vehicles insured by the firm in lines of business 4 and 16 with a deemed policy limit below or equal to GBP 21,200,000.

2\.

The number of motor vehicles covered by the proportional reinsurance obligations of the firm must be weighted by the relative share of the firm's obligations in respect of the sum insured of the motor vehicles.

3\.

The deemed policy limit referred to in 3A18.1 must be:

1. (1) the overall limit of the motor vehicle liability insurance policy or, where no such overall limit is specified in the terms and conditions of the policy, the sum of the limits for damage to property and for personal injury; or
2. (2) where the policy limit is specified as a maximum per victim, based on the assumption of ten victims.

Export Article as

3A19 Marine Risk Sub-Module

1\.

A firm must calculate the capital requirement for marine risk in accordance with the following formula:

$$\sqrt{\{SCR\}_{\{vessel\}}^2 + \{SCR\}_{\{platform\}}^2}$$

where:

1. (1) $\{SCR\}_{\{vessel\}}$ is the capital requirement for the risk of a vessel collision; and
2. (2) $\{SCR\}_{\{platform\}}$ is the capital requirement for the risk of a platform explosion.

2\.

A firm must calculate the capital requirement for the risk of a vessel collision as equal to the loss in its basic own funds that would result from an instantaneous loss of an amount calculated in accordance with the following formula:

$$\llbracket L_{\text{vessel}} = \{\max\}_v \left(\{\text{SI}\}_{\text{hull}, v} + \{\text{SI}\}_{\text{liab}, v} + \{\text{SI}\}_{\text{pollution}, v} \right) \rrbracket$$

where:

1. (1) the maximum relates to all sea, lake, river, and canal vessels insured by the firm in respect of vessel collision in lines of business 6, 18 and 27 where the insured value of the vessel is at least GBP 220,000;
2. (2) $\text{SI}_{\text{hull}, v}$ is the sum insured by the firm, after deduction of the amounts that the firm can recover from reinsurance contracts and special purpose vehicles, for marine hull insurance and reinsurance in relation to vessel v ;
3. (3) $\text{SI}_{\text{liab}, v}$ is the sum insured by the firm, after deduction of the amounts that the firm can recover from reinsurance contracts and special purpose vehicles, for marine liability insurance and reinsurance in relation to vessel v ; and
4. (4) $\text{SI}_{\text{pollution}, v}$ is the sum insured by the firm, after deduction of the amounts that the firm can recover from reinsurance contracts and special purpose vehicles, for oil pollution insurance and reinsurance in relation to vessel v .

3\.

For the purposes of determining $\text{SI}_{\text{hull}, v}$, $\text{SI}_{\text{liab}, v}$, and $\text{SI}_{\text{pollution}, v}$, a firm must only take into account reinsurance contracts and special purpose vehicles that would pay out in the event of insurance claims related to vessel v and must not take into account reinsurance contracts and special purpose vehicles where payout is dependent on insurance claims not related to vessel v .

4\.

Where the deduction of amounts recoverable would lead to a capital requirement for the risk of a vessel collision that insufficiently captures the risk of a vessel collision that the firm is exposed to, the firm must calculate $\text{SI}_{\text{hull}, v}$, $\text{SI}_{\text{liab}, v}$, or $\text{SI}_{\text{pollution}, v}$ without deduction of amounts recoverable.

5\.

A firm must calculate the capital requirement for the risk of a platform explosion as equal to the loss in its basic own funds that would result from an instantaneous loss of an amount calculated in accordance with the following formula:

$$\llbracket L_{\text{platform}} = \underset{p}{\{\max\}} \left(\{\text{SI}\}_p \right) \rrbracket$$

where:

1. (1) the maximum relates to all oil and gas offshore platforms insured by the firm in respect of platform explosion in lines of business 6, 18, and 27; and
2. (2) SI_p is the accumulated sum insured by the firm, after deduction of the amounts that the firm can recover from reinsurance contracts and special purpose vehicles, for the following insurance and reinsurance obligations in relation to platform p :
 1. (a) obligations to compensate for property damage;
 2. (b) obligations to compensate for the expenses for the removal of wreckage;
 3. (c) obligations to compensate for loss of production income;
 4. (d) obligations to compensate for the expenses for capping of the well or making the well secure; and
 5. (e) liability insurance and reinsurance obligations.

6\.

For the purposes of determining $_{SI\ p}$, a firm must only take into account reinsurance contracts and special purpose vehicles that would pay out in the event of insurance claims related to platform $_p$ and must not take into account reinsurance contracts and special purpose vehicles where payout is dependent on insurance claims that are not related to platform $_p$.

7\.

Where the deduction of amounts recoverable would lead to a capital requirement for the risk of a platform explosion that insufficiently captures the risk of a platform explosion that the firm is exposed to, the firm must calculate $_{SI\ p}$ without the deduction of amounts recoverable.

Export Article as

3A20 Aviation Risk Sub-Module

1\.

A firm must calculate the capital requirement for aviation risk as equal to the loss in its basic own funds that would result from an instantaneous loss of an amount calculated in accordance with the following formula:

$$\mathbb{L}_{\{aviation\}} = \underset{a}{\max} \left(\mathbb{S}I_a \right)$$

where:

1. (1) the maximum relates to all aircrafts insured by the firm in lines of business 6, 18, and 27; and
2. (2) $_{SI\ a}$ is the sum insured by the firm, after deduction of the amounts that the firm can recover from reinsurance contracts and special purpose vehicles, for aviation hull insurance and reinsurance and aviation liability insurance and reinsurance in relation to aircraft $_a$.

2\.

For the purposes of 3A20, a firm must only take into account reinsurance contracts and special purpose vehicles that would pay out in the event of insurance claims related to aircraft $_a$ and must not take into account reinsurance contracts and special purpose vehicles where payout is dependent on insurance claims that are not related to aircraft $_a$.

3\.

Where the deduction of amounts recoverable would lead to a capital requirement for aviation risk that insufficiently captures the aviation risk that the firm is exposed to, the firm must, calculate $_{SI\ a}$ without the deduction of amounts recoverable.

Export Article as

3A21 Fire Risk Sub-Module

1\.

A firm must calculate the capital requirement for fire risk as equal to the loss in its basic own funds that would result from an instantaneous loss of an amount equal to the sum insured by the firm with respect to the largest fire risk concentration.

2\.

The largest fire risk concentration of a firm is the set of buildings with the highest sum insured, after deduction of the amounts that the firm can recover from reinsurance contracts and special purpose vehicles, that meets all of the following requirements:

1. (1) the firm has insurance or reinsurance obligations in lines of business 7 and 19, in relation to each building that cover damage due to fire or explosion, including as a result of terrorist attacks; and
2. (2) all buildings are partly or fully located within a radius of 200 metres.

3\.

In determining the sum insured for a set of buildings, a firm must only take into account reinsurance contracts and special purpose vehicles that would pay out in the event of insurance claims related to that set of buildings and must not take into account reinsurance contracts and special purpose vehicles where payout is dependent on insurance claims that are not related to that set of buildings.

4\.

Where the deduction of amounts recoverable would lead to a capital requirement for fire risk that insufficiently captures the fire risk that the firm is exposed to, the firm must calculate the sum insured for a set of buildings without the deduction of amounts recoverable.

5\.

For the purposes of 3A21.2 to 3A21.4, the set of buildings may be covered by one or several contracts of insurance or reinsurance contracts.

Export Article as

3A22 Liability Risk Sub-Module

1\.

A firm must calculate the capital requirement for liability risk in accordance with the following formula:

$$\sqrt{\sum_{i,j} \{ \text{Corr}(\text{liability}, i, j) \} \cdot \text{SCR}_{\text{left}}(\text{liability}, i) \cdot \text{SCR}_{\text{left}}(\text{liability}, j) } \sqrt{\text{liability}, j}$$

where:

1. (1) the sum includes all possible combinations of liability risk groups $_i, j_$ as set out in Annex XI;
2. (2) $_ \text{Corr}(\text{liability}, i, j) _$ denotes the correlation coefficient for liability risk of liability risk groups $_i_$ and $_j_$ as set out in Annex XI; and
3. (3) $_ \text{SCR}(\text{liability}, i) _$ denotes the capital requirement for liability risk of liability risk group $_i_$.

2\.

For all liability risk groups set out in Annex XI, a firm must calculate the capital requirement for liability risk of a particular liability risk group $_i_$ as equal to the loss in its basic own funds that would result from an instantaneous loss of an amount that, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, is calculated in accordance with the following formula:

$$\sqrt{L_{\text{left}}(\text{liability}, i) = f_{\text{left}}(\text{liability}, i) \cdot P_{\text{left}}(\text{liability}, i)}$$

where:

1. (1) $f(\text{liability}, i)$ denotes the risk factor for liability risk group i as set out in Annex XI; and
2. (2) $P(\text{liability}, i)$ denotes the premiums earned by the firm during the following 12 months in relation to insurance and reinsurance obligations in liability risk group i ; for this purpose premiums must be gross, without deduction of premiums for reinsurance contracts.

3\.

The calculation of the loss in basic own funds referred to in 3A22.2 must be based on the following assumptions:

1. (1) the loss of liability risk group i is caused by n_i claims and the losses caused by these claims are representative for the business of the firm in liability risk group i and sum up to the loss of liability risk group i ; and
2. (2) the number of claims n_i is equal to the lowest integer that exceeds the following amount:

$$\left\lceil \frac{f(\text{liability}, i) \cdot P(\text{liability}, i)}{1.15 \cdot \text{Lim}(i, 1)} \right\rceil$$

where:

1. (a) $f(\text{liability}, i)$ and $P(\text{liability}, i)$ are defined as in 3A22.2; and
2. (b) $\text{Lim}(i, 1)$ denotes the highest liability limit of indemnity provided by the firm in liability risk group i ;
1. (3) where the firm provides unlimited cover in liability risk group i , the number of claims n_i is equal to one.

Export Article as

3A23 Credit And Suretyship Risk Sub-Module

1\.

A firm must calculate the capital requirement for credit and suretyship risk in accordance with the following formula:

$$\sqrt{\text{SCR}_{\text{default}}^2 + \text{SCR}_{\text{recession}}^2}$$

where:

1. (1) $\text{SCR}_{\text{default}}$ is the capital requirement for the risk of a large credit default; and
2. (2) $\text{SCR}_{\text{recession}}$ is the capital requirement for recession risk.

2\.

A firm must calculate the capital requirement for the risk of a large credit default as equal to the loss in its basic own funds that would result from an instantaneous default of the two largest exposures relating to obligations included in lines of business 9 and 21 of the firm and must base this calculation on the assumption that the loss-given-default, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, of each exposure is 10% of the sum insured in relation to the exposure.

3\.

The two largest credit insurance exposures referred to in 3A23.2 must be determined based on a comparison of the net loss-given-default of the credit insurance exposures, being the loss-given-default after deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles.

4\.

A firm must calculate the capital requirement for recession risk as equal to the loss in its basic own funds that would

result from an instantaneous loss of an amount that, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, is equal to 100% of the premiums earned by the firm during the following 12 months in lines of business 9 and 21.

Export Article as

3A24 Sub-Module For Other Non-Life Catastrophe Risk

1\.

A firm must calculate the capital requirement for other non-life catastrophe risk as equal to the loss in its basic own funds that would result from an instantaneous loss, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, that is equal to the following amount:

$$\sqrt{\left(c_1 \cdot P_1 + c_2 \cdot P_2\right)^2 + \left(c_3 \cdot P_3\right)^2 + \left(c_4 \cdot P_4\right)^2 + \left(c_5 \cdot P_5\right)^2}$$

where:

1. (1) P_1 , P_2 , P_3 , P_4 , and P_5 denote estimates of the gross premium, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, expected to be earned by the firm during the following 12 months in relation to the groups of insurance and reinsurance obligations 1 to 5 set out in Annex XII; and
2. (2) c_1 , c_2 , c_3 , c_4 , and c_5 denote the risk factors for the groups of insurance and reinsurance obligations 1 to 5 set out in Annex XII.

Export Article as

3B

Life Underwriting Risk Module

3B1 Life Mortality Risk Sub-Module

1\.

A firm must calculate the capital requirement for mortality risk as equal to the loss in its basic own funds that would result from an instantaneous permanent increase of 15% in the mortality rates used for the calculation of technical provisions.

2\.

A firm must only apply the increase in mortality rates referred to in 3B1.1 to those insurance policies for which an increase in mortality rates leads to an increase in technical provisions without the risk margin and in identifying such policies, the firm may make the following assumptions:

1. (1) multiple insurance policies in respect of the same insured person may be treated as if they were one insurance policy; and
2. (2) where the calculation of technical provisions is based on groups of policies as referred to in Technical Provisions ? Further Requirements 20, the identification of the policies for which technical provisions increase under an increase in mortality rates may also be based on those groups of policies instead of single policies, provided that it yields a result that is not materially different.

3\.

With regard to reinsurance obligations, the identification of the policies for which technical provisions increase under an increase in mortality rates must only apply to the underlying insurance policies and must be carried out in accordance with 3B1.2.

Export Article as

3B2 Life Longevity Risk Sub-Module

1\.

A firm must calculate the capital requirement for longevity risk as equal to the loss in its basic own funds that would result from an instantaneous permanent decrease of 20% in the mortality rates used for the calculation of technical provisions.

2\.

A firm must only apply the decrease in mortality rates referred to in 3B2.1 to those insurance policies for which a decrease in mortality rates leads to an increase in technical provisions without the risk margin and in identifying such policies, the firm may make the following assumptions:

1. (1) multiple insurance policies in respect of the same insured person may be treated as if they were one insurance policy; and
2. (2) where the calculation of technical provisions is based on groups of policies as referred to in Technical Provisions ? Further Requirements 20, the identification of the policies for which technical provisions increase under a decrease in mortality rates may also be based on those groups of policies instead of single policies, provided that it yields a result that is not materially different.

3\.

With regard to reinsurance obligations, the identification of the policies for which technical provisions increase under a decrease in mortality rates must only apply to the underlying insurance policies and must be carried out in accordance with 3B2.2.

Export Article as

3B3 Life Disability Morbidity Risk Sub-Module

1\.

A firm must calculate the capital requirement for disability-morbidity risk as equal to the loss in its basic own funds that would result from the combination of the following instantaneous permanent changes:

1. (1) an increase of 35% in the disability and morbidity rates that are used in the calculation of technical provisions to reflect the disability and morbidity experience in the following 12 months;
2. (2) an increase of 25% in the disability and morbidity rates that are used in the calculation of technical provisions to reflect the disability and morbidity experience for all months after the following 12 months; and
3. (3) a decrease of 20% in the disability and morbidity recovery rates used in the calculation of technical provisions in respect of the following 12 months and for all years thereafter.

Export Article as

3B4 Life-Expense Risk Sub-Module

1\.

A firm must calculate the capital requirement for life expense risk as equal to the loss in its basic own funds that would result from the combination of the following instantaneous permanent changes:

1. (1) an increase of 10% in the amount of expenses taken into account in the calculation of technical provisions; and
2. (2) an increase of one percentage point to the expense inflation rate (expressed as a percentage) used in the calculation of technical provisions.

2\.

With regard to reinsurance obligations, a firm must apply those changes to its own expenses and, where relevant, to the expenses of the ceding undertakings.

Export Article as

3B5 Life Revision Risk Sub-Module

1\.

A firm must calculate the capital requirement for life revision risk as equal to the loss in its basic own funds that would result from an instantaneous permanent increase of 3% in the amount of annuity benefits only on annuity insurance and reinsurance obligations where the benefits payable under the underlying insurance policies could increase as a result of changes in the legal environment or in the state of health of the person insured.

Export Article as

3B6 Life Lapse Risk Sub-Module

1\.

A firm must calculate the capital requirement for lapse risk as equal to the highest of the following capital requirements:

1. (1) the capital requirement for the risk of a permanent increase in lapse rates;
2. (2) the capital requirement for the risk of a permanent decrease in lapse rates; and
3. (3) the capital requirement for mass lapse risk.

2\.

A firm must calculate the capital requirement for the risk of a permanent increase in lapse rates as equal to the loss in its basic own funds that would result from an instantaneous permanent increase of 50% in the option exercise rates of the relevant options (as set out in 3B6.4 and 3B6.5), provided that the increased option exercise rates must not exceed 100% and the increase in option exercise rates must only apply to relevant options for which the exercise of the option would result in an increase in technical provisions without the risk margin.

3\.

A firm must calculate the capital requirement for the risk of a permanent decrease in lapse rates as equal to the loss in its basic own funds that would result from an instantaneous permanent decrease of 50% in the option exercise rates of the relevant options (as set out in 3B6.4 and 3B6.5), provided that the decrease in option exercise rates must not exceed 20 percentage points and the decrease in option exercise rates must only apply to relevant options for which the

exercise of the option would result in a decrease in technical provisions without the risk margin.

4\.

The relevant options for the purposes of 3B6.2 and 3B6.3 are the following:

1. (1) all legal or contractual policyholder rights to fully or partly terminate, surrender, decrease, restrict or suspend insurance cover or permit the insurance policy to lapse; and
2. (2) all legal or contractual policyholder rights to fully or partially establish, renew, increase, extend or resume the insurance or reinsurance cover.

For the purposes of 3B6.4(2) the change in the option exercise rate referred to in 3B6.2 and 3B6.3 must be applied to the rate reflecting that the relevant option is not exercised.

5\.

In relation to reinsurance contracts the relevant options for the purposes of 3B6.2 and 3B6.3 are the following:

1. (1) the rights referred to in 3B6.4 of the policyholders of the reinsurance contracts;
2. (2) the rights referred to in 3B6.4 of the policyholders of the contracts of insurance underlying the reinsurance contracts; and
3. (3) where the reinsurance contract covers contracts of insurance or reinsurance contracts that will be written in the future, the right of the potential policyholders not to conclude those contracts of insurance or reinsurance contracts.

6\.

A firm must calculate the capital requirement for mass lapse risk as equal to the loss in its basic own funds that would result from a combination of the following instantaneous events:

1. (1) the discontinuance of 70% of the insurance policies falling within the scope of operations referred to in Regulated Activities Order Schedule 1, Part II, class VII for which discontinuance would result in an increase in technical provisions without the risk margin and where one of the following requirements are met:
 1. (a) the policyholder is not a natural person and discontinuance of the policy is not subject to approval by the beneficiaries of the pension fund; or
 2. (b) the policyholder is a natural person acting for the benefit of the beneficiaries of the policy, except where there is a family relationship between that natural person and the beneficiaries, or where the policy is effected for private estate planning or inheritance purposes and the number of beneficiaries under the policy does not exceed 20;
2. (2) the discontinuance of 40% of the insurance policies other than those falling within 3B6.6(1) for which discontinuance would result in an increase in technical provisions without the risk margin; and
3. (3) where reinsurance contracts cover contracts of insurance or reinsurance contracts that will be written in the future, the decrease of 40% of the number of those future contracts of insurance or reinsurance contracts used in the calculation of technical provisions.

7\.

A firm must apply the events referred to in 3B6.6 uniformly to all relevant contracts of insurance and reinsurance contracts and, in respect of any such reinsurance contracts, the firm must apply the event referred to in 3B6.6(1) to the underlying contracts of insurance.

8\.

For the purposes of determining the loss in its basic own funds under the events referred to in 3B6.6(1) and (2) the firm must base the calculation on the type of discontinuance that most negatively affects its basic own funds on a per policy

basis.

9\.

Where the highest of the capital requirements referred to in 3B6.1(1), (2) and (3) and the highest of the corresponding capital requirements calculated in accordance with 6.3(2) are not based on the same scenario, the capital requirement for lapse risk must be the capital requirement referred to in 3B6.1(1), (2) and (3) for which the underlying scenario results in the highest corresponding capital requirement calculated in accordance with 6.3(2).

Export Article as

3B7 Life-Catastrophe Risk Sub-Module

1\.

A firm must calculate the capital requirement for life-catastrophe risk as equal to the loss in its basic own funds that would result from an instantaneous increase of 0.15 percentage points in the mortality rates (expressed as percentages) that are used in the calculation of technical provisions to reflect the mortality experience in the following 12 months.

2\.

A firm must only apply the increase in mortality rates referred to in 3B7.1 to those insurance policies for which an increase in mortality rates that are used to reflect the mortality experience in the following 12 months leads to an increase in technical provisions without the risk margin and in identifying such policies, the firm may make the following assumptions:

1. (1) multiple insurance policies in respect of the same insured person may be treated as if they were one insurance policy; and
2. (2) where the calculation of technical provisions is based on groups of policies as referred to in Technical Provisions ? Further Requirements 20 the identification of the policies for which technical provisions increase under an increase in mortality rates may also be based on those groups of policies instead of single policies, provided that it yields a result that is not materially different.

3\.

With regard to reinsurance policies, the identification of the policies for which technical provisions increase under an increase in mortality rates must only apply to the underlying insurance policies and must be carried out in accordance with 3B7.2.

Export Article as

3C

Health Underwriting Risk Module

3C1 NSLT Health Underwriting Risk Sub-Module

1\.

The NSLT health underwriting risk sub-module must consist of the following sub-modules:

1. (1) the NSLT health premium and reserve risk sub-module; and
2. (2) the NSLT health lapse risk sub-module.

2\.

A firm must calculate the capital requirement for NSLT health underwriting risk in accordance with the following formula:

$$\sqrt{\{{\text{SCR}}\}_{\text{NSLT}} = \sqrt{\{{\text{SCR}}\}_{\text{left(NSLT,pr)}}^2 + \{{\text{SCR}}\}_{\text{left(NSLT,lapse)}}^2}}$$

where:

1. (1) $\text{SCR}(\text{NSLT}, \text{pr})$ denotes the capital requirement for NSLT health premium and reserve risk; and
2. (2) $\text{SCR}(\text{NSLT}, \text{lapse})$ denotes the capital requirement for NSLT health lapse risk.

Export Article as

3C2 NSLT Health Premium And Reserve Risk Sub-Module

1\.

A firm must calculate the capital requirement for NSLT health premium and reserve risk in accordance with the following formula:

$$\sqrt{\{{\text{SCR}}\}_{\text{left(NSLT,pr)}} = 3 \cdot \sigma_{\text{NSLT}} \cdot V_{\text{NSLT}}}$$

where:

1. (1) σ_{NSLT} denotes the standard deviation for NSLT health premium and reserve risk determined in accordance with 3C5; and
2. (2) V_{NSLT} denotes the volume measure for NSLT health premium and reserve risk determined in accordance with 3C3.

Export Article as

3C3 Volume Measure For NSLT Health Premium And Reserve Risk

1\.

A firm must calculate the volume measure for NSLT health premium and reserve risk as equal to the sum of the volume measures for premium and reserve risk of the segments set out in 3C4.

2\.

For all segments set out in 3C4 a firm must calculate the volume measure of a particular segment s in accordance with the following formula:

$$V_s = \left(V_{\text{left(prem,s)}} + V_{\text{left(res,s)}} \right) \cdot \left(0.75 + 0.25 \cdot \text{DIV}_s \right)$$

where:

1. (1) $V(\text{prem}, s)$ denotes the volume measure for premium risk of segment s ;
2. (2) $V(\text{res}, s)$ denotes the volume measure for reserve risk of segment s ; and
3. (3) DIV_s denotes the factor for geographical diversification of segment s .

3\.

For all segments set out in 3C4 a firm must calculate the volume measure for premium risk of a particular segment s

in accordance with the following formula:

$$V_{\text{left}(prem,s\text{right})} = \max\{P_s; P_{\text{left}(last,s\text{right})}\} + \{FP\}_{\text{left}(existing,s\text{right})} + \{FP\}_{\text{left}(future,s\text{right})}$$

where:

1. (1) P_s denotes an estimate of the premiums to be earned by the firm for the segment s during the following 12 months;
2. (2) $P_{\text{left}(last,s\text{right})}$ denotes the premiums earned by the firm for the segment s during the last 12 months;
3. (3) $\{FP\}_{\text{left}(existing,s\text{right})}$ denotes the expected present value of premiums to be earned by the firm for the segment s after the following 12 months for existing contracts of insurance; and
4. (4) $\{FP\}_{\text{left}(future,s\text{right})}$ denotes the following amount with respect to contracts of insurance where the initial recognition date falls in the following 12 months:
 1. (a) for all such contracts of insurance with an initial term of one year or less, the expected present value of premiums to be earned by the firm for the segment s , but excluding the premiums to be earned during the 12 months after the initial recognition date; and
 2. (b) for all such contracts of insurance with an initial term of more than one year, the amount equal to 30% of the expected present value of premiums to be earned by the firm for the segment s after the following 12 months.

4\.

For all segments set out in 3C4, a firm may, as an alternative to the calculation set out in 3C3.3, choose to calculate the volume measure for premium risk of a particular segment s in accordance with the following formula:

$$V_{\text{left}(prem,s\text{right})} = P_s + \{FP\}_{\text{left}(existing,s\text{right})} + \{FP\}_{\text{left}(future,s\text{right})}$$

provided that all of the following requirements are met:

1. (1) the governing body of the firm has decided that its earned premiums for the segment s during the following 12 months will not exceed P_s ;
2. (2) the firm has established effective control mechanisms to ensure that the limits on earned premiums referred to in (1) will be met; and
3. (3) the firm has informed the PRA in writing about the decision referred to in (1) and the reasons for it.

5\.

For the purposes of 3C3.4, the terms P_s , $\{FP\}_{\text{left}(existing,s\text{right})}$ and $\{FP\}_{\text{left}(future,s\text{right})}$ must be determined in accordance with 3C3.3(a), (c) and (d).

7\.

For all segments set out in 3C4, a firm must calculate the volume measure for reserve risk of a particular segment as equal to the best estimate for the provision for claims outstanding for the segment, after deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, provided that:

1. (1) the reinsurance contracts or special purpose vehicles comply with 3G2, 3G3, 3G5 and 3G7; and
2. (2) the volume measure must not be a negative amount.

8\.

For all segments set out in 3C4, the default factor for geographical diversification must be either equal to 1 or calculated in accordance with 3A5.

Export Article as

3C4 Segmentation of NSLT Health Insurance Obligations and NSLT Health Reinsurance Obligations and Standard Deviations for the NSLT Health Premium and Reserve Risk Sub-module

| Segment | Lines of business that the segment consists of | Standard deviation for gross premium risk of the segment | Standard deviation for reserve risk of the segment

---|---|---|---

1 | Medical expense insurance and proportional reinsurance | 1 and 13 | 5% | 5.7%

2 | Income protection insurance and proportional reinsurance

| 2 and 14

| 8.5%

| 14%

3 | Workers' compensation insurance and proportional reinsurance

| 3 and 15

| 9.6% | 11%

4 | Non-proportional health reinsurance | 25 | 17% | 17%

Export Article as

3C5 Standard Deviation For NSLT Health Premium And Reserve Risk

1\.

A firm must calculate the standard deviation for NSLT health premium and reserve risk in accordance with the following formula:

$$\sigma_{\text{NSLT}} = \frac{1}{V_{\text{NSLT}}} \cdot \sqrt{\sum_{s,t} \text{CorrHS}_{s,t} \cdot \sigma_s \cdot \sigma_t \cdot V_s \cdot V_t}$$

where:

1. (1) V_{NSLT} denotes the volume measure for NSLT health premium and reserve risk;
2. (2) the sum covers all possible combinations (s, t) of the segments set out in 3C4;
3. (3) $\text{CorrHS}_{s,t}$ denotes the correlation coefficient for NSLT health premium and reserve risk for segment s and segment t set out in 3C6;
4. (4) σ_s and σ_t denote standard deviations for NSLT health premium and reserve risk of segments s and t respectively; and
5. (5) V_s and V_t denote volume measures for premium and reserve risk of segments s and t , referred to in 3C4, respectively.

2\.

For all segments set out in 3C4, a firm must calculate the standard deviation for NSLT health premium and reserve risk of a particular segment s in accordance with the following formula:

$$\sigma_s = \frac{\sqrt{\sigma_{\text{prem}}^2 + \sigma_{\text{res}}^2 + 2 \cdot \sigma_{\text{prem}} \cdot \sigma_{\text{res}} \cdot \text{Corr}_{\text{prem,res}}}}{V_{\text{prem}} + V_{\text{res}}}$$

where:

1. (1) $\sigma_{\text{prem}, s}$ denotes the standard deviation for NSLT health risk of segment s determined in accordance with 3C5.3;
2. (2) $\sigma_{\text{res}, s}$ denotes the standard deviation for NSLT health reserve risk of segment s as set out in 3C4; and
3. (3) $V_{\text{prem}, s}$ denotes the volume measure for premium risk of segment s referred to in 3C3;
4. (4) $V_{\text{res}, s}$ denotes the volume measure for reserve risk of segment s referred to in 3C3.

3\.

For all segments set out in 3C4, a firm must calculate the standard deviation for NSLT health premium risk of a particular segment as equal to the product of the standard deviation for NSLT health gross premium risk of the segment set out in 3C4 and the adjustment factor for non-proportional reinsurance, which, for all segments set out in 3C4 must be equal to 100%.

Export Article as

3C6 Correlation Matrix For NSLT Health Premium And Reserve Risk

1\.

The correlation coefficient $\text{CorrHS}(s, t)$ referred to in 3C5.1 must be equal to the item set out in row s and in column t of the following correlation matrix. The headings of the rows and columns denote the numbers of the segments set out 3C4:

s				
t	1	2	3	4
---	---	---	---	---
1	1	0.5	0.5	0.5
2	0.5	1	0.5	0.5
3	0.5	0.5	1	0.5
4	0.5	0.5	0.5	1

Export Article as

3C7 NSLT Health Lapse Risk Sub-Module

1\.

A firm must calculate the capital requirement for NSLT health lapse risk as equal to the loss in its basic own funds that would result from the combination of the following instantaneous events:

1. (1) the discontinuance of 40% of the insurance policies for which discontinuance would result in an increase of technical provisions without the risk margin; and
2. (2) where reinsurance contracts cover contracts of insurance or reinsurance contracts that will be written in the future, the decrease of 40% of the number of those future contracts of insurance or reinsurance contracts used in the calculation of technical provisions.

2\.

A firm must apply the events referred to in 3C7.1 uniformly to relevant all contracts of insurance and reinsurance

contracts and, in respect of any such reinsurance contracts, the firm must apply the event referred to in 3C7.1(1) to the underlying contracts of insurance.

3\.

For the purposes of determining the loss in its basic own funds under the event referred to in 3C7.1(1), the firm must base the calculation on the type of discontinuance that most negatively affects its basic own funds on a per policy basis.

Export Article as

3C8 SLT Health Underwriting Risk Sub-Module

1\.

The SLT health underwriting risk sub-module must consist of all of the following sub-modules:

1. (1) the health mortality risk sub-module;
2. (2) the health longevity risk sub-module;
3. (3) the health disability-morbidity risk sub-module;
4. (4) the health expense risk sub-module;
5. (5) the health revision risk sub-module; and
6. (6) the SLT health lapse risk sub-module.

2\.

A firm must calculate the capital requirement for SLT health underwriting risk in accordance with the following formula:

$$\sqrt{\sum_{i,j} \{CorrSLTH\}_{\left(i,j\right)} \cdot \{SCR\}_i \cdot \{SCR\}_j}$$

where:

1. (1) the sum denotes all possible combinations $_{(i,j)}$ of the sub-modules set out in 3C8.1;
2. (2) $_{CorrSLTH(i,j)}$ denotes the correlation coefficient for SLT health underwriting risk for sub-modules $_{i}$ and $_{j}$; and
3. (3) $_{SCR i}$ and $_{SCR j}$ denote the capital requirements for risk sub-modules $_{i}$ and $_{j}$ respectively.

3\.

The correlation coefficient $_{CorrSLTH(i,j)}$ referred to in 3C8.2 must be equal to the value set out in row $_{i}$ and in column $_{j}$ of the following correlation matrix:

Health mortality Health longevity
Health disability-morbidity Health expense
Health revision
SLT health lapse

---|---|---|---|---|---

Health mortality

| 1 | -0.25 | 0.25 | 0.25 | 0 | 0

Health longevity

| -0.25 | 1 | 0 | 0.25 | 0.25 | 0.25

Health disability-morbidity

| 0.25 | 0 | 1 | 0.5 | 0 | 0

Health expense

| 0.25 | 0.25 | 0.5 | 1 | 0.5 | 0.5

Health revision

| 0 | 0.25 | 0 | 0.5 | 1 | 0

SLT health lapse

| 0 | 0.25 | 0 | 0.5 | 0 | 1

[Export Article as](#)

3C9 Health Mortality Risk Sub-Module

1\.

A firm must calculate the capital requirement for health mortality risk as equal to the loss in its basic own funds that would result from an instantaneous permanent increase of 15% in the mortality rates used for the calculation of technical provisions.

2\.

A firm must only apply the increase in mortality rates referred to in 3C9.1 to those insurance policies for which an increase in mortality rates leads to an increase in technical provisions without the risk margin and in identifying such policies, the firm may make the following assumptions:

1. (1) multiple insurance policies in respect of the same insured person may be treated as if they were one insurance policy; and
2. (2) where the calculation of technical provisions is based on groups of policies as referred to in Technical Provisions ? Further Requirements 20, the identification of the policies for which technical provisions increase under an increase in mortality rates may also be based on those groups of policies instead of single policies, provided that it yields a result that is not materially different.

3\.

With regard to reinsurance obligations, the identification of the policies for which technical provisions increase under an increase in mortality rates must only apply to the underlying insurance policies and must be carried out in accordance with 3C9.2.

[Export Article as](#)

3C10 Health Longevity Risk Sub-Module

1\.

A firm must calculate the capital requirement for health longevity risk as equal to the loss in its basic own funds that would result from an instantaneous permanent decrease of 20% in the mortality rates used for the calculation of technical provisions.

2\.

A firm must only apply the decrease in mortality rates referred to in 3C10.1 to those insurance policies for which a decrease in mortality rates leads to an increase in technical provisions without the risk margin and in identifying such

policies, the firm may make the following assumptions:

1. (1) multiple insurance policies in respect of the same insured person may be treated as if they were one insurance policy; and
2. (2) where the calculation of technical provisions is based on groups of policies as referred to in Technical Provisions ? Further Requirements 20, the identification of the policies for which technical provisions increase under a decrease in mortality rates may also be based on those groups of policies instead of single policies, provided that it yields a result that is not materially different.
- 3\.

With regard to reinsurance obligations, the identification of the policies for which technical provisions increase under a decrease in mortality rates must only apply to the underlying insurance policies and must be carried out in accordance with 3C10.2.

Export Article as

3C11 Health Disability-Morbidity Risk Sub-Module

1\.

A firm must calculate the capital requirement for health disability-morbidity risk as the sum of the following:

1. (1) the capital requirement for medical expense disability-morbidity risk; and
2. (2) the capital requirement for income protection disability-morbidity risk.

2\.

A firm must apply:

1. (1) the scenarios underlying the calculation of the capital requirement for medical expense disability-morbidity risk only to medical expense insurance obligations and medical expense reinsurance obligations where the underlying business is pursued on a similar technical basis to that of life insurance; and
2. (2) the scenarios underlying the calculation of the capital requirement for income protection disability-morbidity risk only to income protection insurance obligations and income protection reinsurance obligations where the underlying business is pursued on a similar technical basis to that of life insurance.

Export Article as

3C12 Capital Requirement For Medical Expense Disability-Morbidity Risk

1\.

A firm must calculate the capital requirement for medical expense disability-morbidity risk as equal to the higher of the following capital requirements:

1. (1) the capital requirement for the increase of medical payments; and
2. (2) the capital requirement for the decrease of medical payments.

2\.

A firm must calculate the capital requirement for the increase of medical payments as equal to the loss in its basic own funds that would result from the following combination of instantaneous permanent changes:

1. (1) an increase of 5% in the amount of medical payments taken into account in the calculation of technical provisions; and
 2. (2) an increase of one percentage point in the inflation rate of medical payments (expressed as a percentage) used for the calculation of technical provisions.
- 3\.

A firm must calculate the capital requirement for the decrease of medical payments as equal to the loss in its basic own funds that would result from the following combination of instantaneous permanent changes:

1. (1) a decrease of 5% in the amount of medical payments taken into account in the calculation of technical provisions; and
2. (2) a decrease of one percentage point in the inflation rate of medical payments (expressed as a percentage) used for the calculation of technical provisions.

Export Article as

3C13 Capital Requirement For Income Protection Disability-Morbidity Risk

1\.

A firm must calculate the capital requirement for income protection disability-morbidity risk as equal to the loss in its basic own funds that would result from the following combination of instantaneous permanent changes:

1. (1) an increase of 35% in the disability and morbidity rates that are used in the calculation of technical provisions to reflect the disability and morbidity in the following 12 months;
2. (2) an increase of 25% in the disability and morbidity rates that are used in the calculation of technical provisions to reflect the disability and morbidity in the years after the following 12 months;
3. (3) where the disability and morbidity recovery rates used in the calculation of technical provisions are lower than 50%, a decrease of 20% in those rates; and
4. (4) where the disability and morbidity persistency rates used in the calculation of technical provisions are equal to or lower than 50%, an increase of 20% in those rates.

Export Article as

3C14 Health Expense Risk Sub-Module

1\.

A firm must calculate the capital requirement for health expense risk as equal to the loss in its basic own funds that would result from the following combination of instantaneous permanent changes:

1. (1) an increase of 10% in the amount of expenses taken into account in the calculation of technical provisions; and
2. (2) an increase of one percentage point in the expense inflation rate (expressed as a percentage) used for the calculation of technical provisions.

With regard to reinsurance obligations, a firm must apply those changes to its own expenses and, where relevant, to the expenses of the ceding undertakings.

Export Article as

3C15 Health Revision Risk Sub-Module

1\.

A firm must calculate the capital requirement for health revision risk as equal to the loss in its basic own funds that would result from an instantaneous permanent increase of 4% in the amount of annuity benefits, only on annuity insurance and reinsurance obligations where the benefits payable under the underlying insurance policies could increase as a result of changes in inflation, the legal environment or the state of health of the person insured.

Export Article as

3C16 SLT Health Lapse Risk Sub-Module

1\.

A firm must calculate the capital requirement for SLT health lapse risk as equal to the higher of the following capital requirements:

1. (1) capital requirement for the risk of a permanent increase in SLT health lapse rates;
2. (2) capital requirement for the risk of a permanent decrease in SLT health lapse rates; and
3. (3) capital requirement for SLT health mass lapse risk.

2\.

A firm must calculate the capital requirement for the risk of a permanent increase in SLT health lapse rates as equal to the loss in its basic own funds that would result from an instantaneous permanent increase of 50% in the exercise rates of the relevant options (as set out in 3C16.4 and 3C16.5), provided that the increased option exercise rates must not exceed 100% and the increase in option exercise rates must only apply to relevant options for which the exercise would result in an increase in technical provisions without the risk margin.

3\.

A firm must calculate the capital requirement for the risk of a permanent decrease in SLT health lapse rates as equal to the loss in its basic own funds that would result from an instantaneous permanent decrease of 50% in the option exercise rates of the relevant options (as set out in 3C16.4 and 3C16.5), provided that, the decrease in option exercise rates must not exceed 20 percentage points and the decrease in option exercise rates must only apply to relevant options for which the exercise would result in a decrease in technical provisions without the risk margin.

4\.

The relevant options for the purposes of 3C16.2 and 3C16.3 must be the following:

1. (1) all legal or contractual policyholder rights to fully or partly terminate, surrender, decrease, restrict or suspend the insurance or reinsurance cover or permit the insurance policy to lapse; and
2. (2) all legal or contractual policyholder rights to fully or partially establish, renew, increase, extend or resume the insurance or reinsurance cover.

For the purposes of 3C16.4(2), the change in the option exercise rate referred to in 3C16.2 and 3C16.3 should be applied to the rate reflecting that the relevant option is not exercised.

5\.

In relation to reinsurance contracts, the relevant options for the purposes of 3C16.2 and 3C16.3 must be the following:

1. (1) the rights referred to in 3C16.4 of the policyholders of the reinsurance contracts;
2. (2) the rights set out in 3C16.4 of the policyholders of the contracts of insurance underlying the reinsurance contracts;

and

3. (3) where reinsurance contracts cover contracts of insurance or reinsurance contracts that will be written in the future, the right of the potential policyholders not to conclude those contracts of insurance or reinsurance contracts.

6\.

A firm must calculate the capital requirement for SLT health mass lapse risk as equal to the loss in its basic own funds that would result from a combination of the following instantaneous events:

1. (1) the discontinuance of 40% of the insurance policies for which discontinuance would result in an increase in technical provisions without the risk margin; and
2. (2) where reinsurance contracts cover contracts of insurance or reinsurance contracts that will be written in the future, the decrease of 40% of the number of those future contracts of insurance or reinsurance contracts used in the calculation of technical provisions.

7\.

A firm must apply the events referred to in 3C16.6 uniformly to all relevant contracts of insurance and reinsurance contracts and in respect of any such reinsurance contracts, the firm must apply the event referred to in 3C16.6(1) to the underlying contracts of insurance.

8\.

For the purposes of determining the loss in its basic own funds under the event referred to in 3C16.6(1), the firm must base the calculation on the type of discontinuance that most negatively affects its basic own funds on a per policy basis.

9\.

Where the highest of the capital requirements referred to in 3C16.1(1), (2) and (3) and the highest of the corresponding capital requirements calculated in accordance with 6.3(2) are not based on the same scenario, the capital requirement for lapse risk must be the capital requirement referred to in 3C16.1(1), (2) and (3) for which the underlying scenario results in the highest corresponding capital requirement calculated in accordance with 6.3(2).

Export Article as

3C17 Health Catastrophe Risk Sub-Module

1\.

A firm must calculate the capital requirement for the health catastrophe risk sub-module in accordance with the following formula:

$$\sqrt{\{SCR\}_{healthCAT} = \sqrt{\{SCR\}_{ma}^2 + \{SCR\}_{ac}^2 + \{SCR\}_p^2}}$$

where:

1. (1) $_SCR_{ma}$ denotes the capital requirement for the mass accident risk sub-module;
2. (2) $_SCR_{ac}$ denotes the capital requirement for the accident concentration risk sub-module; and
3. (3) $_SCR_p$ denotes the capital requirement for the pandemic risk sub-module.

2\.

A firm must apply:

1. (1) the mass accident risk sub-module to health insurance obligations and health reinsurance obligations other than workers? compensation insurance obligations and workers? compensation reinsurance obligations;
2. (2) the accident concentration risk sub-module to workers? compensation insurance obligations and workers? compensation reinsurance obligations and to group income protection insurance obligations and group income protection reinsurance obligations; and
3. (3) the pandemic risk sub-module to health insurance obligations and health reinsurance obligations other than workers? compensation insurance obligations and workers? compensation reinsurance obligations.

Export Article as

3C18 Mass Accident Risk Sub-Module

1\.

A firm must calculate the capital requirement for the mass accident risk sub-module in accordance with the following formula:

$$\sqrt{\sum_s \{SCR\}_{\left\{ma, s\right\}}^2}$$

where:

1. (1) the sum includes all countries set out in Annex XVI; and
2. (2) $\{SCR\}_{\left\{ma, s\right\}}$ denotes the capital requirement for mass accident risk of country s .

2\.

For all countries set out in Annex XVI, a firm must calculate the capital requirement for mass accident risk of a particular country s as equal to the loss in its basic own funds that would result from an instantaneous loss of an amount that, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, is calculated in accordance with the following formula:

$$L_{\left\{ma, s\right\}} = r_s \cdot \sum_e x_e \cdot E_{\left\{e, s\right\}}$$

where:

1. (1) r_s denotes the ratio of persons affected by the mass accident in country s as set out in Annex XVI;
2. (2) the sum includes the event types e set out in Annex XVI;
3. (3) x_e denotes the ratio of persons who will receive benefits attributable to event type e as a result of the accident as set out in Annex XVI; and
4. (4) $E_{\left\{e, s\right\}}$ denotes the total value of benefits payable by the firm in respect of event type e in country s .

3\.

For all event types set out in Annex XVI and all countries set out in Annex XVI, a firm must calculate its sum insured for a particular event type e in a particular country s in accordance with the following formula:

$$E_{\left\{e, s\right\}} = \sum_i \{SI\}_{\left\{e, i\right\}}$$

where:

1. (1) the sum includes all insured persons i of the firm who are insured against event type e and are inhabitants of country s ; and
2. (2) $\{SI\}_{\left\{e, i\right\}}$ denotes the value of the benefits payable by the firm for the insured person i in case of event type e .

4\.

For the purposes of 3C18.3(2), a firm must calculate the value of the benefits as the sum insured or where the contract of insurance provides for recurring benefit payments the best estimate of the benefit payments in case of event type $_e_$. Where the benefits of a contract of insurance depend on the nature or extent of any injury resulting from event type $_e_$, the calculation of the value of the benefits must be based on the maximum benefits payable under the contract of insurance that are consistent with the event. For medical expense insurance obligations and medical expense reinsurance obligations the value of the benefits must be based on an estimate of the average amounts paid in case of event type $_e_$, assuming the insured person is disabled for the duration specified and taking into account the specific guarantees included within the obligations.

5\.

Subject to 7.2, a firm may calculate the value of benefits payable for the insured person referred to in 3C18.3(2) based on homogenous risk groups, provided that the grouping of policies complies with Technical Provisions ? Further Requirements 20.

Export Article as

3C19 Accident Concentration Risk Sub-Module

1\.

A firm must calculate the capital requirement for the accident concentration risk sub-module in accordance with the following formula:

$$\sqrt{\sum_{c \in \mathcal{C}} \left(\text{SCR}_{ac} \right)^2}$$

where:

1. (1) the sum includes all countries $_c_$; and
2. (2) $_SCR(ac, c)_$ denotes the capital requirement for accident concentration risk of country $_c_$.

2\.

For all countries a firm must calculate the capital requirement for accident concentration risk of country $_c_$ as equal to the loss in its basic own funds that would result from an instantaneous loss of an amount that, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, is calculated in accordance with the following formula:

$$L_{ac} = C_c \cdot \sum_{e \in \mathcal{E}} x_e \cdot CE_{ec}$$

where:

1. (1) $_C c_$ denotes the largest accident risk concentration of the firm in country $_c_$;
2. (2) the sum includes the event types $_e_$ set out in Annex XVI;
3. (3) $_x_e_$ denotes the ratio of persons who will receive benefits attributable to event type $_e_$ as a result of the accident as set out in Annex XVI; and
4. (4) $_CE(e, c)_$ denotes the average value of benefits payable by the firm for event type $_e_$ for the largest accident risk concentration in country $_c_$.

3\.

For all countries, a firm must calculate the highest accident risk concentration of a firm in country $_c_$ as equal to the

highest number of persons for which all of the following requirements are met:

1. (1) the firm has a workers' compensation insurance obligation or a workers' compensation reinsurance obligation or a group income protection insurance obligation or a group income protection reinsurance obligation in relation to each of the persons;
 2. (2) the obligations in relation to each of the persons cover at least one of the events set out in Annex XVI; and
 3. (3) the persons are working in the same building, which is situated in country _c_.
- 4\.

For all event types and countries, a firm must __calculate its average sum insured for event type _e_ for the largest accident risk concentration in country _c_ in accordance with the following formula:

$$\frac{1}{N_e} \sum_{i=1}^{N_e} SI(e, i)$$

where:

1. (1) _N e_ denotes the number of insured persons who are insured by the firm against event type _e_ and who belong to the largest accident risk concentration of the firm in country _c_ ;
 2. (2) the sum includes all the insured persons referred to in (1); and
 3. (3) _SI(e, i)_ denotes the value of the benefits payable by the firm for the insured person _i_ in case of event type _e_.
- 5\.

For the purposes of 3C19.4(3), a firm must calculate the value of the benefits as the sum insured or where the contract of insurance provides for recurring benefit payments the best estimate of the benefit payments in case of event type _e_. Where the benefits of an insurance policy depend on the nature or extent of the injury resulting from event type _e_, the calculation of the value of the benefits must be based on the maximum benefits payable under the policy, that are consistent with the event. For medical expense insurance obligations and medical expense reinsurance obligations the value of the benefits must be based on an estimate of the average amounts paid in case of event type _e_, assuming the insured person is disabled for the duration specified and taking into account the specific guarantees included within the obligations.

6\.

Subject to 7.2, a firm may calculate the value of the benefits referred to in 3C19.4(3) based on homogenous risk groups, provided that the grouping of policies complies with the requirements set out in Technical Provisions ? Further Requirements 20.

Export Article as

3C20 Pandemic Risk Sub-Module

1\.

A firm must calculate the capital requirement for the pandemic risk sub-module as equal to the loss in its basic own funds that would result from an instantaneous loss of an amount that, without deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles, is calculated in accordance with the following formula:

$$L_p = 0.000075 \cdot E + 0.4 \cdot \sum_{c} N_c \cdot M_c$$

where:

1. (1) E denotes the income protection pandemic exposure of the firm;
2. (2) the sum includes all countries c ;
3. (3) N_c denotes the number of insured persons of the firm who meet all of the following requirements:
 1. (a) the insured persons are inhabitants of country c ; and
 2. (b) the insured persons are covered by a contract of insurance that includes medical expense insurance obligations or medical expense reinsurance obligations, other than workers' compensation insurance obligations or workers' compensation reinsurance obligations, that cover medical expenses resulting from an infectious disease; and
4. (4) M_c denotes the expected average amount payable by the firm per insured person of country c in case of a pandemic.

2\.

A firm must calculate its income protection pandemic exposure in accordance with the following formula:

$$E = \sum_i E_i$$

where:

1. (1) the sum includes all insured persons i covered by a contract of insurance that includes income protection insurance obligations or income protection reinsurance obligations other than workers' compensation insurance obligations or workers' compensation reinsurance obligations;
2. (2) E_i denotes the value of the benefits payable by the firm for the insured person i in case of a permanent work disability caused by an infectious disease. The value of the benefits must be the sum insured or, where the contract of insurance provides for recurring benefit payments, the best estimate of the benefit payments assuming that the insured person is permanently disabled and will not recover.

3\.

For all countries, a firm must calculate the expected average amount payable by the firm per insured person of a particular country c in case of a pandemic in accordance with the following formula:

$$M_c = \sum_h H_h \cdot CH_{(h,c)}$$

where:

1. (1) the sum includes the types of healthcare utilisation h set out in Annex XVI;
2. (2) H_h denotes the ratio of insured persons with clinical symptoms utilising healthcare type h as set out in Annex XVI; and
3. (3) $CH_{(h,c)}$ denotes the best estimate of the amounts payable by the firm for an insured person in country c in relation to medical expense insurance obligations or medical expense reinsurance obligations, other than workers' compensation insurance obligations or workers' compensation reinsurance obligations, for healthcare utilisation type h in the event of a pandemic.

Export Article as

3D

Market Risk Module

3D1 Lists Of Regional Governments And Local Authorities

1\.

A firm must treat exposures to the Scottish Government, the Welsh Government and the Northern Ireland Executive as exposures to the central government of the UK for the calculation of the market risk module of the standard formula.

Export Article as

Qualifying Infrastructure Investments

3D2 Qualifying Infrastructure Investments

1\.

The requirements that must be met for an investment in an infrastructure entity to constitute a qualifying infrastructure investment are as follows:

1. (1) the cash-flows generated by the infrastructure assets allow for all financial obligations to be met under sustained stresses that are relevant for the risks of the project;
2. (2) the cash-flows that the infrastructure entity generates for debt providers and equity investors are predictable;
3. (3) the infrastructure assets and infrastructure entity are governed by a regulatory or contractual framework that provides debt providers and equity investors with a high degree of protection including the following:
 1. (a) the contractual framework must include provisions that effectively protect debt providers and equity investors against losses resulting from the termination of the project by the party which agrees to purchase the goods or services provided by the infrastructure project, unless one of the following requirements is met:
 1. (i) the revenues of the infrastructure entity are funded by payments from a large number of users; or
 2. (ii) the revenues are subject to a rate-of-return regulation; and
 2. (b) the infrastructure entity has sufficient reserve funds or other financial arrangements to cover the contingency funding and working capital requirements of the project.
4. (4) where investments are in bonds or loans, this contractual framework must also include the following:
 1. (a) debt providers have security or the benefit of security to the extent permitted by applicable law in all assets and contracts that are critical to the operation of the project;
 2. (b) the use of net operating cash-flows after mandatory payments from the project for purposes other than servicing debt obligations is restricted; and
3. (c) restrictions on activities that may be detrimental to debt providers, including that new debt cannot be issued without the consent of existing debt providers in the form agreed with them, unless such new debt issuance is permitted under the documentation for the existing debt;
5. (5) notwithstanding (4)(a), for investments in bonds or loans, where a firm can demonstrate that security in all assets and contracts is not essential for debt providers to effectively protect or recover the vast majority of their investment, other security mechanisms may be used, provided they comprise at least one of the following:
 1. (a) pledge of shares;
 2. (b) step-in rights;
 3. (c) lien over bank accounts;
 4. (d) control over cash-flows; or
 5. (e) provisions for assignment of contracts;
6. (6) where investments are in bonds or loans, the firm is able to hold the investment to maturity and, subject to 3D2.3, has notified the PRA of this in writing before it treats an investment as a qualifying infrastructure investment;
7. (7) where investments are in bonds or loans for which a credit assessment by a nominated external credit assessment institution is not available, the investment instrument and other pari passu instruments are senior to all other claims other than statutory claims and claims from liquidity facility providers, trustees and derivatives counterparties; and
8. (8) where investments are in equities, or bonds or loans for which a credit assessment by a nominated external credit assessment institution is not available, the following criteria are met:

1. (a) the infrastructure assets and infrastructure entity are located in the OECD;
 2. (b) where the infrastructure project is in the construction phase the following criteria must be fulfilled by the equity investor, or where there is more than one equity investor, the following criteria must be fulfilled by a group of equity investors as a whole:
 1. (i) the equity investors have a history of successfully overseeing infrastructure projects and the relevant expertise;
 2. (ii) the equity investors have a low risk of default, or there is a low risk of material losses for the infrastructure entity as a result of their default; and
 3. (iii) the equity investors are incentivised to protect the interests of investors;
 3. (c) where there are construction risks, safeguards exist to ensure completion of the project according to the agreed specification, budget or completion date;
 4. (d) where operating risks are material, they are properly managed;
 5. (e) the infrastructure entity uses tested technology and design;
 6. (f) the capital structure of the infrastructure entity allows it to service its debt;
 7. (g) the refinancing risk for the infrastructure entity is low; and
 8. (h) the infrastructure entity uses derivatives only for risk-mitigation purposes.
- 2\.

For the purposes of 3D2.1(2), a firm must not treat the cash-flows generated for debt providers and equity investors as predictable unless all except an immaterial part of the revenues satisfy the following requirements:

1. (1) one of the following criteria is met:
 1. (a) the revenues are availability-based;
 2. (b) the revenues are subject to a rate-of-return regulation;
 3. (c) the revenues are subject to a take-or-pay contract; or
 4. (d) the level of output or the usage and the price must independently meet one of the following criteria:
 1. (i) it is regulated;
 2. (ii) it is contractually fixed; or
 3. (iii) it is sufficiently predictable as a result of low demand risk; and
 2. (2) where the revenues of the infrastructure entity are not funded by payments from a large number of users, the party which agrees to purchase the goods or services provided by the infrastructure project entity must be one of the following:
 1. (a) an entity listed in 3D24.2;
 2. (b) a body listed in 3D1;
 3. (c) an entity with an external credit assessment institution rating with a credit quality step of at least 3; or
 4. (d) an entity that is replaceable without a significant change in the level and timing of revenues.
- 3\.

Where a firm treated an investment as a qualifying infrastructure investment in accordance with Article 164a of Commission Delegated Regulation (EU) 2015/35 immediately before 31 December 2024 and from 31 December 2024 treats that investment as a qualifying infrastructure investment, the firm must notify in writing the PRA by 31 January 2025.

Export Article as

3D3 Qualifying Infrastructure Corporate Investments

1\.

The requirements that must be met for an investment in an infrastructure entity to constitute a qualifying infrastructure corporate investment are as follows:

1. (1) the substantial majority of the infrastructure entity's revenues is derived from owning, financing, developing or operating infrastructure assets located in the OECD;
2. (2) the revenues generated by the infrastructure assets satisfy one of the criteria set out in 3D2.2(1);
3. (3) where the revenues of the infrastructure entity are not funded by payments from a large number of users, the party which agrees to purchase the goods or services provided by the infrastructure entity must be one of the entities listed in 3D2.2(2);
4. (4) the revenues must be diversified in terms of activities, location, or payers, unless the revenues are subject to a rate-of-return regulation in accordance with 3D2.1(3)(a)(ii) or a take-or-pay contract or the revenues are availability based;
5. (5) where investments are in bonds or loans, the firm is able to hold the investment to maturity and, subject to 3D3.2, has notified the PRA of this in writing before it treats an investment as a qualifying infrastructure corporate investment;
6. (6) where no credit assessment from a nominated external credit assessment institution is available for the infrastructure entity:
 1. (a) the capital structure of the infrastructure entity must allow it to service all its debt under conservative assumptions based on an analysis of the relevant financial ratios; and
 2. (b) the infrastructure entity must have been active for at least three years or, in the case of an acquired business, it must have been in operation for at least three years; and
7. (7) where a credit assessment from a nominated external credit assessment institution is available for the infrastructure entity, such credit assessment has a credit quality step between 0 and 3.

2\.

Where a firm treated an investment as a qualifying infrastructure corporate investment in accordance with Article 164b of Commission Delegated Regulation (EU) 2015/35 immediately before 31 December 2024 and from 31 December 2024 treats that investment as a qualifying infrastructure corporate investment, the firm must notify the PRA in writing by 31 January 2025.

Export Article as

Interest Rate Risk Sub-Module

3D4 General Provisions

1\.

A firm must calculate the capital requirement for interest-rate risk as equal to the higher of the following:

1. (1) the sum, over all currencies, of the capital requirements for the risk of an increase in the term structure of interest rates as set out in 3D5; and
2. (2) the sum, over all currencies, of the capital requirements for the risk of a decrease in the term structure of interest rates as set out in 3D6.

2\.

Where the higher of the capital requirements referred to in 3D4.1(1) and (2) and the higher of the corresponding capital requirements calculated in accordance with 6.3(2) are not based on the same scenario, the capital requirement for interest-rate risk must be the capital requirement referred to in 3D4.1(1) and (2) for which the underlying scenario results

in the highest corresponding capital requirement calculated in accordance with 6.3(2).

Export Article as

3D5 Increase In The Term Structure Of Interest Rates

1\.

A firm must calculate the capital requirement for the risk of an increase in the term structure of interest rates for a given currency as equal to the loss in its basic own funds that would result from an instantaneous increase in basic risk-free interest rates for that currency at different maturities in accordance with the following table:

****Maturity**

(in years) | Increase**

---|---

1 | 70%

2 | 70%

3 | 64%

4 | 59%

5 | 55%

6 | 52%

7 | 49%

8 | 47%

9 | 44%

10 | 42%

11 | 39%

12 | 37%

13 | 35%

14 | 34%

15 | 33%

16 | 31%

17 | 30%

18 | 29%

19 | 27%

20 | 26%

90 | 20%

2\.

For maturities not specified in the table above, the value of the increase must be linearly interpolated, provided that:

1. (1) for maturities shorter than 1 year, the increase must be 70%; and
2. (2) for maturities longer than 90 years, the increase must be 20%.

3\.

In any case, the increase of basic risk-free interest rates at any maturity must be at least one percentage point.

4\.

The impact of the increase in the basic relevant risk-free rate term structure on the value of participations in financial institutions and credit institutions as referred to in Own Funds 3K.6 must only be taken into account on the value of the participations that are not deducted from own funds pursuant to Own Funds 3K and the part deducted from own funds must only be taken into account to the extent that such impact increases the basic own funds.

Export Article as

3D6 Decrease In The Term Structure Of Interest Rates

1\.

A firm must calculate the capital requirement for the risk of a decrease in the term structure of interest rates for a given currency as equal to the loss in its basic own funds that would result from an instantaneous decrease in basic risk-free interest rates for that currency at different maturities in accordance with the following table:

Maturity

(in years) | Decrease

---|---

1 | 75%

2 | 65%

3 | 56%

4 | 50%

5 | 46%

6 | 42%

7 | 39%

8 | 36%

9 | 33%

10 | 31%

11 | 30%

12 | 29%

13 | 28%

14 | 28%

15 | 27%

16 | 28%

17 | 28%

18 | 28%

19 | 29%

20 | 29%

90 | 20%

2\.

For maturities not specified in the table above, the value of the decrease must be linearly interpolated, provided that:

1. (1) for maturities shorter than 1 year, the decrease must be 75%; and
2. (2) for maturities longer than 90 years, the decrease must be 20%.

3\.

Notwithstanding 3D6.1 and 3D6.2, for negative basic risk-free interest rates the decrease must be nil.

4\.

The impact on the value of participations as referred to in Own Funds 3K.6 in financial institutions and credit institutions of the decrease in the basic relevant risk-free interest rate term structure must only be taken into account on the value of the participations that are not deducted from own funds pursuant to Own Funds 3K and the part deducted from own funds must only be taken into account to the extent that such impact increases the basic own funds.

Export Article as

Equity Risk Sub-Module

3D7 General Provisions

1\.

The equity risk sub-module must include a risk sub-module for type 1 equities, a risk sub-module for type 2 equities, a risk sub-module for qualifying infrastructure equities and a risk sub-module for qualifying infrastructure corporate equities.

2\.

A firm must treat as type 1 equities:

1. (1) those listed in 3D7.8; and
2. (2) equities listed in regulated markets in countries which are members of the OECD, or traded on multilateral trading facilities, as defined in Article 3 of the _RAO_, whose registered office or head office is in an EU Member State.

3\.

A firm must treat as type 2 equities:

1. (1) equities other than those referred to in 3D7.2, commodities and other alternative investments; and
2. (2) all assets other than those covered in the interest-rate risk sub-module, the property risk sub-module or the spread risk sub-module, including the assets and indirect exposures referred to in 2.3(1) and (2) where a look-through approach is not possible and the firm does not make use of the provisions in 2.3(3) and (4).

4\.

A firm must treat as qualifying infrastructure equities equity investments in infrastructure entities that meet the requirements set out in 3D2.

5\.

A firm must treat as qualifying infrastructure corporate equities equity investments in infrastructure entities that meet the requirements set out in 3D3.

6\.

A firm must calculate the capital requirement for equity risk in accordance with the following formula:

$$\sqrt{\{SCR\}_{equity}} = \sqrt{\{SCR\}_{equ1}^2 + 2 \cdot 0.75 \cdot \{SCR\}_{equ1} \cdot \left(\{SCR\}_{equ2} + \{SCR\}_{quinf} + \{SCR\}_{q} \right)}$$

$$\sqrt{\left(\text{SCR}_{\text{equ2}} + \text{SCR}_{\text{quinf}} + \text{SCR}_{\text{quinf}}\right)^2}$$

where:

1. (a) SCR_{equ1} denotes the capital requirement for type 1 equities;
2. (b) SCR_{equ2} denotes the capital requirement for type 2 equities;
3. (c) $\text{SCR}_{\text{quinf}}$ denotes the capital requirement for qualifying infrastructure equities; and
4. (d) $\text{SCR}_{\text{quinf}}^{\text{c}}$ denotes the capital requirement for qualifying infrastructure corporate equities.

7\.

The impact of the instantaneous decreases set out in 3D9 and 3D10 on the value of participations as referred to in Own Funds 3K.6 in financial institutions and credit institutions must only be taken into account on the value of the participations that are not deducted from own funds pursuant to Own Funds 3K.

8\.

A firm must treat the following equities as type 1:

1. (1) equities, other than qualifying infrastructure equities or qualifying infrastructure corporate equities, held within collective investment undertakings which are qualifying social entrepreneurship funds as referred to in Article 3(b) of Regulation (EU) No 346/2013 of the European Parliament and of the Council where the look-through approach is possible for all exposures within the collective investment undertakings, or units or shares of those funds where the look-through approach is not possible for all exposures within the collective investment undertakings;
2. (2) equities, other than qualifying infrastructure equities or qualifying infrastructure corporate equities, held within collective investment undertakings which are qualifying venture capital funds as referred to in Article 3(b) of Regulation (EU) No 345/2013 of the European Parliament and of the Council where the look-through approach is possible for all exposures within the collective investment undertakings, or units or shares of those funds where the look-through approach is not possible for all exposures within the collective investment undertakings;
3. (3) as regards closed-ended alternative investment funds which are established in the UK or, if they are not established in the UK, which are marketed in the UK in accordance with regulations 49, 50 and 54 of the Alternative Investment Fund Managers Regulations 2013/1773 in the form such regulations will take when regulation 3 and Schedule 1 of the Alternative Investment Fund Managers (Amendment) Regulations 2013/1797 come into force and which, in either case, have no leverage in accordance with the commitment method set out in Article 8 of Commission Delegated Regulation (EU) No 231/2013:
 1. (a) equities, other than qualifying infrastructure equities or qualifying infrastructure corporate equities, held within such funds where the look-through approach is possible for all exposures within the alternative investment funds; and
 2. (b) units or shares of such funds where the look-through approach is not possible for all exposures within the alternative investment funds; and
4. (4) qualifying unlisted equity portfolios as defined in 3D8.

Export Article as

3D8 Qualifying Unlisted Equity Portfolios

1\.

For the purposes of 3D7.8(4), a qualifying unlisted equity portfolio is a set of equity investments that meets all of the following requirements:

1. (1) the set of investments consists solely of investments in the ordinary shares of companies;
2. (2) the ordinary shares of each of the companies concerned are not listed in any regulated market;

3. (3) each company has its head office in the UK;
4. (4) more than 50% of the annual revenue of each company is denominated in currencies of countries which are members of the OECD;
5. (5) more than 50% of the staff employed by each company have their principal place of work in the UK;
6. (6) each company fulfils at least one of the following requirements for each of the last three financial years ending prior to the date on which the SCR is being calculated:
 1. (a) the annual turnover of the company exceeds GBP 8,800,000;
 2. (b) the balance sheet total of the company exceeds GBP 8,800,000; or
 3. (c) the number of staff employed by the company exceeds 50;
7. (7) the value of the investment in each company represents no more than 10% of the total value of the set of investments;
8. (8) none of the companies is a UK Solvency II undertaking, a credit institution, an investment firm, a financial institution, an alternative investment fund manager, a UCITS management company, an institution for occupational retirement provision or a non-regulated undertaking carrying out financial activities; and
9. (9) the beta of the set of investments does not exceed 0.796.

2\.

For the purposes of 3D8.1(9), the beta of a set of investments is the average of the betas for each of the investments in that set of investments, weighted by the book values of those investments and a firm must determine the beta of an investment in a company in accordance with the following formula:

$$\beta = 0.9478 - 0.0034 \cdot GM + 0.0139 \cdot \frac{\text{Debt}}{\text{CFO}} - 0.0015 \cdot ROCE$$

where:

1. (a) β is the beta of the equity investment in the company;
2. (b) GM is the average gross margin for the company over the last five financial years ending prior to the date on which the SCR is being calculated;
3. (c) $Debt$ is the total debt of the company at the end of the most recent financial year for which figures are available;
4. (d) CFO is the average net cash-flow for the company from operations over the last five financial years ending prior to the date on which the SCR is being calculated; and
5. (e) $ROCE$ is the average return on common equity for the company over the last five financial years ending prior to the date on which the SCR is being calculated. Common equity for these purposes shall mean capital and reserves as referred to in Schedule 1 to the Large and Medium-sized Companies and Groups (Accounts and Reports) Regulations 2008/410 as amended from time to time excluding preference shares and the related share premium account.

Export Article as

3D9 Standard Equity Risk Sub-Module

1\.

A firm must calculate the capital requirement for type 1 equities referred to in 3D7 as equal to the loss in its basic own funds that would result from the following instantaneous decreases:

1. (1) an instantaneous decrease equal to 22% in the value of type 1 equity investments in related undertakings where these investments are of a strategic nature, in accordance with 3D10;
2. (2) an instantaneous decrease equal to 22% in the value of type 1 equity investments that are treated as long-term equity investments in accordance with 3D11; and

3. (3) an instantaneous decrease equal to the sum of 39% and the symmetric adjustment, in the value of type 1 equities other than those referred to in (1) and (2).

2\.

A firm must calculate the capital requirement for type 2 equities referred to in 3D7 as equal to the loss in its basic own funds that would result from the following instantaneous decreases:

1. (1) an instantaneous decrease equal to 22% in the value of type 2 equity investments in related undertakings where these investments are of a strategic nature, in accordance with 3D10;
2. (2) an instantaneous decrease equal to 22% in the value of type 2 equity investments that are treated as long-term equity investments in accordance with 3D11; and
3. (3) an instantaneous decrease equal to the sum of 49% and the symmetric adjustment, in the value of type 2 equities other than those referred to in (1) and (2).

3\.

A firm must calculate the capital requirement for qualifying infrastructure equities referred to in 3D7 as equal to the loss in its basic own funds that would result from the following instantaneous decreases:

1. (1) an instantaneous decrease equal to 22% in the value of qualifying infrastructure equity investments in related undertakings, where those investments are of a strategic nature, in accordance with 3D10;
2. (2) an instantaneous decrease equal to 22% in the value of qualifying infrastructure equity investments that are treated as long-term equity investments in accordance with 3D11; and
3. (3) an instantaneous decrease equal to the sum of 30% and 77% of the symmetric adjustment in the value of qualifying infrastructure equity investments, other than those referred to in (1) and (2).

4\.

A firm must calculate the capital requirement for qualifying infrastructure corporate equities referred to in 3D7 as equal to the loss in its basic own funds that would result from the following instantaneous decreases:

1. (1) an instantaneous decrease equal to 22% in the value of qualifying infrastructure corporate equity investments in related undertakings where those investments are of a strategic nature, in accordance with 3D10;
2. (2) an instantaneous decrease equal to 22% in the value of qualifying infrastructure corporate equity investments that are treated as long-term equity investments in accordance with 3D11; and
3. (3) an instantaneous decrease equal to the sum of 36% and 92% of the symmetric adjustment in the value of qualifying infrastructure corporate equities, other than those referred to in (1) and (2).

Export Article as

3D10 Strategic Equity Investments

1\.

For the purposes of 3D9.1(1), 3D9.2(1), 3D9.3(1) and 3D9.4(1), equity investments of a strategic nature means equity investments for which the firm holding a participation demonstrates the following:

1. (1) that the value of the equity investment is likely to be materially less volatile for the following 12 months than the value of other equities over the same period as a result of both the nature of the investment and the influence exercised by the firm holding a participation in the related undertaking; and
2. (2) that the nature of the investment is strategic, taking into account all relevant factors, including:

1. (a) the existence of a clear decisive strategy to continue holding the participation for a long period;
2. (b) the consistency of the strategy referred to in (a) with the main policies guiding or limiting the actions of the undertaking;
3. (c) the firm's ability to continue holding the participation in the related undertaking;
4. (d) the existence of a durable link; and
5. (e) where the firm holding a participation is part of a group, the consistency of such strategy with the main policies guiding or limiting the actions of the group.

Export Article as

3D11 Long-Term Equity Investments

1\.

A firm may treat a sub-set of equity investments as long-term equity investments if all of the following requirements are met and, subject to 3D11.4, the firm has notified the PRA in writing that it meets these requirements:

1. (1) the sub-set of equity investments as well as the holding period of each equity investment within the sub-set are clearly identified;
2. (2) the sub-set of equity investments is included within a portfolio of assets which is assigned to cover the best estimate of a portfolio of insurance or reinsurance obligations corresponding to one or several clearly identified businesses, and the firm maintains that assignment over the lifetime of the obligations;
3. (3) the portfolio of insurance or reinsurance obligations, and the assigned portfolio of assets referred to in (2) are identified, managed and organised separately from the other activities of the firm, and the assigned portfolio of assets cannot be used to cover losses arising from other activities of the firm;
4. (4) the technical provisions within the portfolio of insurance or reinsurance obligations referred to in (2) only represent a part of the total technical provisions of the firm;
5. (5) the average holding period of equity investments in the sub-set exceeds five years, or where the average holding period of the sub-set is lower than five years, the firm does not sell any equity investments within the sub-set until the average holding period exceeds five years;
6. (6) the sub-set of equity investments consists only of equities that are listed in the UK or of unlisted equities of companies that have their head offices in UK;
7. (7) the solvency and liquidity position of the firm, as well as its strategies, processes and reporting procedures with respect to asset-liability management, are such as to ensure, on an ongoing basis and under stressed conditions, that it is able to avoid forced sales of each equity investments within the sub-set for at least 10 years; and
8. (8) the risk management, asset-liability management and investment policies of the firm reflects the firm's intention to hold the sub-set of equity investments for a period that is compatible with the requirement of (5) and its ability to meet the requirement of (7).

2\.

Where equities are held within collective investment undertakings or within alternative investment funds that meet the requirements of 3D7.8(1) to (3), the requirements set out in 3D11.1 may be assessed at the level of the funds and not of the underlying assets held within those funds.

3\.

A firm that treats a sub-set of equity investments as long-term equity investments in accordance with 3D11.1 must not revert back to an approach that does not include long-term equity investments, and if the firm is no longer able to comply with the requirements set out in 3D11.1, it must immediately inform the PRA in writing and cease to apply 3D9.1(2), 3D9.2(2), 3D9.3(2) and 3D9.4(2) to any of its equity investments for a period of 36 months.

4\.

Where a firm treated a sub-set of equity investments as long-term equity investments in accordance with Article 171a of Commission Delegated Regulation (EU) 2015/35 immediately before 31 December 2024 and from 31 December 2024 treats that sub-set of equity investments as long-term equity investments, the firm must notify the PRA in writing by 31 January 2025.

Export Article as

3D12 Symmetric Adjustment Of The Equity Capital Charge

1\.

The equity index upon which the symmetric adjustment to the standard equity capital charge is to be based must comply with all of the following requirements:

1. (1) the equity index measures the market price of a diversified portfolio of equities which is representative of the nature of equities typically held by UK Solvency II undertakings;
2. (2) the level of the equity index is publicly available; and
3. (3) the frequency of published levels of the equity index is sufficient to enable the current level of the index and its average value over the last 36 months to be determined.

2\.

Subject to 3D12.4, a firm must calculate the symmetric adjustment in accordance with the following formula:

$$\backslash[SA = \frac{1}{2} \cdot \left(\frac{CI - AI}{AI} - 8\% \right)\backslash]$$

where:

1. (a) $_CI_$ denotes the current level of the equity index; and
2. (b) $_AI_$ denotes the weighted average of the daily levels of the equity index over the last 36 months.

3\.

For the purposes of calculating the weighted average of the daily levels of the equity index, the weights for all daily levels must be equal and the days during the last 36 months in respect of which the index was not determined must not be included in the average.

4\.

The symmetric adjustment must not be lower than -10% or higher than 10%.

Export Article as

3D13 Calculation Of The Equity Index

1\.

For the purpose of this Chapter, the following definitions apply:

1. (1) 'last level' means the last value of the equity index for the day of reference published by the provider of the equity index; and
2. (2) 'working day' means every day other than Saturdays and Sundays.

2\.

The level of the equity index referred to in 3D12 must be determined for each working day.

3\.

The level of the equity index for a particular working day must be the sum of the contributions of all equity indices included in 3D14 on that working day.

4\.

For each of the equity indices set out in 3D14, its contribution for a particular working day must be the product of its normalised level for the working day and the respective weight for the equity index as set out in 3D14.

5\.

For each of the equity indices set out in 3D14, its normalised level for a particular working day must be its last level on that working day divided by its last level on the first day of the 36 month period ending on the working day for which the level of the equity index as defined in 3D12.1 is being calculated, provided that where the last level of an equity index is not available for a specific day, the most recent last level before that day must be used.

Export Article as

3D14 Calculation Of The Equity Index

1\.

The equity indices referred to in 3D13 are as follows:

Equity indices (Price indices) | Weights

---|---

FTSE All-Share Index | 0.48

Nikkei 225 | 0.07

S&P 500 | 0.30

FTSE Developed Europe ex UK (local currency) | 0.15

Export Article as

Property Risk Sub-Module

3D15 Property Risk Sub-Module

1\.

A firm must calculate the capital requirement for property risk as equal to the loss in its basic own funds that would result from an instantaneous decrease of 25% in the value of immovable property.

Export Article as

Spread Risk Sub-Module

3D16 Scope Of The Spread Risk Sub-Module

1\.

A firm must calculate the capital requirement for spread risk in accordance with the following formula:

$$\backslash\{ \{ \text{SCR} \}_\text{spread} \} = \{ \text{SCR} \}_\text{bonds} + \{ \text{SCR} \}_\text{securitisation} + \{ \text{SCR} \}_\text{cd} \backslash\}$$

where:

1. (a) $\text{SCR}_{\text{bonds}}$ denotes the capital requirement for spread risk on bonds and loans;
2. (b) $\text{SCR}_{\text{securitisation}}$ denotes the capital requirement for spread risk on securitisation positions; and
3. (c) SCR_{cd} denotes the capital requirement for spread risk on credit derivatives.

Export Article as

3D17 Spread Risk On Bonds And Loans

1\.

A firm must calculate the capital requirement for spread risk on bonds and loans $\text{SCR}_{\text{bonds}}$ as equal to the loss in its basic own funds that would result from an instantaneous relative decrease of stress_i in the value of each bond or loan i other than mortgage loans that meet the requirements in 3E3, including bank deposits other than cash at bank referred to in 3.14(2).

2\.

A firm must calculate the risk factor stress_i by reference to the modified duration of the bond or loan i denominated in years (dur_i) provided that dur_i must never be lower than 1. For variable interest rate bonds or loans, dur_i must be equivalent to the modified duration of a fixed interest rate bond or loan of the same maturity and with coupon payments equal to the forward interest rate.

3\.

A firm must assign bonds or loans for which a credit assessment by a nominated external credit assessment institution is available a risk factor stress_i depending on the credit quality step and the modified duration dur_i of the bond or loan i according to the following table:

$\text{Credit quality step}_i$ | 0 | 1 | 2 | 3 | 4 | 5 and 6

---|---|---|---|---|---

**Duration

dur_i | stress_i | a_i | b_i | a_i^{**} | b_i | a_i | b_i | a_i | b_i | a_i | b_i

up to 5 | $\backslash\{b_i \cdot \text{dur}_i\}$ | ? | 0.9% | ? | 1.1% | ? | 1.4% | ? | 2.5% | ? | 4.5% | ? | 7.5%

More than 5 and up to 10 | $\backslash\{a_i + b_i \cdot (\text{dur}_i - 5)\}$ | 4.5% | 0.5% | 5.5% | 0.6% | 7.0% | 0.7% | 12.5% | 1.5% | 22.5% | 2.5% | 37.5% | 4.2%

More than 10 and up to 15 | $\backslash\{a_i + b_i \cdot (\text{dur}_i - 10)\}$ | 7.0% | 0.5% | 8.5% | 0.5% | 10.5% | 0.5% | 20.0% | 1.0% | 35.0% | 1.8% | 58.5% | 0.5%

More than 15 and up to 20 | $\backslash\{a_i + b_i \cdot (\text{dur}_i - 15)\}$ | 9.5% | 0.5% | 11.0% | 0.5% | 13.0% | 0.5% | 25.0% | 1.0% | 44.0% | 0.5% | 61.0% | 0.5%

More than 20 | $\backslash\{\min\{a_i + b_i \cdot (\text{dur}_i - 20), 1\}\}$ | 12.0% | 0.5% | 13.5% | 0.5% | 15.5% | 0.5% | 30.0% | 0.5% | 46.6% | 0.5% | 63.5% | 0.5%

4\.

A firm must assign bonds and loans for which a credit assessment by a nominated external credit assessment institution is not available and for which debtors have not posted collateral by way of a collateral arrangement that meets the criteria set out in 3G8 a risk factor stress_i depending on the duration dur_i of the bond or loan i according to the following table:

Duration (dur_i) | stress_i

---|---

up to 5 | $\frac{3}{100} \cdot \text{dur}_i$

More than 5 and up to 10 | $\frac{15 + 1.7}{100} \cdot (\text{dur}_i - 5)$

More than 10 and up to 20 | $\frac{23.5}{100} + \frac{1.2}{100} \cdot (\text{dur}_i - 10)$

More than 20 | $\max\left(\frac{35.5}{100} + \frac{0.5}{100} \cdot (\text{dur}_i - 20), 1\right)$

5\.

Notwithstanding 3D17.4, bonds and loans that are assigned to a credit quality step in accordance with 3D18.1 or 3D18.2 or 3D20.1 must be assigned a risk factor stress_i depending on the credit quality step and the modified duration dur_i of the bond or loan i assigned in accordance with the table set out in 3D17.3.

6\.

A firm must assign bonds and loans for which a credit assessment by a nominated external credit assessment institution is not available and for which debtors have posted collateral by way of a collateral arrangement, where the collateral of those bonds and loans meet the criteria set out in 3G8, a risk factor stress_i according to the following:

1. (1) where the risk-adjusted value of collateral is higher than or equal to the value of the bond or loan i , stress_i must be equal to half of the risk factor that would be determined in accordance with 3D17.4;
2. (2) where the risk-adjusted value of collateral is lower than the value of the bond or loan i , and where the risk factor determined in accordance with 3D17.4 would result in a value of the bond or loan i that is lower than the risk-adjusted value of the collateral, stress_i must be equal to the average of the following:
 1. (a) the risk factor determined in accordance with 3D17.4; and
 2. (b) the difference between the value of the bond or loan i and the risk-adjusted value of the collateral, divided by the value of the bond or loan i ; and
3. (3) where the risk-adjusted value of collateral is lower than the value of the bond or loan i , and where the risk factor determined in accordance with 3D17.4 would result in a value of the bond or loan i that is higher than or equal to the risk-adjusted value of the collateral, stress_i must be determined in accordance with 3D17.4.

A firm must calculate the risk-adjusted value of the collateral in accordance with 7.34, 3E10 and 3E11.

7\.

A firm must take into account the impact of the instantaneous decrease in the value of participations in financial institutions and credit institutions, as referred to in Own Funds 3K.6, only on the value of the participations that are not deducted from own funds pursuant to Own Funds 3K.

Export Article as

3D18 Internal Assessment Of Credit Quality Steps Of Bonds And Loans

1\.

A firm may assign a bond or loan for which a credit assessment by a nominated external credit assessment institution is not available and for which debtors have not posted collateral by way of a collateral arrangement that meets the criteria

set out in 3G8 to credit quality step 2 if all of the criteria set out in 3D18.3 and 3D18.4 are met with respect to the bond or loan.

2\.

A firm may assign a bond or loan for which a credit assessment by a nominated external credit assessment institution is not available and for which debtors have not posted collateral by way of a collateral arrangement that meets the criteria set out in 3G8, other than a bond or loan assigned to credit quality step 2 under 3D18.1, to credit quality step 3 if all of the criteria set out in 3D18.3 and 3D18.5 are met with respect to the bond or loan.

3\.

The criteria in this rule are as follows:

1. (1) the firm's own internal credit assessment of the bond or loan meets the requirements listed in 3D19;
2. (2) the bond or loan is issued by a company which does not belong to the same corporate group as the firm;
3. (3) the bond or loan is not issued by a company which is a UK Solvency II undertaking, an infrastructure entity, a credit institution, an investment firm, a financial institution, an alternative investment fund manager, a UCITS management company, an institution for occupational retirement provision or a non-regulated undertaking carrying out financial activities;
4. (4) no claims on the issuing company of the bond or loan rank senior to the bond or loan, except for the following claims:
 1. (a) statutory claims and claims from liquidity facility providers provided that those statutory claims and claims from liquidity facility providers are in aggregate not material relative to the overall senior debt of the issuing company;
 2. (b) claims from trustees; and
 3. (c) claims from derivatives counterparties;
5. (5) the bond or loan provides a fixed redemption payment on or before the date of maturity, in addition to regular fixed or floating rate interest payments;
6. (6) the contractual terms and conditions of the bond or loan provide for the following:
 1. (a) the borrower is obliged to provide audited financial data to the lender at least annually;
 2. (b) the borrower is obliged to notify the lender of any events that could materially affect the credit risk of the bond or loan;
 3. (c) the borrower is not entitled to change the terms and conditions of the bond or loan unilaterally, nor to make other changes to its business that would materially affect the credit risk of the bond or loan;
 4. (d) the issuer is prohibited from issuing new debt without the prior agreement of the firm;
 5. (e) what constitutes a default event is defined in a way that is specific to the issue and the issuer; and
 6. (f) what is to happen on a change of control; and
7. (7) the bond or loan is issued by a company that meets all of the following criteria:
 1. (a) the company is a limited liability company;
 2. (b) the company has its head office in the UK;
 3. (c) more than 50% of the annual revenue of the company is denominated in currencies of countries which are members of the OECD;
 4. (d) the company has operated without any credit event over at least the last 10 years;
 5. (e) at least one of the following requirements is fulfilled with respect to each of the last three financial years ending prior to the date on which the SCR is being calculated:
 1. (i) the annual turnover of the company exceeds GBP 8,800,000;
 2. (ii) the balance sheet total of the company exceeds GBP 8,800,000; or
 3. (iii) the number of staff employed by the company exceeds 50;

6. (f) the sum of the company's annual earnings before interest, tax, depreciation and amortisation ('EBITDA') over the last five financial years is greater than 0;
7. (g) the total debt of the company at the end of the most recent financial year for which figures are available is no higher than 6.5 times the average of the company's annual free cash-flows over the last five financial years;
8. (h) the average of the company's EBITDA over the last five financial years is no lower than 6.5 times the company's interest expense for the most recent financial year for which figures are available; and
9. (i) the net debt of the company at the end of the most recent financial year for which figures are available is no higher than 1.5 times the company's total equity at the end of that financial year.

4\.

The yield on the bond or loan, and the yield on any bonds and loans with similar contractual terms and conditions issued by the same company in the previous three financial years, is no higher than the higher of the following values:

1. (1) the average of the yields on the two indices determined in accordance with 3D18.6; and
2. (2) the sum of 0.5% and the yield on the index that meets the requirement in 3D18.6(4).

5\.

The yield on the bond or loan, and the yield on bonds and loans with similar contractual terms and conditions issued by the same company in the previous three financial years, is no higher than the higher of the following values:

1. (1) the average of the yields on the two indices determined in accordance with 3D18.7; and
2. (2) the sum of 0.5% and the yield on the index that meets the requirement in [3D18.7](/pra-rules/solvency-capital-requirement---standard-formula#8a71cb0effa741e1b84ee396b839671b)(2).

6\.

For the purposes of 3D18.4, a firm must use, for the bond or loan referred to in 3D18.1, the yield, as at the time of issuance of that bond or loan, on two indices that meet all of the following requirements:

1. (1) both indices are broad indexes of traded bonds for which an external credit assessment is available;
2. (2) the constituent traded bonds in the two indices are denominated in the same currency as the bond or loan;
3. (3) the constituent traded bonds in the two indices have a similar maturity date as the bond or loan;
4. (4) one of the two indices consists of traded bonds of credit quality step 2; and
5. (5) one of the two indices consists of traded bonds of credit quality step 4.

7\.

For the purposes of 3D18.5, a firm must use, for the bond or loan referred to in 3D18.2, the yield, as at the time of issuance of that bond or loan, on two indices that meet all of the following requirements:

1. (1) both indices meet the requirements set out in 3D18.6(1), (2) and (3);
2. (2) one of the two indices consists of traded bonds of credit quality step 3; and
3. (3) one of the two indices consists of traded bonds of credit quality step 4.

8\.

For the purposes of 3D18.4, where the bond or loan referred to in 3D18.1 has features, other than those related to credit risk or illiquidity, which materially differ from the features of the constituent traded bonds in the two indices determined in accordance with 3D18.6, a firm must adjust the yield on the bond or loan to reflect those differences.

9\.

For the purposes of 3D18.5, where the bond or loan referred to in 3D18.2 has features, other than those related to credit risk or illiquidity, which materially differ from the features of the constituent traded bonds in the two indices determined in accordance with 3D18.7, a firm must adjust the yield on the bond or loan to reflect those differences.

Export Article as

3D19 Requirements For A Firm's Own Internal Credit Assessment Of Bonds And Loans

1\.

For the purposes of 3D18.3(1), a firm must comply with the following requirements in respect of its own internal credit assessment of a bond or loan:

1. (1) the bond or loan is allocated a credit quality step on the basis of the firm's own internal credit assessment;
2. (2) the firm's own internal credit assessment, and the allocation of a credit quality step to the bond or loan on the basis of that assessment, are reliable and properly reflect the spread risk of the bond or loan spread risk sub-module, and, subject to 3D19.2, the firm has notified the PRA of this in writing before it assigns a bond or loan to a credit quality step in accordance with 3D18.1 or 3D18.2;
3. (3) the firm's own internal credit assessment takes into account all factors which could have a material effect on the credit risk associated with the bond or loan, including the following factors:
 1. (a) the competitive position of the issuer;
 2. (b) the quality of the issuer's management;
 3. (c) the financial policies of the issuer;
 4. (d) country risk;
 5. (e) the effect of any covenants that are in place;
 6. (f) the issuer's financial performance history, including the number of years that it has been operating;
 7. (g) the issuer's size and the level of diversity in its activities;
 8. (h) the quantitative impact on the issuer's risk profile and financial ratios of its having issued the bond or loan;
 9. (i) the issuer's ownership structure; and
 10. (j) the complexity of the issuer's business model;
4. (4) the firm's own internal credit assessment uses all relevant quantitative and qualitative information;
5. (5) the firm's own internal credit assessment, the allocation of a credit quality step on the basis of that assessment and the information used to support the own internal credit assessment are documented;
6. (6) the firm's own internal credit assessment takes into account the characteristics of comparable assets for which a credit assessment by a nominated external credit assessment institution is available;
7. (7) the firm's own internal credit assessment takes into account trends in the issuer's financial performance;
8. (8) the firm's own internal credit assessment is procedurally independent from the decision to underwrite; and
9. (9) the firm regularly reviews its own internal credit assessment.

2\.

Where a firm assigned a bond or loan to a credit quality step in accordance with Article 176a(1) or (2) of Commission Delegated Regulation (EU) 2015/35 immediately before 31 December 2024 and from 31 December 2024 assigns the bond or loan to a credit quality step in accordance with 3D18.1 or 3D18.2, the firm must notify the PRA in writing by 31 January 2025.

Export Article as

3D20 Assessment Of Credit Quality Steps Of Bonds And Loans Based On An Approved Internal

Model

1\.

This Chapter applies in the following circumstances:

1. (1) a firm has concluded an agreement (?co-investment agreement?) to invest in bonds and loans jointly with another entity;
2. (2) that other entity (?the co-investor?) is one or other of the following:
 1. (a) an institution which uses the Internal Ratings Based Approach referred to in Article 143(1) of the CRR; or
 2. (b) a UK Solvency II undertaking which uses an internal model to calculate its SCR;
3. (3) pursuant to the co-investment agreement, the firm and the co-investor invest jointly in bonds and loans for which a credit assessment by a nominated external credit assessment institution is not available and for which debtors have not posted collateral by way of a collateral arrangement that meets the criteria set out in 3G8; and
4. (4) the co-investment agreement provides that the co-investor shares with the firm the probabilities of default produced by its Internal Ratings Based Approach or, as applicable, the credit quality steps produced by its internal model for the bonds or loans referred to in (3) for the purpose of using that information for the calculation of the SCR of the firm.

2\.

If all of the criteria set out in 3D20.3 to 3D20.6 are met, a firm must assign the bonds and loans referred to in 3D20.1(3) to credit quality steps determined as follows:

1. (1) in a case where the co-investor falls within 3D20.1(2)(a), credit quality steps must be determined on the basis of the most recent probabilities of default that the Internal Ratings Based Approach has produced; and
2. (2) in a case where the co-investor falls within 3D20.1(2)(b), credit quality steps must be the credit quality steps produced by the internal model.

3\.

The criteria in this rule are as follows:

1. (1) the issuer of each bond or loan does not belong to the same corporate group as the firm;
2. (2) the issuer is not a UK Solvency II undertaking, an infrastructure entity, a credit institution, an investment firm, a financial institution, an alternative investment fund manager, a UCITS management company, an institution for occupational retirement provision or a non-regulated undertaking carrying out financial activities;
3. (3) the issuer has its head office in the UK;
4. (4) more than 50% of the issuer?s annual revenue is denominated in currencies of countries which are members of the OECD; and
5. (5) at least one of the following requirements is met for each of the last three financial years ending prior to the date on which the SCR is being calculated:
 1. (a) the annual turnover of the issuer exceeds GBP 8,800,000;
 2. (b) the balance sheet total of the issuer exceeds GBP 8,800,000; or
 3. (c) the number of staff employed by the issuer exceeds 50.

4\.

The criteria in this rule are as follows:

1. (1) the co-investment agreement defines the types of bonds and loans to be underwritten, and the applicable

assessment criteria;

2. (2) the co-investor provides the firm with sufficient details of the underwriting process, including the criteria used, the organisational structure of the co-investor and the controls conducted by the co-investor;
3. (3) the co-investor provides the firm with data on all applications for bonds and loans to be underwritten;
4. (4) the co-investor provides the firm with details of all decisions to approve or reject applications for bonds and loans to be underwritten;
5. (5) the co-investor retains an exposure of at least 20% of the nominal value of each bond and loan;
6. (6) the underwriting process is the same as the underwriting process followed by the co-investor for its other investments in comparable bonds and loans;
7. (7) the firm invests in all bonds and loans of the types referred to in (1) for which the co-investor decides to approve the bond or loan application; and
8. (8) the co-investor provides the firm with information that allows the firm to understand the Internal Ratings Based Approach or, as applicable, internal model and its limitations, as well as its adequacy and appropriateness, in particular:
 1. (a) a description of the Internal Ratings Based Approach or, as applicable, internal model, including the inputs and risk factors, the quantification of risk parameters and the underlying methods, and the general methodology applied;
 2. (b) a description of the scope of the use of the Internal Ratings Based Approach or, as applicable, internal model; and
 3. (c) a description of the model validation process and of other processes which allow the model's performance to be monitored, the appropriateness of its specification to be reviewed over time, and the results of the Internal Ratings Based Approach or, as applicable, internal model to be tested against experience.

5\.

In a case where the co-investor falls within 3D20.1(2)(a):

1. (1) the firm clearly documents to which credit quality step the probability of default produced by the institution's Internal Ratings Based Approach corresponds;
2. (2) the mapping of probabilities of default to credit quality step carried out by the firm ensures that, for the bond or loan in question, the resulting level of capital requirement for the spread risk sub-module is appropriate;
3. (3) the mapping is based on Table 1 in Annex I to Commission Implementing Regulation (EU) 2016/1799;
4. (4) adjustments are made in a prudent manner to the probabilities of default before the mapping is carried out, taking into account the qualitative factors set out in Article 7 of Commission Implementing Regulation (EU) 2016/1799; and
5. (5) an adjustment to the probabilities of default is made in either of the following situations:
 1. (a) the time horizon covered by the Internal Ratings Based Approach deviates significantly from the 3-year time horizon set out in Article 4(2) of Commission Implementing Regulation (EU) 2016/1799; and
 2. (b) the definition of default used in the Internal Ratings Based Approach deviates significantly from the one set out in Article 4(4) of that Commission Implementing Regulation.

6\.

In a case where the co-investor falls within 3D20.1(2)(b), its internal model ensures that, for the bond or loan in question, the resulting level of capital requirement for the spread risk sub-module is appropriate.

Export Article as

3D21 Spread Risk On Securitisation Positions: Calculation Of The Capital Requirement

1\.

A firm must calculate the capital requirement `_SCR securitisation_` for spread risk on securitisation positions as equal to the loss in its basic own funds that would result from an instantaneous relative decrease of `_stress I_` in the value of

each securitisation position i .

2\.

The risk factor $\text{stress } i$ must be calculated by reference to the modified duration denominated in years ($\text{dur } i$), and $\text{dur } i$ must not be lower than 1 year.

3\.

In respect of senior securitisation positions in STS securitisations which fulfil the requirements set out in Article 243 of the CRR and for which a credit assessment by a nominated external credit assessment institution is available, a firm must assign a risk factor $\text{stress } i$ depending on the credit quality step and the modified duration of the securitisation position i , as set out in the following table:

Credit quality step | 0 | 1 | 2 | 3 | 4 | 5 and 6

---|---|---|---|---|---

****Duration**

$(\text{dur } i) | \text{stress } i | a_i | b_i | a_i | b_i | a_i | b_i | a_i | b_i | a_i | b_i$

up to 5 | $\lfloor b_i \cdot \text{dur}_i \rfloor$ | ? | 1.0% | ? | 1.2% | ? | 1.6% | ?

| 2.8% | ? | 5.6% | ? | 9.4%

More than 5 and up to 10 | $\lfloor a_i + b_i \cdot (\text{dur}_i - 5) \rfloor$ | 5.0% | 0.6% | 6.0% | 0.7% | 8.0% | 0.8% | 14.0% | 1.7% | 28.0% | 3.1% | 47.0% | 5.3%

More than 10 and up to 15 | $\lfloor a_i + b_i \cdot (\text{dur}_i - 10) \rfloor$ | 8.0% | 0.6% | 9.5% | 0.5% | 12.0% | 0.6% | 22.5% | 1.1% | 43.5% | 2.2% | 73.5% | 0.6%

More than 15 and up to 20 | $\lfloor a_i + b_i \cdot (\text{dur}_i - 15) \rfloor$ | 11.0% | 0.6% | 12.0% | 0.5% | 15.0% | 0.6% | 28.0% | 1.1% | 54.5% | 0.6% | 76.5% | 0.6%

More than 20 | $\lfloor \min \{ a_i + b_i \cdot (\text{dur}_i - 20); 1 \} \rfloor$ | 14.0% | 0.6% | 14.5% | 0.5% | 18.0% | 0.6% | 33.5% | 0.6% | 57.5% | 0.6% | 79.5% | 0.6%

4\.

In respect of securitisation positions in STS securitisations that are not senior securitisation positions, which fulfil the requirements set out in Article 243 of the CRR and for which a credit assessment by a nominated external credit assessment institution is available, a firm must assign a risk factor $\text{stress } i$ depending on the credit quality step and the modified duration of the securitisation position i , as set out in the following table:

Credit quality step | 0 | 1 | 2 | 3 | 4 | 5 and 6

---|---|---|---|---|---

****Duration**

$| \text{stress } i | a_i | b_i | a_i | b_i | a_i | b_i | a_i | b_i | a_i | b_i$

$(\text{dur } i)$

up to 5 | $\lfloor \text{min} \{ b_i \cdot \text{dur}_i; 1 \} \rfloor$ | ? | 2.8% | ? | 3.4% | ? | 4.6% | ? | 7.9% | ? | 15.8% | ? | 26.7%

More than 5 and up to 10 | $\lfloor \text{min} \{ a_i + b_i \cdot (\text{dur}_i - 5); 1 \} \rfloor$ | 14.0%

| 1.6% | 17.0% | 1.9% | 23.0% | 2.3% | 39.5% | 4.7% | 79.0% | 8.8% | 100.0% | 0.0%

More than 10 and up to 15 | $\lfloor a_i + b_i \cdot (\text{dur}_i - 10) \rfloor$ | 22.0% | 1.6% | 26.5% | 1.5% | 34.5% | 1.6% | 63.0% | 3.2% | 100.0% | 0.0% | 100.0% | 0.0%

More than 15 and up to 20 | $\lfloor a_i + b_i \cdot (\text{dur}_i - 15) \rfloor$ | 30.0% | 1.6% | 34.0% | 1.5% | 42.5% | 1.6% | 79.0% | 3.2% | 100.0% | 0.0% | 100.0% | 0.0%

More than 20 | $\lfloor \min \{ a_i + b_i \cdot (\text{dur}_i - 20); 1 \} \rfloor$ | 38.0% | 1.6% | 41.5% | 1.5% | 50.5% | 1.6% |

95.0% | 1.6% | 100.0% | 0.0% | 100.0% | 0.0%

5\.

In respect of senior securitisation positions in STS securitisations which fulfil the criteria set out in Article 243 of the CRR and for which no credit assessment by a nominated external credit assessment institution is available, a firm must assign a risk factor $\text{stress } i$ depending on the modified duration of the securitisation position i , as set out in the following table:

****Duration**

$\text{stress } i \mid a \mid b$

---|---|---|---

($\text{dur } i$)

up to 5 | $\lfloor b \cdot \text{dur}_i \rfloor \mid ? \mid 4.6\%$

More than 5 and up to 10 | $\lfloor a + b \cdot (\text{dur}_i - 5) \rfloor \mid 23\% \mid 2.5\%$

More than 10 and up to 15 | $\lfloor a + b \cdot (\text{dur}_i - 10) \rfloor \mid 35.5\% \mid 1.8\%$

More than 15 and up to 20 | $\lfloor a + b \cdot (\text{dur}_i - 15) \rfloor \mid 44.5\% \mid 0.5\%$

More than 20 | $\lfloor \min(\lfloor a + b \cdot (\text{dur}_i - 20) \rfloor, 1) \rfloor \mid 47.0\% \mid 0.5\%$

6\.

In respect of securitisation positions in STS securitisations that are not senior securitisation positions, which fulfil the criteria set out in Article 243 of the CRR and for which no credit assessment by a nominated external credit assessment institution is available, a firm must assign a risk factor $\text{stress } i$ equivalent to credit quality step 5 and depending on the modified duration of the exposure, as set out in the table in 3D21.3.

7\.

In respect of resecuritisation positions for which a credit assessment by a nominated external credit assessment institution is available, a firm must assign a risk factor $\text{stress } i$ in accordance with the following formula:

$$\text{stress } i = \min(\text{b } i \cdot \text{dur } i; 1)$$

where the value of $\text{b } i$ depends on the credit quality step of resecuritisation position i , as set out in the following table:

Credit quality step | 0 | 1 | 2 | 3 | 4 | 5 | 6

---|---|---|---|---|---|---

$\lfloor b_i \rfloor \mid 33\% \mid 40\% \mid 51\% \mid 91\%$

$\mid 100\% \mid 100\%$

$\mid 100\%$

8\.

In respect of securitisation positions not covered by 3D21.3 to 3D21.7, for which a credit assessment by a nominated external credit assessment institution is available, a firm must assign a risk factor $\text{stress } i$ in accordance with the following formula:

$$\text{stress } i = \min(\text{b } i \cdot \text{dur } i; 1)$$

where the value of $\text{b } i$ depends on the credit quality step of securitisation position i , as set out in the following table:

Credit quality step | 0 | 1 | 2 | 3 | 4 | 5 | 6

---|---|---|---|---|---|---

\\[b_i\\] | 12.5 % | 13.4 %

| 16.6 %

| 19.7 %

| 82% | 100%

| 100%

9\\.

In respect of securitisation positions not covered by 3D21.3 to 3D21.8, a firm must assign a risk factor `_stress i_` of 100%.

Export Article as

3D22 Spread Risk On Securitisation Positions: Transitional Provisions

1\\.

Notwithstanding 3D21.3, in respect of securitisations issued before 1 January 2019 that qualify as type 1 securitisations in accordance with Article 177(2) of Commission Delegated Regulation (EU) 2015/35 in the version that was in force on 31 December 2018, a firm must (subject to 3D22.2) assign a risk factor `_stress i_` in accordance with 3D21.3 even where those securitisations are not STS securitisations which fulfil the requirements set out in Article 243 of the CRR.

2\\.

3D22.1 applies only in circumstances where no new underlying exposures were added or substituted after 31 December 2018.

3\\.

Notwithstanding 3D21.3, in respect of securitisations issued before 18 January 2015 that qualify as type 1 securitisations in accordance with Article 177(4) in the version of Commission Delegated Regulation (EU) 2015/35 that was in force on 31 December 2018, a firm must assign a risk factor `_stress i_` in accordance with Articles 177 and 178 in the version in force on 31 December 2018.

4\\.

Notwithstanding 3D21.3, in respect of securitisations issued before 1 January 2019 that qualify as type 1 securitisations in accordance with Article 177(5) in the version of Commission Delegated Regulation (EU) 2015/35 that was in force on 31 December 2018, a firm must, until 31 December 2025, assign a risk factor `_stress i_` in accordance with Articles 177 and 178 in the version in force on 31 December 2018.

5\\.

For the purposes of 3D22.3 and 3D22.4, Article 177 (in the version of Commission Delegated Regulation (EU) 2015/35 which was in force on 31 December 2018) continues to have effect notwithstanding its deletion by Article 1(3) of Commission Delegated Regulation (EU) 2018/1221, and has effect for those purposes with the following modifications:

1. (1) paragraph 2 is to be read as if:

1. (a) a reference to Regulation (EU) No 575/2013 were a reference to the version of that Regulation which was in force on 31 December 2018;

2. (b) in point (b) 'the EEA or' were omitted;

3. (c) in point (h)(i):

1. (i) for 'national law of the Member State where the loans were originated' there were substituted 'loans were originated in the UK and the law of the UK';
2. (ii) ', and that Member State has notified this law to the Commission and EIOPA' were omitted;
4. (d) point (h)(ii) were omitted;
5. (e) in point (h)(iv) for the words from 'agricultural' to 'tracked' there were substituted 'tractors as defined in point (8) of Article 3 of Regulation (EU) No 167/2013 of the European Parliament and of the Council (as it had effect immediately before IP completion day), powered two-wheelers or powered tricycles as defined in points (68) and (69) of Article 3 of Regulation (EU) No 168/2013 of the European Parliament and of the Council (as it had effect immediately before IP completion day) or tracked';
6. (f) in points (r) and (s) for the words 'countries that are not members of the Union', both times it occurs, substitute 'a country other than the UK'; and
7. (g) in point (t):
 1. (i) the words from 'and discloses information' to 'stress tests' were omitted;
 2. (ii) for 'Union', in both places it occurs, there were substituted 'UK';
 2. (2) paragraph 4 is to be read as if for 'the entry into force of this Regulation' there were substituted '18 January 2015'; and
 3. (3) paragraph 5 is to be read as if, in points (a) and (c), for 'the date of entry into force of this Regulation' there were substituted '18 January 2015'.

Export Article as

3D23 Spread Risk On Credit Derivatives

1\.

A firm must calculate the capital requirement `_SCR cd_` for spread risk on credit derivatives other than those referred to in 3D23.4 as equal to the higher of the following capital requirements:

1. (1) the loss in its basic own funds that would result from an instantaneous increase in absolute terms of the credit spread of the instruments underlying the credit derivatives; and
2. (2) the loss in its basic own funds that would result from an instantaneous relative decrease of the credit spread of the instruments underlying the credit derivatives by 75%.

2\.

For the purposes of 3D23.1(1), a firm must calculate the instantaneous increase of the credit spread of the instruments underlying the credit derivatives for which a credit assessment by a nominated external credit assessment institution is available according to the following table:

Credit quality step | 0 | 1 | 2 | 3 | 4 | 5 | 6

---|---|---|---|---|---|---

Instantaneous increase in spread (in percentage points) | 1.3 | 1.5 | 2.6 | 4.5 | 8.4 | 16.2 | 16.2

3\.

For the purposes of 3D23.1(1), a firm must calculate the instantaneous increase of the credit spread of the instruments underlying the credit derivatives for which a credit assessment by a nominated external credit assessment institution is not available as 5 percentage points.

4\.

A firm must not subject credit derivatives which are part of the firm's risk mitigation policy to a capital requirement for spread risk, provided that the firm holds either the instruments underlying the credit derivative or another exposure with respect to which the basis risk between that exposure and the instruments underlying the credit derivative is not material in any circumstances.

5\.

Where the higher of the capital requirements referred to in 3D23.1(1) and (2) and the higher of the corresponding capital requirements calculated in accordance with 6.3(2) are not based on the same scenario, the capital requirement for spread risk on credit derivatives must be the capital requirement referred to in 3D23.1 for which the underlying scenario results in the highest corresponding capital requirement calculated in accordance with 6.3(2).

Export Article as

3D24 Specific Exposures

1\.

In respect of exposures in the form of covered bonds which have been assigned to credit quality step 0 or 1, a firm must assign a risk factor stress_i according to the following table: **Credit quality step Duration (dur_i)

| 0 | 1**

---|---|---

up to 5 | 0.7% ? dur_i | 0.9% ? dur_i

More than 5 years | $\max\{3.5\% + 0.5\% \cdot \max\{\text{dur}_i - 5, 0\}; 1\}$ | $\max\{4.5\% + 0.5\% \cdot \max\{\text{dur}_i - 5, 0\}; 1\}$

2\.

A firm must assign to exposures in the form of bonds and loans to the following a risk factor stress_i of 0%:

1. (1) UK central government and Bank of England denominated and funded in pounds sterling;
2. (2) multilateral development banks referred to in paragraph 2 of Article 117 of the CRR; and
3. (3) international organisations referred to in Article 118 of the CRR;

3\.

A firm must assign a risk factor stress_i of 0% to exposures in the form of bonds and loans that are fully, unconditionally and irrevocably guaranteed by one of the counterparties mentioned in 3D24.2(1) to (3), where the guarantee meets the requirements set out in 3G9.

4\.

For the purposes of 3D24.2(1), a firm must treat exposures in the form of bonds and loans that are fully, unconditionally and irrevocably guaranteed by bodies listed in 3D1, where the guarantee meets the requirements set out in 3G9, as exposures to the central government.

5\.

In respect of exposures in the form of bonds and loans to central governments and central banks other than those referred to in 3D24.2(1), denominated and funded in the domestic currency of that central government and central bank, and for which a credit assessment by a nominated external credit assessment institution is available, a firm must assign a risk factor stress_i depending on the credit quality step and the duration of the exposure according to the following table:

11\.

A firm must assign to exposures in the form of bonds and loans to a UK Solvency II undertaking which does not meet its MCR a risk factor stress_i according to the following table:

Duration (dur_i) | risk factor stress_i

---|---

up to 5 | 7.5% dur_i

More than 5 and up to 10 | 37.50% + 4.20% dur_i (5)

More than 10 and up to 15

| 58.50% + 0.50% dur_i (10)

More than 15 and up to 20 | 61% + 0.50% dur_i (15)

More than 20 | $\max\{63.5\% + 0.5\% \cdot (\text{dur}_i - 20); 1\}$

12\.

3D24.8 to 3D24.11 only applies as of the first date of public disclosure, by the UK Solvency II undertaking corresponding to the exposure, of the SFCR, and before that date:

1. (1) if a credit assessment by a nominated external credit assessment institution is available for the exposures, 3D17 applies;
2. (2) in all other cases, a firm must assign to the exposures the same risk factor as the ones that would result from the application of 3D24.8 to 3D24.10 to exposures to a UK Solvency II undertaking whose solvency ratio is 100%.

13\.

In respect of exposures in the form of bonds and loans to a third country insurance undertaking or a third country reinsurance undertaking for which a credit assessment by a nominated external credit assessment institution is not available, situated in a third country which is an overseas jurisdiction designated under regulation 11 in relation to regulation 13 of the IRPR regulations in respect of the insurance group capital requirements calculation, and which complies with the solvency requirements of that third country, a firm must assign the same risk factor as the ones that would result from the application of 3D24.8 to 3D24.10 to exposures to a UK Solvency II undertaking whose solvency ratio is 100%.

14\.

In respect of exposures in the form of bonds and loans to credit institutions and financial institutions which comply with the solvency requirements set out in the PRA Rulebook, the CRR or technical standards as amended from time to time, for which a credit assessment by a nominated external credit assessment institution is not available, a firm must assign the same risk factor as the ones that would result from the application of 3D24.8 to 3D24.10 to exposures to a UK Solvency II undertaking whose solvency ratio is 100%.

15\.

A firm must calculate the capital requirement for spread risk on credit derivatives where the underlying financial instrument is a bond or a loan to any exposure listed in 3D24.2 as nil.

16\.

In respect of exposures in the form of bonds and loans that fulfil the criteria set out in 3D24.17, a firm must assign a risk factor stress_i depending on the credit quality step and the duration of the exposure, according to the following table:

Credit quality step | 0 | 1 | 2 | 3

**Duration

17\.

The criteria for exposures that are assigned a risk factor in accordance with 3D24.16 are:

- 18\.

In respect of exposures in the form of bonds and loans that meet the criteria set out in 3D24.17(1) and (2), but do not meet the criteria set out in 3D24.17(3), a firm must assign a risk factor `_stress i_` equivalent to credit quality step 3 and the duration of the exposure in accordance with the table set out in 3D24.16.

19\.

In respect of exposures in the form of bonds and loans that fulfil the criteria set out in 3D24.20, a firm must assign a risk factor stress i depending on the credit quality step and the duration of the exposure according to the following table:

Credit quality step | 0 | 1 | 2 | 3

**Duration

20\

The criteria for exposures that are assigned a risk factor in accordance with 3D24.19 are:

1. (1) the exposure relates to a qualifying infrastructure corporate investment that meets the criteria set out in 3D3;

2. (2) the exposure is not an asset that fulfils the following conditions:

1. (a) it is assigned to a matching adjustment portfolio; and
2. (b) it has been assigned a credit quality step between 0 and 2;
3. (3) a credit assessment by a nominated external credit assessment institution is available for the infrastructure entity; and
4. (4) the exposure has been assigned a credit quality step between 0 and 3.

21\.

In respect of exposures in the form of bonds and loans that meet the criteria set out in 3D24.20(1) and (2), but do not meet the criteria set out in 3D24.20(3), a firm must assign a risk factor `_stress i_` equivalent to credit quality step 3 and the duration of the exposure in accordance with the table set out in 3D24.19.

Export Article as

3D25 Application of the Spread Risk Scenarios to Matching Adjustment Portfolios

1

Where a firm applies the matching adjustment, it must carry out the scenario-based calculation for spread risk as follows:

1. (1) the relevant portfolio of assets must be subject to the instantaneous decrease in value for spread risk set out in 3D17, 3D21 and 3D24; and
2. (2) the firm must recalculate the technical provisions to take into account the impact on the amount of the matching adjustment of the instantaneous decrease in value of the relevant portfolio of assets and, in particular, the firm must increase the fundamental spread calculated in respect of assigned assets by an absolute amount that is calculated as the product of the following:
 1. (a) the absolute increase in spread that, multiplied by the modified duration of the relevant asset, would result in the relevant risk factor `_stress i_`, referred to in 3D17, 3D21 and 3D24; and
 2. (b) a reduction factor, depending on the credit quality as set out in the following table:

Credit quality step | 0 | 1 | 2 | 3 | 4 | 5 | 6

---|---|---|---|---|---|---

Reduction factor | 45% | 50% | 60% | 75% | 100% | 100% | 100%

2

In respect of the assigned assets for which no credit assessment by a nominated external credit assessment institution is available, and for qualifying infrastructure assets and for qualifying infrastructure corporate assets that have been assigned credit quality step 3, a firm must apply a reduction factor of 100%.

Export Article as

Market Risk Concentrations Sub-Module

3D26 Single Name Exposure

1

A firm must calculate the capital requirement for market risk concentrations on the basis of single name exposures. For this purpose:

1. (1) exposures to undertakings which belong to the same corporate group must be treated as a single name exposure; and
2. (2) immovable properties which are located in the same building must be treated as a single immovable property.

2

A firm must calculate the exposure at default to a counterparty as the sum of its exposures to this counterparty.

3

A firm must calculate the exposure at default to a single name exposure as the sum of the exposures at default to all counterparties that belong to the single name exposure.

4

A firm must calculate the weighted average credit quality step on a single name exposure as equal to the rounded-up average of the credit quality steps of all exposures to all counterparties that belong to the single name exposure, weighted by the value of each exposure.

5

For the purposes of 3D26.4, a firm must assign to exposures for which a credit assessment by a nominated external credit assessment institution is available a credit quality step in accordance with 1A to 1C.

6

For the purposes of 3D26.4, in respect of exposures to a UK Solvency II undertaking for which a credit assessment by a nominated external credit assessment institution is not available and where the UK Solvency II undertaking meets its MCR, a firm must assign a credit quality step depending on the UK Solvency II undertaking's solvency ratio using the following mapping between solvency ratios and credit quality steps:

Solvency Ratio | 196% | 175% | 122% | 100% | 95%

---|---|---|---|---

Credit quality step | 1 | 2 | 3 | 3.82 | 5

7

Where the solvency ratio falls in between the solvency ratios set out in the table above, the credit quality step must be linearly interpolated from the closest credit quality steps corresponding to the closest solvency ratios set out in the table above, provided that:

1. (1) where the solvency ratio is lower than 95%, the credit quality step must be 5; and
2. (2) where the solvency ratio is higher than 196%, the credit quality step must be 1.

8

For the purposes of 3D26.6 to 3D26.7, 'solvency ratio' denotes the ratio of the eligible own funds to cover the SCR and the SCR, using the latest available values.

9

For the purposes of 3D26.4, a firm must assign credit quality step 6 to exposures to a UK Solvency II undertaking for which a credit assessment by a nominated external credit assessment institution is not available and where the UK Solvency II undertaking does not meet its MCR.

10

3D26.6 to 3D26.9 only applies as of the first date of public disclosure, by the UK Solvency II undertaking corresponding to the exposure, of the SFCR and before that date, a firm must assign the exposures to credit quality step 3.82.

11

For the purposes of 3D26.4, a firm must assign exposures to a third country insurance undertaking or a third country reinsurance undertaking for which a credit assessment by a nominated external credit assessment institution is not available, situated in a third country which is an overseas jurisdiction designated under regulation 11 in relation to regulation 13 of the IRPR regulations in respect of the insurance group capital requirements calculation, and which complies with the solvency requirements of that third country, to credit quality step 3.82.

12

For the purposes of 3D26.4, a firm must assign exposures to credit institutions and financial institutions, which comply with the solvency requirements set out in the PRA Rulebook, the CRR or technical standards as amended from time to time, for which a credit assessment by a nominated external credit assessment institution is not available, to credit quality step 3.82.

13

For the purpose of 3D26.4, firm must assign exposures other than those to which a credit quality step is assigned under 3D26.5 to 3D26.12 to credit quality step 5.

Export Article as

3D27 Calculation of the Capital Requirements for Market Risk Concentrations

1

A firm must calculate the capital requirement for market risk concentrations in accordance with the following formula:

$$\sqrt{\sum_{i=1}^n \text{Conc}_i^2}$$

where:

1. (a) the sum covers all single name exposures Conc_i ; and
2. (b) Conc_i denotes the capital requirement for market risk concentrations on a single name exposure Conc_i .

2

For each single name exposure Conc_i , a firm must calculate the capital requirement for market risk concentrations Conc_i as equal to the loss in its basic own funds that would result from an instantaneous decrease in the value of the assets corresponding to the single name exposure Conc_i calculated in accordance with the following formula:

$$X_{S_i} \cdot g_i$$

where:

1. (a) X_{S_i} is the excess exposure referred to in 3D28; and
2. (b) g_i is the risk factor for market risk concentrations referred to in 3D30 and 3D31;

Export Article as

3D28 Excess Exposure

1

A firm must calculate the excess exposure on a single name exposure $_i$ in accordance with the following formula:

$$\max\{XS_i - \max(0; E_i - CT_i) \cdot \text{Assets}, 0\}$$

where:

1. (a) E_i denotes the exposure at default to single name exposure $_i$ that is included in the calculation base of the market risk concentrations sub-module;
2. (b) Assets denotes the calculation base of the market risk concentrations sub-module; and
3. (c) CT_i denotes the relative excess exposure threshold referred to in 3D29.

2

The calculation base of the market risk concentrations sub-module Assets must be equal to the value of all assets held by the firm, excluding the following:

1. (1) assets held in respect of long-term insurance contracts where the investment risk is fully borne by the policyholders;
2. (2) exposures to a counterparty which belongs to the same group as the firm, provided that all of the following requirements are met:
 1. (a) the counterparty is a UK Solvency II undertaking, an insurance holding company, a mixed financial holding company or an ancillary services undertaking;
 2. (b) the counterparty is fully consolidated in accordance with Group Supervision 11.1A(1);
 3. (c) the counterparty is subject to the same risk evaluation, measurement and control procedures as the firm;
 4. (d) the counterparty is established in the UK; and
 5. (e) there is no current or foreseen material practical or legal impediment to the prompt transfer of own funds or repayment of liabilities from the counterparty to the firm;
3. (3) the value of the participations as referred to in Own Funds 3K.6 in financial institutions and credit institutions that is deducted from own funds pursuant to Own Funds 3K;
4. (4) exposures included in the scope of the counterparty default risk module;
5. (5) deferred tax assets; and
6. (6) intangible assets.

3

A firm must reduce the exposure at default on a single name exposure $_i$ by the amount of the exposure at default to counterparties belonging to that single name exposure and for which the risk factor for market risk concentrations referred to in 3D30 and 3D31 is 0%.

Export Article as

3D29 Relative Excess Exposure Thresholds

1

In respect of each single name exposure $_i$, a firm must assign, in accordance with the following table, a relative excess exposure threshold depending on the weighted average credit quality step of the single name exposure $_i$,

calculated in accordance with 3D26.4.

Weighted average credit quality step of single name exposure $_i_$ | 0 | 1 | 2 | 3 | 4 | 5 | 6

---|---|---|---|---|---|---

Relative excess exposure threshold $_CT_i_$ | 3% | 3% | 3% | 1.5% | 1.5% | 1.5% | 1.5%

Export Article as

3D30 Risk Factor for Market Risk Concentrations

1

In respect of each single name exposure $_i_$, a firm must assign, in accordance with the following table, a risk factor $_g_i_$ for market risk concentrations depending on the weighted average credit quality step of the single name exposure $_i_$, calculated in accordance with 3D26.4.

Weighted average credit quality step of single name exposure $_i_$ | 0 | 1 | 2 | 3 | 4 | 5 | 6

---|---|---|---|---|---|---

Risk factor $_g_i_$ | 12% | 12% | 21% | 27% | 73% | 73% | 73%

Export Article as

3D31 Specific Exposures

1

A firm must assign to exposures in the form of covered bonds a relative excess exposure threshold $_CT_i_$ of 15%, provided that the corresponding exposures in the form of covered bonds have been assigned to credit quality step 0 or 1 and must treat exposures in the form of covered bonds as single name exposures, regardless of other exposures to the same counterparty as the issuer of the covered bonds, which constitute a distinct single name exposure.

2

A firm must assign to exposures to a single immovable property a relative excess exposure threshold $_CT_i_$ of 10% and a risk factor $_g_i_$ for market risk concentrations of 12%.

3

A firm must assign to the following exposures a risk factor $_g_i_$ for market risk concentrations of 0%:

1. (1) the UK central government and Bank of England denominated and funded in pounds sterling;
2. (2) multilateral development banks referred to in Article 117(2) of the CRR; and
3. (3) international organisations referred to in Article 118 of the CRR.

4

In respect of exposures that are fully, unconditionally and irrevocably guaranteed by one of the counterparties mentioned in 3D31.3(1) to (3), where the guarantee meets the requirements set out in 3G9, a firm must also assign a risk factor $_g_i_$ for market risk concentrations of 0%.

5

For the purposes of 3D31.3(1), a firm must treat exposures that are fully, unconditionally and irrevocably guaranteed by

bodies listed in 3D1, where the guarantee meets the requirements set out in 3G9, as exposures to the central government.

6\.

In respect of exposures to central governments and central banks other than those referred to in 3D31.3(1), denominated and funded in the domestic currency of that central government and central bank, a firm must assign a risk factor $_g_i_$ for market risk concentrations depending on their weighted average credit quality steps, in accordance with the following table:

Weighted average credit quality step of single name exposure $_i_$ | 0 | 1 | 2 | 3 | 4 | 5 | 6

---|---|---|---|---|---|---

Risk factor $_g_i_$ | 0% | 0% | 12% | 21% | 27% | 73% | 73%

7

In respect of exposures to the UK's regional governments and local authorities not listed in 3D1, a firm must assign a risk factor $_g_i_$ for market risk concentrations corresponding to weighted average credit quality step 2 in accordance with 3D31.6.

8

In respect of exposures that are fully, unconditionally and irrevocably guaranteed by the UK's regional government or local authority that is not listed in 3D1, where the guarantee meets the requirements set out in 3G9, a firm must assign a risk factor $_g_i_$ for market risk concentration corresponding to weighted average credit quality step 2 in accordance with 3D31.6.

9

A firm must assign to exposures in the form of bank deposits a risk factor $_g_i_$ for market risk concentration of 0%, provided they meet all of the following requirements:

1. (1) the full value of the exposure is covered by a government guarantee scheme in the UK;
2. (2) the guarantee covers the firm without any restriction; and
3. (3) there is no double counting of such guarantee in the calculation of the SCR.

Export Article as

Currency Risk Sub-Module

3D32 Currency Risk Sub-Module

1

A firm must calculate the capital requirement for currency risk as equal to the sum of the capital requirements for currency risk for each foreign currency. For these purposes, a firm must treat investments as follows:

1. (1) type 1 equities referred to in 3D7.2 and type 2 equities referred to in 3D7.3 which are listed in stock exchanges operating with different currencies as sensitive to the currency of its main listing;
2. (2) type 2 equities referred to in 3D7.3 which are not listed as sensitive to the currency of the country in which the issuer has its main operations; and
3. (3) immovable property as sensitive to the currency of the country in which it is located.

2

For the purposes of this Chapter, foreign currencies are currencies other than the currency used for the preparation of the firm's financial statements (the local currency).

3

For each foreign currency, a firm must calculate the capital requirement for currency risk as equal to the higher of the following capital requirements:

1. (1) the capital requirement for the risk of an increase in value of the foreign currency against the local currency; and
2. (2) the capital requirement for the risk of a decrease in value of the foreign currency against the local currency.

4

A firm must calculate the capital requirement for the risk of an increase in value of a foreign currency against the local currency as equal to the loss in its basic own funds that would result from an instantaneous increase of 25% in the value of the foreign currency against the local currency.

5

A firm must calculate the capital requirement for the risk of a decrease in value of a foreign currency against the local currency as equal to the loss in its basic own funds that would result from an instantaneous decrease of 25% in the value of the foreign currency against the local currency.

6

For currencies which are pegged to the euro, a firm may adjust the 25% factor referred to in 3D32.4 and 3D32.5 in accordance with 3D33 and 3D34, provided that all of the following requirements are met:

1. (1) the pegging arrangement must ensure that the relative changes in the exchange rate over a one-year period do not exceed the relative adjustments to the 25% factor, in the event of extreme market events, that correspond to the confidence level set out in Solvency Capital Requirement ? General Provisions 3.3 and 3.4; and
2. (2) one of the following criteria is complied with:
 1. (a) participation of the currency in the European Exchange Rate Mechanism (ERM II);
 2. (b) existence of a decision from the European Council which recognises pegging arrangements between this currency and the euro; or
 3. (c) establishment of the pegging arrangement by the law of country establishing the country's currency.

7

For the purposes of 3D32.6(1), the financial resources of the parties that guarantee the pegging must be taken into account.

8

The impact of an increase or a decrease in the value of a foreign currency against the local currency on the value of participations in financial institutions and credit institutions as defined in Own Funds 3K.6 must only be taken into account in respect of the value of the participations that are not deducted from own funds pursuant to Own Funds 3K, and the part deducted from own funds must be taken into account only to the extent such impact increases the firm's basic own funds.

9

Where the higher of the capital requirements referred to in 3D32.3(1) and (2) and the highest of the corresponding capital requirements calculated in accordance with 6.3(2) are not based on the same scenario, a firm must apply as the capital requirement for currency risk on a given currency the capital requirement referred to in 3D32.3(1) or (2) for which the underlying scenario results in the highest corresponding capital requirement calculated in accordance with 6.3(2).

Export Article as

3D33 Adjusted Factors for Currency Risk Where the Local or Foreign Currency is the Euro

1

Where the local or foreign currency is the euro, for the purposes of 3D32.4 and 3D32.5, a firm must replace the 25% factor with:

1. (1) 0.39% where the other currency is the Danish krone (DKK);
2. (2) 1.81% where the other currency is the lev (BGN);
3. (3) 2.18% where the other currency is the West African CFA franc (BCEAO) (XOF);
4. (4) 1.96% where the other currency is the Central African CFA franc (BEAC) (XAF); and
5. (5) 2.00% where the other currency is the Comorian franc (KMF).

Export Article as

3D34 Adjusted Factors for Currency Risk Where the Local and the Foreign Currency are Pegged to the Euro

1

For the purposes of 3D32.4 and 3D32.5, a firm must replace the 25% factor with:

1. (1) 2.24% where the two currencies are the DKK and the BGN;
2. (2) 2.62% where the two currencies are the DKK and the XOF;
3. (3) 2.40% where the two currencies are the DKK and the XAF;
4. (4) 2.44% where the two currencies are the DKK and the KMF;
5. (5) 4.06% where the two currencies are the BGN and the XOF;
6. (6) 3.85% where the two currencies are the BGN and the XAF;
7. (7) 3.89% where the two currencies are the BGN and the KMF;
8. (8) 4.23% where the two currencies are the XOF and the XAF;
9. (9) 4.27% where the two currencies are the XOF and the KMF; and
10. (10) 4.04% where the two currencies are the XAF and the KMF.

Export Article as

3E

Counterparty Default Risk Module

3E1 Lists of Regional Governments and Local Authorities

1\.

A firm must treat exposures to the Scottish Government, the Welsh Government and the Northern Ireland Executive as exposures to the central government of the UK for the calculation of the counterparty default risk module of the standard formula.

[Export Article as](#)

3E2 Single Name Exposures

1\.

A firm must calculate the capital requirement for counterparty default risk on the basis of single name exposures. For that purpose exposures to undertakings which belong to the same corporate group must be treated as a single name exposure.

2\.

A firm may treat exposures which belong to different members of the same legal or contractual pooling arrangement as different single name exposures where the probability of default of the single name exposure is calculated in accordance with 3E12 and the loss-given-default is calculated as follows:

1. (1) in accordance with 3E6, if it is a pool exposure of type A,
2. (2) in accordance with 3E7, if it is a pool exposure of type B; or
3. (3) in accordance with 3E8, if it is a pool exposure of type C.

Alternatively exposures to the undertakings which belong to the same pooling arrangement must be treated as a single name exposure.

[Export Article as](#)

3E3 Mortgage Loans

1\.

A firm must treat retail loans secured by mortgages on residential property (mortgage loans) as type 2 exposures under the counterparty default risk provided the requirements in 3E3.2 to 3E3.13 are met.

2\.

The exposure must be either to a natural person or persons or to a small or medium-sized enterprise.

3\.

The exposure must be one of a significant number of exposures with similar characteristics such that the risks associated with such lending are substantially reduced.

4\.

The total amount owed to the firm and, where relevant, to all related undertakings, including any exposure in default, by the counterparty or other connected third party, must not, to the knowledge of the firm, exceed GBP 880,000 and the firm must take reasonable steps to acquire this knowledge.

5\.

The residential property is or will be occupied or let by the owner.

6\.

The value of the property does not materially depend upon the credit quality of the borrower.

7\.

The risk of the borrower does not materially depend upon the performance of the underlying property, but on the underlying capacity of the borrower to repay the debt from other sources, and as a consequence, the repayment of the facility does not materially depend on any cash-flow generated by the underlying property serving as collateral provided by way of a collateral arrangement. For those other sources, the firm must determine maximum loan-to-income ratio as part of its lending policy and obtain suitable evidence of the relevant income when granting the loan.

8\.

All of the following requirements on legal certainty must be met:

1. (1) a mortgage or charge is enforceable in all jurisdictions which are relevant at the time of the conclusion of the credit agreement and must be properly filed on a timely basis;
2. (2) all legal requirements for establishing the pledge have been fulfilled; and
3. (3) the protection agreement and the legal process underpinning it enable the firm to realise the value of the protection within a reasonable timeframe.

9\.

All of the following requirements on the monitoring of property values and on property valuation must be met:

1. (1) the firm must monitor the value of the property on a frequent basis and at a minimum once every three years, provided that the firm must carry out more frequent monitoring where the market is subject to significant changes in conditions; and
2. (2) the firm must review the property valuation when information available to the firm indicates that the value of the property may have declined materially relative to general market prices and that review must be external and independent and carried out by a valuer who possesses the necessary qualifications, ability and experience to execute a valuation and who is independent from the credit decision process.

10\.

For the purposes of 3E3.9, a firm may use statistical methods to monitor the value of the property and to identify property that needs revaluation.

11\.

A firm must clearly document the types of residential property it accepts as collateral and its lending policies in this regard and must require the independent valuer of the market value of the property, as referred to in 3E11.2, to document that market value in a transparent and clear manner.

12\.

A firm must have in place procedures to monitor that the property taken as credit protection is adequately insured against the risk of damage.

13\.

A firm must report all of the following data on losses stemming from mortgage loans to the PRA:

1. (1) losses stemming from loans that have been classified as type 2 exposures in accordance with 3.15 in any given

year; and

2. (2) overall losses in any given year.

Export Article as

3E4 Loss-Given-Default

1\.

A firm must calculate the loss-given-default on a single name exposure as equal to the sum of the loss-given-default on each of the exposures to counterparties belonging to the single name exposure on the following basis:

1. (1) the loss-given-default must be net of the liabilities towards counterparties belonging to the single name exposure provided that:
 1. (a) those liabilities and exposures are set off in the case of default of the counterparties; and
 2. (b) 3G2 and 3G3 are complied with in relation to that right of set-off; and
2. (2) no offsetting shall be allowed for if the liabilities are expected to be met before the credit exposure is cleared.

2\.

Where a firm has concluded contractual netting agreements covering several derivatives that represent credit exposure to the same counterparty, it may calculate the loss-given-default on those derivatives, as set out in 3E4.5 to 3E4.8, on the basis of the combined economic effect of all of those derivatives that are covered by the same contractual netting agreement, provided that 3G2 and 3G3 are complied with in relation to the netting.

3\.

A firm must calculate the loss-given-default on a reinsurance arrangement or insurance securitisation in accordance with the following formula:

$$\text{LGD} = \max\left(50\% \cdot \left(\text{Recoverables} + 50\% \cdot \text{RM}_{\text{re}}\right) - F \cdot \text{Collateral}; 0\right)$$

where:

1. (a) *_Recoverables_* denotes the best estimate of amounts recoverable from the reinsurance arrangement or insurance securitisation and the corresponding debtors;
2. (b) *RM_{re}* denotes the risk mitigating effect on underwriting risk of the reinsurance arrangement or securitisation;
3. (c) *_Collateral_* denotes the risk-adjusted value of collateral provided by way of a collateral arrangement in relation to the reinsurance arrangement or securitisation; and
4. (d) *_F_* denotes a factor to take into account the economic effect of the collateral arrangement in relation to the reinsurance arrangement or securitisation in case of any credit event related to the counterparty, determined in accordance with 3E10.7.

4\.

Where the reinsurance arrangement is with a UK Solvency II undertaking, a third country insurance undertaking or a third country reinsurance undertaking and 60% or more of that counterparty's assets are subject to collateral arrangements, a firm must calculate the loss-given-default in accordance with the following formula:

$$\text{LGD} = \max\left(90\% \cdot \left(\text{Recoverables} + 50\% \cdot \text{RM}_{\text{re}}\right) - F' \cdot \text{Collateral}; 0\right)$$

where:

$F?$ denotes a factor to take into account the economic effect of the collateral arrangement in relation to the reinsurance arrangement or securitisation in the case of a credit event related to the counterparty, determined in accordance with 3E10.7.

5\.

A firm must calculate the loss-given-default on a derivative falling within 3E5.1 in accordance with the following formula:

$$LGD = \max\left(18\% \cdot \left(\text{Derivative} + 50\% \cdot \text{RM}_{\text{fin}}\right) - 50\% \cdot F' \cdot \text{Value}; 0\right)$$

where:

1. (a) Derivative denotes the value of the derivative determined in accordance with Valuation 2.1 to 2.2;
2. (b) RM_{fin} denotes the risk-mitigating effect on market risk of the derivative;
3. (c) Value denotes the value of the assets held as collateral, provided by way of a collateral arrangement, determined in accordance with Valuation 2.1 to 2.2; and
4. (d) $F?$ denotes a factor to take into account the economic effect of the collateral arrangement in relation to the derivative in case of a credit event related to the counterparty, determined in accordance with 3E10.7.

6\.

Notwithstanding 3E4.5, a firm must calculate the loss-given-default on a derivative falling within 3E5.2 in accordance with the following formula:

$$LGD = \max\left\{\left(16\% \cdot \left(\text{Derivative} + 50\% \cdot \text{RM}_{\text{fin}}\right) - 50\% \cdot F'' \cdot \text{Value}\right); 0\right\}$$

where:

1. (a) Derivative denotes the value of the derivative in accordance with Valuation 2.1 to 2.2;
2. (b) RM_{fin} denotes the risk-mitigating effect on market risk of the derivative;
3. (c) Value denotes the value of the assets held as collateral, provided by way of a collateral arrangement, in accordance with Valuation 2.1 to 2.2; and
4. (d) $F??$ denotes a factor to take into account the economic effect of the collateral arrangement in relation to the derivative in case of a credit event related to the counterparty, determined in accordance with 3E10.7.

7\.

A firm must calculate the loss-given-default on derivatives other than those referred to in 3E4.5 and 3E4.6 in accordance with the following formula, provided that the derivative contract meets the requirements of Article 11 of Regulation (EU) 648/2012:

$$LGD = \max\left(90\% \cdot \left(\text{Derivative} + 50\% \cdot \text{RM}_{\text{fin}}\right) - 50\% \cdot F''' \cdot \text{Value}; 0\right)$$

where:

1. (a) Derivative denotes the value of the derivative determined in accordance with Valuation 2.1 to 2.2;
2. (b) RM_{fin} denotes the risk-mitigating effect on market risk of the derivative;
3. (c) Value denotes the value of the assets held as collateral, provided by way of a collateral arrangement, determined in accordance with Valuation 2.1 to 2.2; and
4. (d) $F???$ denotes a factor to take into account the economic effect of the collateral arrangement in relation to the derivative in case of a credit event related to the counterparty, determined in accordance with 3E10.7.

8\.

A firm must calculate the loss-given-default on derivatives not covered by 3E4.5, 3E4.6 and 3E4.7 in accordance with the following formula:

$$LGD = \max\left(90\% \cdot \left(\text{Derivative} + \text{RM}_{\text{fin}}\right) - F^{\text{prime}} \cdot \text{Collateral}; 0\right)$$

where:

1. (a) *_Derivative_* denotes the value of the derivative determined in accordance with Valuation 2.1 to 2.2;
2. (b) *_RM_{fin}_* denotes risk-mitigating effect on market risk of the derivative;
3. (c) *_Collateral_* denotes the risk-adjusted value of collateral provided by way of a collateral arrangement in relation to the derivative; and
4. (d) *_F^{prime}_* denotes a factor to take into account the economic effect of the collateral arrangement in relation to the derivative in case of a credit event related to the counterparty, determined in accordance with 3E10.7.

9\.

Where the loss-given-default on derivatives is to be calculated on the basis referred to in 3E4.2, the following rules apply for the purposes of 3E4.5 to 3E4.8:

1. (1) the value of the derivative must be the sum of the values of the derivatives covered by the contractual netting arrangement;
2. (2) the risk-mitigating effect must be determined at the level of the combination of derivatives covered by the contractual netting arrangement; and
3. (3) the risk-adjusted value of collateral provided by way of a collateral arrangement must be determined at the level of the combination of derivatives covered by the contractual netting arrangement.

10\.

A firm must calculate the loss-given-default on a mortgage loan in accordance with the following formula:

$$LGD = \max\left(\text{Loan} - \left(80\% \cdot \text{Mortgage} + \text{Guarantee}\right); 0\right)$$

where:

1. (a) *_Loan_* denotes the value of the mortgage loan determined in accordance with Valuation 2.1 to 2.2;
2. (b) *_Mortgage_* denotes the risk-adjusted value of the mortgage; and
3. (c) *_Guarantee_* denotes the amount that the guarantor would be required to pay to the firm if the obligor of the mortgage loan were to default at a time when the value of the property held as mortgage were equal to 80% of the risk-adjusted value of the mortgage.

11\.

For the purposes of 3E4.10(c), a guarantee must only be recognised if it is provided by a counterparty mentioned in 3D24.2(1) to (3) and it complies with the requirements set out in 3G2, 3G3 and 3G9.1(1) to (5).

12\.

A firm must calculate the loss-given-default on a legally binding commitment as referred to in 3.14(5) as the difference between its nominal value and its value in accordance with Valuation 2.1 to 2.2.

13\.

The loss-given-default on cash at bank as referred to in Schedule 3 to the Large and Medium-sized Companies and Groups (Accounts and Reports) Regulations 2008/410 as amended from time to time, of a deposit with a ceding undertaking, of an item listed in 3.14(4) or 3.15(5), or of a receivable from an intermediary or policyholder debtor, as well as any other exposure not listed elsewhere in this Chapter must be equal to its value in accordance with Valuation 2.1 to 2.2.

Export Article as

3E5 Exposure to Clearing Members

1\.

For the purposes of 3E4.5, a derivative falls within this rule if the following requirements are met:

1. (1) the derivative is a CCP-related transaction in which the firm is the client;
2. (2) the positions and assets of the firm related to that transaction are distinguished and segregated, at the level of both the clearing member and the CCP, from the positions and assets of both the clearing member and the other clients of that clearing member and as a result of that distinction and segregation those positions and assets are bankruptcy remote in the event of the default or insolvency of the clearing member or one or more of its other clients;
3. (3) the laws, regulations, rules and contractual arrangements applicable to or binding the firm or the CCP facilitate the transfer of the client's positions relating to that transaction and of the corresponding collateral provided by way of a collateral arrangement to another clearing member within the applicable margin period of risk in the event of default or insolvency of the original clearing member and in such circumstances, the client's positions and the collateral must be transferred at market value, unless the client requests to close out the position at market value;
4. (4) the firm has available an independent, written and reasoned legal opinion that concludes that, in the event of legal challenge, the relevant courts and administrative authorities would find that the client would bear no losses on account of the insolvency of the clearing member or of any the clients of that clearing member under any of the following laws:
 1. (a) the laws of the jurisdiction of the firm, its clearing member or the CCP;
 2. (b) the law governing the transaction;
 3. (c) the law governing the collateral; and
 4. (d) the law governing any contract or agreement necessary to meet the requirement set out in (2); and
5. (5) the CCP is a qualifying central counterparty as defined in Article 4(1)(88) of the CRR.

2\.

For the purposes of 3E4.6, a derivative falls within this rule if the requirements set out in 3E5.1 are met, with the exception that the firm is not required to be protected from losses in the event that the clearing member and another client of the clearing member jointly default.

Export Article as

3E6 Loss-Given-Default for Pool Exposures of Type A

1\.

For pool exposures of type A which a firm is permitted to treat as separate single name exposures in accordance with 3E2.2, where members are each only liable up to their respective portion of the obligation covered by the pooling arrangement, the firm must calculate the loss-given-default in accordance with 3E4.

2\.

For pool exposures of type A which a firm is permitted to treat as separate single name exposures in accordance with 3E2.2, where members are each liable up to the full amount of the obligation covered by the pooling arrangement, a firm must multiply the loss-given-default calculated in accordance with 3E4 by the risk-share factor, calculated in accordance with the following formula:

$$\text{risk-share factor} = e^{-0.15(\min\{\text{SR}, 196\} - 1)}$$

where:

(a) $\text{SR} = \left(1 - P \right) \cdot \frac{\sum_i \text{EOF}_i}{\sum_i \left(\frac{\text{EOF}_i}{\text{SR}_i} \right) + \sum_j \text{P}_j}$

(b) i denotes all pool members which are UK Solvency II undertakings and j denotes all pool members which are third country insurance undertakings or third country reinsurance undertakings;

(c) $P = \sum_j P_j$

(d) P_j denotes the share of the total risk of the pooling arrangement undertaken by pool member j ; and

(e) for pool members for which a credit assessment by a nominated external credit assessment institution is available, SR_i and SR_j must be assigned in accordance with the following table:

Credit quality step | 0 | 1 | 2 | 3 | 4 | 5 | 6

---|---|---|---|---|---|---

SR_i | 196% | 196% | 175% | 122% | 95% | 75% | 75%

(f) for pool members which are UK Solvency II undertakings and for which a credit assessment by a nominated external credit assessment institution is not available, SR_i and SR_j must be the latest available solvency ratio; and

(g) for pool members situated in a third country and for which a credit assessment by a nominated external credit assessment institution is not available:

(i) SR_i and SR_j must be equal to 100% where the pool member is situated in a third country which is an overseas jurisdiction designated under regulation 11 in relation to regulation 13 of the IRPR regulations in respect of the insurance group capital requirements calculation; and

(ii) SR_i and SR_j must be equal to 75% where the pool member is situated in a third country which is not an overseas jurisdiction designated under regulation 11 in relation to regulation 13 of the IRPR regulations in respect of the insurance group capital requirements calculation.

3\.

Where a firm is ceding risk to a pooling arrangement by the intermediary of a central undertaking, the firm must treat the central undertaking as part of the pooling arrangement and calculate its share of the risk accordingly.

Export Article as

3E7 Loss-Given-Default for Pool Exposures of Type B

1\.

For pool exposures of type B which a firm is permitted to treat as separate single name exposures in accordance with 3E2.2, where members are each liable up to the full amount of the obligation covered by the pooling arrangement, the firm must calculate the loss-given-default in accordance with the following formula:

$$\text{LGD} = \max\left(\frac{1 - \text{RR}_c}{\text{Collateral}} \cdot \left(\frac{P_u}{1 - P_c} \right) \cdot \text{BE}_c + \Delta \text{RM}_c \right) \cdot \text{F}$$

where: |

---|---

- (a) | P_U denotes the firm's share of the risk according to the terms of the pooling arrangement;
- (b) | P_C denotes the counterparty member's share of the risk according to the terms of the pooling arrangement;
- (c) | RR_C is equal to:
 - | (i) | 10% if 60% or more of the assets of the counterparty member are subject to collateral arrangements; or
 - | (ii) | 50% otherwise;
- (d) | BE_C denotes the best estimate of the liability ceded to the counterparty member by the firm, net of any amounts reinsured with counterparties external to the pooling arrangement;
- (e) | $?_{RM_C}$ denotes the counterparty member's contribution to the risk-mitigating effect of the pooling arrangement on the underwriting risk of the firm;
- (f) | Collateral denotes the risk-adjusted value of collateral provided by way of collateral arrangement held by the counterparty member of the pooling arrangement; and
- (g) | F denotes the factor to take into account the economic effect of the collateral provided by way of a collateral arrangement held by the counterparty member, calculated in accordance with 3E10.

2\.

For pool exposures of type B which a firm is permitted to treat as separate single name exposures in accordance with 3E2.2, where members are each only liable up to their respective portion of the obligation covered by the pooling arrangement, the firm must calculate the loss-given-default in accordance with the following formula:

$$LGD = \max \left(\left(1 - RR_{CE} \right) \cdot \left(P_C \cdot BE_{CE} + \Delta_{RM_{CE}} \right) - F \cdot \text{Collateral} \right); 0$$

where: |

---|---

- (a) | P_C denotes the counterparty member's share of the risk according to the terms of the pooling arrangement;
- (b) | RR_C is equal to:
 - | (i) | 10% if 60% or more of the assets of the counterparty member are subject to collateral arrangements; or
 - | (ii) | 50% otherwise;
- (c) | BE_U denotes the best estimate of the liability ceded to the pooling arrangement by the undertaking, net of any amounts reinsured with counterparties external to the pooling arrangement;
- (d) | $?_{RM_C}$ denotes the counterparty member's contribution to the risk-mitigating effect of the pooling arrangement on the underwriting risk of the firm;
- (e) | Collateral denotes the risk-adjusted value of collateral provided by way of collateral arrangement held by the counterparty member of the pooling arrangement; and
- (f) | F denotes the factor to take into account the economic effect of the collateral provided by way of a collateral arrangement held by the counterparty member, calculated in accordance with 3E10.

Export Article as

3E8 Loss-Given-Default for Pool Exposures of Type C

1\.

For pool exposures of type C which a firm is permitted to treat as separate single name exposures in accordance with 3E2.2, the firm must calculate the loss-given-default in accordance with the following formula:

$$\text{LGD} = \max \left(\left(1 - \text{RR}_{\text{CE}} \right) \cdot \left(P_{\text{U}} \cdot \text{BE}_{\text{CE}} + \Delta \text{RM}_{\text{CE}} \right) - F, 0 \right)$$

where: |

---|---

(a) | P_{U} denotes the firm's share of the risk according to the terms of the pooling arrangement;

(b) | RR_{CE} is equal to:

| (i) | 10% if 60% or more of the assets of the external counterparty member are subject to collateral arrangements; or

| (ii) | 50% otherwise;

(c) | BE_{CE} denotes the best estimate of the liability ceded to external counterparty by the pooling arrangement as a whole;

(d) | RM_{CE} denotes the external counterparty's contribution to the risk-mitigating effect of the pooling arrangement on the underwriting risk of the firm;

(e) | Collateral denotes the risk-adjusted value of collateral provided by way of collateral arrangement ___ held by the counterparty member of the pooling arrangement; and

(f) | F denotes the factor to take into account the economic effect of the collateral provided by way of a collateral arrangement ___ held by the counterparty member, calculated in accordance with 3E10.

Export Article as

3E9 Risk-Mitigating Effect

1\.

A firm must calculate the risk-mitigating effect on underwriting risk or market risk of a reinsurance arrangement, securitisation or derivative as the greater of zero and the difference between the following capital requirements:

1. (1) the hypothetical capital requirement for underwriting risk or market risk of the firm, calculated in accordance with Chapter 3 and Sections 3A to 3D, that would apply if the reinsurance arrangement, securitisation or derivative did not exist; and
2. (2) the capital requirement for underwriting risk or market risk of the firm.

Export Article as

3E10 Risk-Adjusted Value of Collateral

1\.

Where the criteria set out in 3G8 are met, a firm must calculate the risk-adjusted value of collateral provided by way of security, as referred to in paragraph (2) of the definition collateral arrangement, as the difference between the value of the assets held as collateral, valued in accordance with Valuation 2.1 to 2.2, and the adjustment for market risk, as referred to in 3E10.5, provided that both of the following requirements are fulfilled:

1. (1) the firm has (or is a beneficiary under a trust where the trustee has) the right to liquidate or retain, in a timely

manner, the collateral in the event of a default, insolvency or bankruptcy or other credit event relating to the counterparty (?the counterparty requirement?); and

2. (2) the firm has (or is a beneficiary under a trust where the trustee has) the right to liquidate or retain, in a timely manner, the collateral in the event of a default, insolvency or bankruptcy or other credit event relating to the custodian or other third party holding the collateral on behalf of the counterparty (?the third party requirement?).

2\.

Where the counterparty requirement is met and the criteria set out in 3G8 are met and the third party requirement is not met, a firm must calculate the risk-adjusted value of a collateral provided by way of security, as referred to in paragraph (2) of the definition collateral arrangement, as equal to 90% of the difference between the value of the assets held as collateral in accordance with Valuation 2.1 to 2.2 and the adjustment for market risk, as referred to in 3E10.5.

3\.

Where either the counterparty requirement is not met or the requirements in 3G8 are not met, a firm must assign a value to the risk-adjusted value of collateral provided by way of security, as referred to in paragraph (2) of the definition collateral arrangements, of zero.

4\.

A firm must calculate the risk-adjusted value of a collateral of which full ownership is transferred, as referred to in paragraph (1) of the definition collateral arrangement, as the difference between the value of the assets held as collateral, valued in accordance with Valuation 2.1 to 2.2, and the adjustment for market risk, as referred to in 3E10.5, provided the requirements in 3G8 are fulfilled.

5\.

A firm must calculate the adjustment for market risk as the difference between the following capital requirements:

1. (1) the hypothetical capital requirement for market risk of the firm that would apply if the assets held as collateral provided by way of a collateral arrangement were not included in the calculation; and
2. (2) the hypothetical capital requirement for market risk of the firm that would apply if the assets held as collateral provided by way of a collateral arrangement were included in the calculation.

6\.

For the purposes of 3E10.5, a firm must calculate the currency risk of the assets held as collateral by comparing the currency of the assets held as collateral against the currency of the corresponding exposure.

7\.

Where, in case of insolvency of the counterparty, the determination of the firm?s proportional share of the counterparty?s insolvency estate in excess of the collateral does not take into account that the firm receives the collateral, the factors $_F$, $F?$, $F??$ and $_F???$ referred to in 3E4.3 to 3E4.8 and 7.35 must all be 100% and in all other cases these factors must be 50%, 18%, 16% and 90% respectively.

Export Article as

3E11 Risk-Adjusted Value of Mortgage

1\.

A firm must calculate the risk-adjusted value of mortgage as the difference between the value of the residential property

held as mortgage, valued in accordance with 3E11.2, and the adjustment for market risk, as referred to in 3E11.3.

2\.

A firm must calculate the value of the residential property held as mortgage as the market value, reduced as appropriate to reflect the results of the monitoring required under 3E3.9 and 3E3.10 and to take account of any prior claims on the property and the external, independent valuation of the property must be the same or less than the market value calculated in accordance with Valuation 2.1 to 2.2.

3\.

A firm must calculate the adjustment for market risk referred to in 3E11.1 as the difference between the following capital requirements:

1. (1) the hypothetical capital requirement for market risk of the firm that would apply if the residential property held as mortgage were not included in the calculation; and
2. (2) the hypothetical capital requirement for market risk of the firm that would apply if the residential property held as mortgage were included in the calculation.

4\.

For the purposes of 3E11.2, a firm must calculate the currency risk of the residential property held as mortgage by comparing the currency of the residential property against the currency of the corresponding loan.

Export Article as

Type 1 Exposures

3E12 Probability of Default

1\.

A firm must calculate the probability of default on a single name exposure as equal to the average of the probabilities of default on each of the exposures to counterparties that belong to the single name exposure, weighted by the loss-given-default in respect of those exposures.

2\.

A firm must assign to a single name exposure i for which a credit assessment by a nominated external credit assessment institution is available, a probability of default PD i in accordance with the following table:

Credit quality step | 0 | 1 | 2 | 3 | 4 | 5 | 6

---|---|---|---|---|---|---

Probability of default PD i | 0.002% | 0.01% | 0.05% | 0.24% | 1.2% | 4.2% | 4.2%

3\.

In respect of single name exposures i to a UK Solvency II undertaking for which a credit assessment by a nominated external credit assessment institution is not available and where this UK Solvency II undertaking meets its MCR, a firm must assign a probability of default PD i depending on the UK Solvency II undertaking's solvency ratio, in accordance with the following table:

Solvency ratio | 196% | 175% | 150% | 125% | 122% | 100% | 95% | 75%

---|---|---|---|---|---|---|---

Probability of default | 0.01% | 0.05% | 0.1% | 0.2% | 0.24% | 0.5% | 1.2% | 4.2%

Where the solvency ratio falls in between the solvency ratios specified in the table above, the value of the probability of default must be linearly interpolated from the closest values of probabilities of default corresponding to the closest solvency ratios specified in the table above, provided that:

(1) where the solvency ratio is lower than 75%, the probability of default must be 4.2%; and

(2) where the solvency ratio is higher than 196%, the probability of default must be 0.01%.

For the purposes of this rule, 'solvency ratio' denotes the ratio of the eligible own funds to cover the SCR and the SCR, using the latest available values.

4\.

A firm must assign a probability of default equal to 4.2% to exposures to a UK Solvency II undertaking that does not meet its MCR.

5\.

3E12.3 and 3E12.4 only apply as of the first date of public disclosure, by the UK Solvency II undertaking corresponding to the exposure, of the SFCR and before that date, the following applies:

1. (1) if a credit assessment by a nominated external credit assessment institution is available for the exposures, 3E12.2 applies;

2. (2) in all other cases, a firm must assign to the exposures the same risk factor as the ones that would result from the application of 3E3.3 to exposures to a UK Solvency II undertaking whose solvency ratio is 100%.

6\.

In respect of exposures to a third country insurance undertaking or a third country reinsurance undertaking for which a credit assessment by a nominated external credit assessment institution is not available, situated in a third country which is an overseas jurisdiction designated under regulation 11 in relation to regulation 13 of the IRPR regulations in respect of the insurance group capital requirements calculation, and which complies with the solvency requirements of that third country, a firm must assign a probability of default equal to 0.5%.

7\.

In respect of exposures to credit institutions and financial institutions which comply with the solvency requirements set out in the PRA Rulebook, the CRR or technical standards as amended from time to time, for which a credit assessment by a nominated external credit assessment institution is not available, a firm must assign a probability of default equal to 0.5%.

8\.

A firm must assign a probability of default equal to 0% to exposures to counterparties referred to in 3D24.2(1) to (3).

9\.

A firm must calculate the probability of default on single name exposures other than those identified in 3E12.2 to 3E12.8 as equal to 4.2%.

10\.

Where a letter of credit, a guarantee or an equivalent arrangement is provided to fully secure an exposure and this arrangement complies with 3G2 to 3G9, a firm may treat the provider of that letter of credit, guarantee or equivalent

arrangement as the counterparty on the secured exposure for the purposes of assessing the probability of default of a single name exposure.

11\.

For the purposes of 3E12.10, a firm must treat exposures fully, unconditionally and irrevocably guaranteed by counterparties listed in 3E1 as exposures to the central government.

12\.

Notwithstanding 3E12.2 to 3E12.11, a firm must assign to exposures referred to 3E4.5 a probability of default equal to 0.002%.

13\.

Notwithstanding 3E12.2 to 3E12.12, a firm must assign to exposures referred to 3E4.6 a probability of default equal to 0.01%.

Export Article as

3E13 Type 1 Exposures

1\.

Where the standard deviation of the loss distribution of type 1 exposures as referred to in 3.13 to 3.19 is lower than or equal to 7% of the total loss-given-default on all type 1 exposures, a firm must calculate the capital requirement for counterparty default risk on type 1 exposures in accordance with the following formula:

$$\mathbb{V}[\{SCR\}_{def,1}] = 3 \cdot \sigma$$

where σ denotes the standard deviation of the loss distribution of type 1 exposures, as defined in 3E13.4.

2\.

Where the standard deviation of the loss distribution of type 1 exposures is higher than 7% of the total loss-given-default on all type 1 exposures and lower or equal to 20% of the total loss-given-default on all type 1 exposures, a firm must calculate the capital requirement for counterparty default risk on type 1 exposures in accordance with the following formula:

$$\mathbb{V}[\{SCR\}_{def,1}] = 5 \cdot \sigma$$

where σ denotes the standard deviation of the loss distribution of type 1 exposures.

3\.

Where the standard deviation of the loss distribution of type 1 exposures is higher than 20% of the total loss-given-default on all type 1 exposures, a firm must calculate the capital requirement for counterparty default risk on type 1 exposures as equal to the total loss-given-default on all type 1 exposures.

4\.

A firm must calculate the standard deviation of the loss distribution of type 1 exposures in accordance with the following formula:

$$\sigma = \sqrt{V}$$

where V denotes the variance of the loss distribution of type 1 exposures.

Export Article as

3E14 Variance of the Loss Distribution of Type Exposures

1\.

The variance of the loss distribution of type 1 exposures as referred to in 3E13.4 must be equal to the sum of V_{inter} and V_{intra} .

2\.

A firm must calculate V_{inter} in accordance with the following formula:

$$V_{inter} = \sum_{j,k} \frac{\{PD\}_j \cdot \{1 - \{PD\}_k\} \cdot \{PD\}_k \cdot \{1 - \{PD\}_j\}}{1.25 \cdot \{PD\}_j + \{PD\}_k} \cdot \{TLGD\}_j \cdot \{TLGD\}_k$$

where:

1. (a) the sum covers all possible combinations (j,k) of probabilities of default on single name exposures in accordance with 3E12; and
2. (b) $TLGD_j$ and $TLGD_k$ denote the sum of loss-given-default on type 1 exposures from counterparties bearing a probability of default PD_j and PD_k respectively.

3\.

A firm must calculate V_{intra} in accordance with the following formula:

$$V_{intra} = \sum_j \frac{1.5 \cdot \{PD\}_j \cdot \{1 - \{PD\}_j\}}{2.5 - \{PD\}_j} \cdot \sum_i \{PD\}_j \cdot \{LGD\}_i^2$$

where:

1. (a) the first sum covers all different probabilities of default on single name exposures in accordance with 3E12;
2. (b) the second sum covers all single name exposures that have a probability of default equal to PD_j ; and
3. (c) LGD_i denotes the loss-given-default on the single name exposure i .

Export Article as

Type 2 Exposures

3E15 Type 2 Exposures

1\.

A firm must calculate the capital requirement for counterparty default risk on type 2 exposures as equal to the loss in its basic own funds that would result from an instantaneous decrease in value of type 2 exposures calculated in accordance with the following formula:

$$[90\% \cdot \{LGD\}_{receivables > 3 \text{ months}}] + \sum_i [15\% \cdot \{LGD\}_i]$$

where:

1. (a) $LGD_{receivables > 3 \text{ months}}$ denote the total loss-given-default on all receivables from intermediaries which have

been due for more than three months;

2. (b) the sum is taken on all type 2 exposures other than receivables from intermediaries which have been due for more than three months; and

3. (c) _LGD_i denotes the loss-given-default on the type 2 exposure _i .

Export Article as

3F

Intangible Asset Module

3F1 Intangible Asset Module

1\.

A firm must calculate the capital requirement for intangible asset risk in accordance with the following formula:

$$\text{_SCR}_{\text{_intangible}} = 0.8 \cdot V_{\text{_intangible}}$$

where $\text{_V}_{\text{_intangibles}}$ denotes the amount of intangible assets as recognised and valued in accordance with Valuation 8.1(2).

Export Article as

3G

Risk Mitigation Techniques

3G1 Methods and Assumptions

1\.

Where a firm transfers underwriting risk using a reinsurance contract or special purpose vehicle that meets the requirements set out in 3G2, 3G5 and 3G7, and where the arrangement provides for protection in several of the scenario-based calculations set out in Sections 3A to 3C, the firm must allocate the risk-mitigating effects of the contractual arrangement to the scenario-based calculations in a manner that, without double-counting, captures the economic effect of the protections provided and, in particular, captures the economic effect of the protections provided in determining the loss in basic own funds in the scenario-based calculations.

2\.

Where a firm transfers underwriting risk using a finite reinsurance contract that meets the requirements set out in 3G2, 3G5 and 3G7:

1. (1) subject to (2), the firm may recognise that contract in the scenario-based calculations set out in Sections 3A to 3C only to the extent underwriting risk is transferred to the counterparty of the contract; and
2. (2) the firm must not take into account finite reinsurance, or similar arrangements where the effective risk transfer is comparable to that of finite reinsurance, for the purposes of determining the volume measures for premium and reserve risk in accordance with 3A2 and 3C3, or for the purposes of calculating undertaking specific parameters in accordance with the Solvency Capital Requirement ? Undertaking Specific Parameters Part.

Export Article as

3G2 Qualitative Criteria

1\.

When calculating the basic SCR, a firm must only take into account a risk-mitigation technique as referred to in Solvency Capital Requirement ? General Provisions 3.5 where all of the following qualitative criteria are met:

1. (1) the contractual arrangements and transfer of risk are legally effective and enforceable in all relevant jurisdictions;
2. (2) the firm has taken all appropriate steps to ensure the effectiveness of the arrangement and to address the risks related to that arrangement;
3. (3) the firm is able to monitor the effectiveness of the arrangement and the related risks on an ongoing basis;
4. (4) the firm has, in the event of a default, insolvency or bankruptcy of a counterparty or other credit event set out in the transaction documentation for the arrangement, a direct claim on that counterparty; and
5. (5) there is no double counting of risk-mitigation effects in own funds and in the calculation of the SCR or within the calculation of the SCR.

2\.

Subject to 3G2.3, a firm must take a risk-mitigation technique into account in the calculation of the basic SCR on the following basis:

1. (1) full recognition of the risk mitigation effect of the risk-mitigation technique where it is in force for at least the next 12 months and meets the qualitative criteria set out in Section 3G; or
2. (2) partial recognition of the risk-mitigation effect of a risk-mitigation technique where it is in force for a period shorter than 12 months and meets the qualitative criteria set out in Section 3G, in proportion to the length of time involved for the shorter of the full term of the risk exposure or the period that the risk-mitigation technique is in force.

3\.

A firm must take a risk-mitigation technique into account in the calculation of the basic SCR on the basis of full recognition of its risk mitigation effect, where contractual arrangements governing the risk-mitigation technique will be in force for a period shorter than the next 12 months and the firm intends to replace that risk-mitigation technique at the time of its expiry with a similar arrangement or where that risk-mitigation technique is subject to an adjustment to reflect changes in the exposure that it covers, provided all of the following qualitative criteria are met:

1. (1) the firm has a written policy on the replacement or adjustment of that risk-mitigation technique, covering situations including the situation where the firm uses several contractual arrangements in combination to transfer risk as referred to in 3G3.5;
2. (2) the replacement or adjustment of the risk-mitigation technique takes place more often than once per week only in cases where, without the replacement or adjustment, an event would have a material adverse impact on the solvency position of the firm;
3. (3) the replacement or adjustment of the risk-mitigation technique is not conditional on any future event which is outside of the control of the firm and where the replacement or adjustment of the risk-mitigation technique is conditional on any future event that is within the control of the firm, the conditions for such replacement or adjustment are clearly documented in the written policy referred to in (1);
4. (4) the replacement or adjustment of the risk-mitigation technique is realistically based on replacements and adjustments undertaken previously by the firm and consistent with the firm's current business practice and business strategy;
5. (5) there is no material risk that the risk-mitigation technique cannot be replaced or adjusted due to an absence of liquidity in the market;
6. (6) the risk that the cost of replacing or adjusting the risk-mitigation technique increases during the following 12

months is reflected in the SCR;

7. (7) the replacement or adjustment of the risk-mitigation technique would not be contrary to requirements that apply to future management actions set out in Technical Provisions ? Further Requirements 8.5;
8. (8) the initial contractual maturity is not shorter than one month in cases where the firm transfers risks through the purchase or issuance of financial instruments; and
9. (9) the initial contractual maturity is not shorter than three months where the firm transfers underwriting risks using reinsurance contracts or special purpose vehicles.

Export Article as

3G3 Effective Transfer of Risk

1\.

A firm must only take a risk-mitigation technique into account in the calculation of the basic SCR where the contractual arrangements governing the risk-mitigation technique ensure that the extent of the cover provided by the risk-mitigation technique and the transfer of risk is clearly defined and incontrovertible.

2\.

The contractual arrangement must not result in material basis risk or in the creation of other risks, unless this is reflected in the calculation of the SCR.

3\.

A firm must treat basis risk as material if it leads to a misstatement of the risk-mitigating effect on the firm's basic SCR that could influence the decision-making or judgement of the intended user of that information, including the supervisory authorities.

4\.

In determining whether the contractual arrangements and transfer of risk are legally effective and enforceable in all relevant jurisdictions in accordance with 3G2.1(1), a firm must take into account the following:

1. (1) whether the contractual arrangement is subject to any condition which could undermine the effective transfer of risk, the fulfilment of which is outside the direct control of the firm; and
2. (2) whether there are any connected transactions which could undermine the effective transfer of risk.

5\.

Where a firm combines several contractual arrangements to transfer risk, each of the contractual arrangements must meet the requirements set out in 3G3.1 and 3G3.4 and the contractual arrangements in combination must meet the requirements set out in 3G3.2 and 3G3.3.

Export Article as

3G4 Material Basis Risk

1\.

Notwithstanding 3G3.2, where a firm transfers underwriting risk using a reinsurance contract or a special purpose vehicle that is subject to material basis risk stemming from a currency mismatch between underwriting risk and the risk-mitigation technique, a firm may take into account the risk-mitigation technique in the calculation of the SCR

according to the standard formula, provided that the risk-mitigation technique complies with 3G2, 3G3.1, 3G3.3, 3G3.4 and 3G5, and the calculation is carried out as follows:

1. (1) the firm must take the basis risk stemming from a currency mismatch between underwriting risk and the risk-mitigation technique into account in the relevant underwriting risk module, sub-module or scenario of the standard formula at the most granular level by adding 25% of the difference between the following to the capital requirement calculated in accordance with the relevant module, sub-module or scenario:

1. (a) the hypothetical capital requirement for the relevant underwriting risk module, sub-module or scenario that would result from a simultaneous occurrence of the scenario set out in 3D32; and

2. (b) the capital requirement for the relevant underwriting risk module, sub-module or scenario; and

2. (2) where the risk-mitigation technique covers more than one module, sub-module or scenario, the firm must apply the calculation referred to in (1) for each of those modules, sub-modules and scenarios and the capital requirement resulting from those calculations must not exceed 25% of the capacity of the non-proportional reinsurance contract or special purpose vehicle.

Export Article as

3G5 Risk-Mitigation Techniques Using Reinsurance Contracts or Special Purpose Vehicles

1\.

Where a firm transfers underwriting risk using a reinsurance contract or special purpose vehicle, the firm must only take the risk-mitigation technique into account in the calculation of the basic SCR if the qualitative criteria set out in 3G2 and 3G3 and those set out in 3G5.2 to 3G5.6 are met.

2\.

In the case of a reinsurance contract the counterparty must be any of the following:

1. (1) a UK Solvency II undertaking which complies with the SCR;

2. (2) a third country insurance undertaking or a third country reinsurance undertaking, situated in a third country which is an overseas jurisdiction designated under regulation 11 in relation to regulation 12 of the IRPR regulations in respect of reinsurance contracts, and which complies with the solvency requirements of that third country; or

3. (3) a third country insurance undertaking or a third country reinsurance undertaking, situated in a third country which is not an overseas jurisdiction designated under regulation 11 in relation to regulation 12 of the IRPR regulations in respect of reinsurance contracts, and which has been assigned to credit quality step 3 or better in accordance with 1A to 1C.

3\.

Where a counterparty to a reinsurance contract is a UK Solvency II undertaking which ceases to comply with its SCR after the reinsurance contract has been entered into, a firm may partially recognise the protection offered by the insurance risk-mitigation technique for a period of no longer than six months after the counterparty ceases to comply with its SCR and, in that case, the effect of the risk-mitigation technique must be reduced by the percentage by which the SCR is breached, provided that:

1. (1) as soon as the counterparty has restored compliance with its SCR, the firm must no longer reduce the effect of the risk-mitigation technique; and

2. (2) where:

1. (a) the counterparty fails to restore compliance with its SCR within that period of six months; or

2. (b) where, before the end of the period of six months, the firm becomes aware that it is unlikely that the counterparty will be able to restore compliance with its SCR within that period,

3. the firm must no longer recognise the effect of the risk-mitigation technique in the calculation of the basic SCR.

4\.

Notwithstanding 3G5.3, where a counterparty to a reinsurance contract is a UK Solvency II undertaking which ceases to comply with its MCR after the reinsurance contract has been entered into, a firm must cease to recognise the effect of the risk-mitigation technique in the calculation of the basic SCR.

5\.

Where risk is transferred to a UK ISPV, a firm must only take the risk-mitigation technique into account in the calculation of the basic SCR where the requirements in Insurance Special Purpose Vehicles Part 2, 2A to 2D and 5A.1 to 5A.4 are met, provided that:

1. (1) where the requirements for a UK ISPV to be fully funded cease to be fully met after the arrangement has been entered into, the firm may only partially recognise the protection offered by the insurance risk-mitigation technique if the firm can demonstrate that compliance with the fully funded requirement will be restored within three months; and
2. (2) for this purpose, the effect of the risk-mitigation technique must be reduced by the percentage of the aggregate maximum risk exposure of the UK ISPV, referred to in Insurance Special Purpose Vehicles 2.1 to 2.5 not covered by the assets of the UK ISPV.

6\.

Where risk is transferred to a special purpose vehicle that is regulated by a third country supervisory authority, a firm must not take the risk-mitigation technique into account in the calculation of the basic SCR unless requirements equivalent to those set out in Insurance Special Purpose Vehicles Part 2, 2A to 2D and 5A.1 to 5A.4 are met by the special purpose vehicle.

Export Article as

3G6 Financial Risk-Mitigation Techniques

1\.

Where a firm transfers risk other than in the cases referred to in 3G5.1, including transfers through the purchase or issuance of financial instruments, the firm may only take that risk-mitigation technique into account in the calculation of the basic SCR if the qualitative criteria provided in 3G6.2 to 3G6.5 are met, in addition to the qualitative criteria set out in 3G2 and 3G3.

2\.

The risk-mitigation technique must be consistent with the firm's written policy on risk management, as referred to in Conditions Governing Business 2.5.

3\.

The firm must be able to value the assets and liabilities that are subject to the risk-mitigation technique and, where the risk-mitigation technique includes the use of financial instruments, the firm must be able to value the financial instruments reliably in accordance with Valuation 2.1 to 2.2.

4\.

Where the risk-mitigation technique includes the use of financial instruments, the financial instruments must have a credit quality which has been assigned to credit quality step 3 or better in accordance with 1A to 1C.

5\.

Where the risk-mitigation technique is not a financial instrument, the counterparties to the risk-mitigation technique must have a credit quality which has been assigned to credit quality step 3 or better in accordance with 1A to 1C.

Export Article as

3G7 Status of the Counterparties

1\.

In the event that the qualitative criteria in 3G5.1, 3G6.4 or 3G6.5 are not met, a firm must only take into account the risk-mitigation technique when calculating the basic SCR where one of the following criteria is met:

1. (1) the risk-mitigation technique meets the qualitative criteria set out in 3G2, 3G3, 3G6.2 and 3G6.3 and collateral arrangements exist that meet the criteria provided in 3G8; and
2. (2) the risk-mitigation technique is accompanied by another risk-mitigation technique that, when viewed in combination with the first technique, meets the qualitative criteria set out in 3G2, 3G3, 3G6.2 and 3G6.3, with the counterparties to that other technique meeting the criteria provided in 3G5.1, 3G6.4 and 3G6.5.

2\.

For the purposes of 3G7.1(1), where the value, of the collateral provided by way of a collateral arrangement, valued in accordance with Valuation 2.1 to 2.2, is less than the total risk exposure, the firm must only take the collateral arrangement into account to the extent that the collateral covers the risk exposure.

Export Article as

3G8 Collateral Arrangements

1\.

In the calculation of the basic SCR, a firm must only recognise collateral arrangements where, in addition to the qualitative criteria in 3G2 and 3G3, the following criteria are met:

1. (1) the firm transferring the risk must have the right to liquidate or retain, in a timely manner, the collateral in the event of a default, insolvency or bankruptcy or other credit event of the counterparty;
2. (2) there is sufficient certainty as to the protection achieved by the collateral because of either of the following:
 1. (a) it is of sufficient credit quality, is of sufficient liquidity and is sufficiently stable in value; or
 2. (b) it is guaranteed by a counterparty, other than a counterparty referred to in 3D31.9 and 3D28.2 who has been assigned a risk factor for concentration risk of 0%;
3. (3) there is no material positive correlation between the credit quality of the counterparty and the value of the collateral; and
4. (4) the collateral is not securities issued by the counterparty or a related undertaking of that counterparty.

2\.

Where a collateral arrangement involves collateral being held by a custodian or other third party, the firm must ensure that all of the following criteria are met:

1. (1) the relevant custodian or other third party segregates the assets held as collateral from its own assets;
2. (2) the segregated assets are held by a deposit-taking institution that has a credit quality which has been assigned to

credit quality step 3 or better in accordance with 1A to 1C;

3. (3) the segregated assets are individually identifiable and can only be changed or substituted with the consent of the firm or a person acting as a trustee in relation to the firm's interest in such assets;
4. (4) the firm has (or is a beneficiary under a trust where the trustee has) the right to liquidate or retain, in a timely manner, the segregated assets in the event of a default, insolvency or bankruptcy or other credit event relating to the custodian or other third party holding the collateral on behalf of the counterparty; and
5. (5) the segregated assets must not be used to pay, or to provide collateral in favour of, any person other than the firm or as directed by the firm.

Export Article as

3G9 Guarantees

1\.

In the calculation of the basic SCR, a firm must only recognise guarantees where explicitly referred to in this Chapter, and where in addition to the qualitative criteria in 3G2 and 3G3, all of the following criteria are met:

1. (1) the credit protection provided by the guarantee is direct;
2. (2) the extent of the credit protection is clearly defined and incontrovertible;
3. (3) the guarantee does not contain any clause, the fulfilment of which is outside the direct control of the lender, that:
 1. (a) would allow the protection provider to cancel the protection unilaterally;
 2. (b) would increase the effective cost of protection as a result of a deterioration in the credit quality of the protected exposure;
 3. (c) could prevent the protection provider from being obliged to pay out in a timely manner in the event that the original obligor fails to make any payments due; or
 4. (d) could allow the maturity of the credit protection to be reduced by the protection provider;
4. (4) on the default, insolvency or bankruptcy or other credit event of the counterparty, the firm has the right to pursue, in a timely manner, the guarantor for any monies due under the claim in respect of which the protection is provided and the payment by the guarantor must not be subject to the firm first having to pursue the obligor;
5. (5) the guarantee is an explicitly documented obligation assumed by the guarantor; and
6. (6) the guarantee fully covers all types of regular payments the obligor is expected to make in respect of the claim.

Export Article as

4

Calculation of the Equity Risk Sub-Module and Application of the Symmetric Adjustment Mechanism

4.1

For the purposes of calculating the equity risk sub-module referred to in 3.11(2)(b), a _firm_ must apply a _symmetric adjustment_ to the _standard equity capital charge_ to cover the risk arising from changes in the level of equity prices.

[Note: Art. 106(1) of the _Solvency II Directive_]

5

Capital Requirement for Operational Risk

5.1

A firm's capital requirement for operational risk must:

1. (1) reflect its operational risks to the extent that they are not already reflected in the risk modules used to calculate its basic SCR; and
2. (2) be calibrated in accordance with Solvency Capital Requirement ? General Provisions 3.3 to 3.4.

[Note: Art. 107(1) of the Solvency II Directive]

5.2

With respect to linked long-term contracts of insurance, the calculation of the capital requirement for operational risk must take into account the amount of annual expenses incurred in respect of those insurance obligations.

[Note: Art. 107(2) of the Solvency II Directive]

5.3

With respect to insurance business operations other than those referred to in 5.2, the capital requirement for operational risk must:

1. (1) take into account the volume of those operations, in terms of earned premiums and technical provisions which are held in respect of that insurance business;
2. (2) not exceed 30% of the basic SCR relating to those operations; and
3. (3) comply with 5.4.

[Note: Art. 107(3) of the Solvency II Directive]

5.4

1. (1) A firm must calculate the capital requirement for the operational risk module in accordance with the following formula:

$$\text{SCR}_{\text{Operational}} = \min(0.3 \cdot \text{BSCR}, \text{Op}) + 0.25 \cdot \text{Exp}_{\text{ul}}$$

1. where:

1. (a) BSCR denotes the basic SCR;
2. (b) Op denotes the basic capital requirement for operational risk as referred to in (2); and
3. (c) Exp_{ul} denotes the amount of expenses incurred during the previous 12 months in respect of long-term insurance contracts where the investment risk is borne by policyholders.

2. (2) A firm must calculate the basic capital requirement for operational risk in accordance with the following formula:

$$\text{Op} = \max(\text{Op}_{\text{premiums}}, \text{Op}_{\text{provisions}})$$

1. where:

1. (a) Op_{premiums} denotes the capital requirement for operational risk based on earned premiums; and
2. (b) Op_{provisions} denotes the capital requirement for operational risk based on technical provisions.
2. (3) A firm must calculate the capital requirement for operational risk based on earned premiums as follows:

$$\text{Op}_{\text{premiums}} = 0.04 \cdot \left(\text{Earn}_{\text{life}} - \text{Earn}_{\text{life-ul}} \right) + 0.03 \cdot \text{Earn}_{\text{non-life}} + \max \left(0, 0.04 \cdot \left(\text{Earn}_{\text{life}} - 1.2 \cdot \text{pEarn}_{\text{life}} \right) - \left(\text{Earn}_{\text{life-ul}} - 1.2 \cdot \text{pEarn}_{\text{life-ul}} \right) \right) + \max \left(0, 0.03 \cdot \text{right} \cdot \left(\text{Earn}_{\text{non-life}} - \text{right} \cdot 1.2 \cdot \text{pEarn}_{\text{non-life}} \right) \right)$$

1. where:

1. (a) `_Earn life_` denotes the premiums earned during the last 12 months for long-term insurance and reinsurance obligations, without deducting premiums for reinsurance contracts;
2. (b) `_Earn life-ul_` denotes the premiums earned during the last 12 months for long-term insurance and reinsurance obligations where the investment risk is borne by the policyholders without deducting premiums for reinsurance contracts;
3. (c) `_Earn non-life_` denotes the premiums earned during the last 12 months for general insurance and reinsurance obligations, without deducting premiums for reinsurance contracts;
4. (d) `_pEarn life_` denotes the premiums earned during the 12 months prior to the last 12 months for long-term insurance and reinsurance obligations, without deducting premiums for reinsurance contracts;
5. (e) `_pEarn life-ul_` denotes the premiums earned during the 12 months prior to the last 12 months for long-term insurance and reinsurance obligations where the investment risk is borne by the policyholders without deducting premiums for reinsurance contracts; and
6. (f) `_pEarn non-life_` denotes the premiums earned during the 12 months prior to the last 12 months for general insurance and reinsurance obligations, without deducting premiums for reinsurance contracts.
7. For the purposes of (3), earned premiums must be gross, without deduction of premiums for reinsurance contracts.
2. (4) A firm must calculate the capital requirement for operational risk based on technical provisions in accordance with the following formula:

$$\mathbb{I}[\{Op\}_{provisions}=0.0045\cdot \max\left(0;\{TP\}_{life}-\{TP\}_{life-ul}\right)+0.03\cdot \max\left(0;\{TP\}_{non-life}\right)\mathbb{I}]$$

1. where:

1. (a) `_TP life_` denotes the technical provisions for long-term insurance and reinsurance obligations;
2. (b) `_TP life-ul_` denotes the technical provisions for long-term insurance obligations where the investment risk is borne by the policyholders; and
3. (c) `_TP non-life_` denotes the technical provisions for general insurance and reinsurance obligations.
4. For the purposes of (4), technical provisions must not include the risk margin, and must be calculated without deduction of recoverables from reinsurance contracts and special purpose vehicles.

6

Adjustment for Loss-Absorbing Capacity of Technical Provisions and Deferred Taxes

6.1

The adjustment for the loss-absorbing capacity of `_technical provisions_` and deferred taxes as referred to in 2.1(3) must:

1. (1) reflect potential compensation of unexpected losses through a simultaneous decrease in `_technical provisions_` or deferred taxes, or a combination of the two;
2. (2) take account of the risk-mitigating effect provided by future discretionary benefits of `_contracts of insurance_`; and
3. (3) represent the sum of:
 1. (a) the adjustment for the loss-absorbing capacity of technical provisions calculated in accordance with 6.3; and
 2. (b) the adjustment for the loss-absorbing capacity of deferred taxes calculated in accordance with 6.4 and, if applicable, 6.5.

[Note: Art. 108 of the `_Solvency II Directive_`]

6.2

For the purposes of 6.1(2):

1. (1) a _firm_ must take account of the risk-mitigating effect provided by future discretionary benefits to the extent that it can establish that a reduction in future discretionary benefits may be used to cover unexpected losses when they arise;
2. (2) the risk-mitigating effect provided by future discretionary benefits must be no higher than the sum of _technical provisions_ and deferred taxes relating to those future discretionary benefits; and
3. (3) the value of future discretionary benefits under adverse circumstances must be compared to the value of those benefits under the underlying assumptions of the _best estimate_ calculation.

[Note: Art. 108 of the _Solvency II Directive_]

6.3

1. (1) A firm must calculate the adjustment for the loss-absorbing capacity of technical provisions in accordance with the following formula:

$$\Delta_{TP} = -\max(\min(BSCR - nBSCR; FDB); 0)$$

where:

1. (a) _BSCR_ denotes the basic SCR;
2. (b) _nBSCR_ denotes the net basic SCR calculated in accordance with (2); and
3. (c) _FDB_ denotes the technical provisions without risk margin in relation to future discretionary benefits.
2. (2) The net basic SCR is the basic SCR calculated with all the following modifications:
 1. (a) where the calculation of a module or sub-module of the basic SCR is based on the impact of a scenario on a firm's basic own funds, the firm must assume that the scenario can change the value of the future discretionary benefits included in technical provisions;
 2. (b) the scenario based calculations of the life underwriting risk module, the SLT health underwriting risk sub-module, the health catastrophe risk sub-module, the market risk module and the counterparty default risk module as well as the scenario-based calculation set out in (c) and (d) must take into account the impact of the scenario on future discretionary benefits included in technical provisions and this must be done on the basis of assumptions on future management actions that comply with Technical Provisions ? Further Requirements 8;
 3. (c) instead of the capital requirement for counterparty default risk on type 1 exposures referred to in 3.13, the calculation must be based on the capital requirement that is equal to the loss in the firm's basic own funds that would result from an instantaneous loss, due to default events relating to type 1 exposures referred to in 3.14, of the amount of the capital requirement for counterparty default risk on type 1 exposures referred to in 3.13; and
 4. (d) where a firm uses a simplified calculation for a specific capital requirement as set out in 7.8, 7.9, 7.10, 7.11, 7.12(1), 7.12(2), 7.14, 7.20, 7.23(1)(a), 7.23(1)(b) or 7.24, the firm must base the calculation on the capital requirement that is equal to the loss in its basic own funds that would result from an instantaneous loss of the amount of the capital requirement referred to in the relevant rule and must assume that the instantaneous loss is due to the risk that the capital requirement referred to in that rule captures.
3. (3) For the purposes of (2)(b), a firm must take into account any legal, regulatory or contractual restrictions in the distribution of future discretionary benefits.

6.4

1. (1) A firm must calculate the adjustment for the loss-absorbing capacity of deferred taxes as equal to the change in the value of its deferred taxes that would result from an instantaneous loss of an amount that is equal to the sum of the following:
 1. (a) the basic SCR;
 2. (b) the adjustment for the loss-absorbing capacity of technical provisions referred to in 6.3; and

3. (c) the capital requirement for operational risk as set out in 5.
2. (2) For the purposes of (1), deferred taxes must be valued in accordance with Valuation 11.1 and 11.2, without prejudice to (3) and 6.5.
3. (3) Where the loss referred to in (1) would result in an increase in the amount of deferred tax assets, a firm must not utilise that increase for the purposes of calculating the adjustment referred to in (1), unless 6.5 applies.
4. (4) A firm may assume the implementation of future management actions following the loss referred to in (1), provided that the provisions set out in Technical Provisions ? Further Requirements 8 are complied with.
5. (5) For the purposes of (1), a decrease in deferred tax liabilities or an increase in deferred tax assets must result in a negative adjustment for the loss-absorbing capacity of deferred taxes.
6. (6) Where the calculation of the adjustment in accordance with (1) results in a positive change of deferred taxes, the adjustment must be nil.
7. (7) Where it is necessary to allocate the loss referred to in (1) to its causes in order to calculate the adjustment for the loss-absorbing capacity of deferred taxes, a firm must:
 1. (a) allocate the loss to the risks that are captured by the basic SCR and the capital requirement for operational risk;
 2. (b) make that allocation consistent with the contribution of the modules and sub-modules of the standard formula to the basic SCR; and
 3. (c) where a firm has an internal model permission to use a partial internal model and where the adjustment for the loss-absorbing capacity of technical provisions and deferred taxes is not within the scope of the partial internal model, make that allocation consistent with the contribution of the modules and sub-modules of the standard formula which are outside of the scope of the partial internal model to the basic SCR.

6.5

1. (1) For a transitional period ending on 30 December 2025, where the loss referred to in 6.4(1) would result in an increase in the amount of deferred tax assets, a firm may utilise that increase for the purposes of calculating the adjustment referred to in 6.4(1), if all of the following requirements are met:
 1. (a) it is probable that future taxable profit will be available against which that increase can be utilised;
 2. (b) the firm has determined that the requirement in (a) is met based on an assessment that:
 1. (i) takes account of all of the matters referred to in (2); and
 2. (ii) uses assumptions that comply with the requirements in (3);
 3. (c) the firm has documentary evidence explaining how the requirements in (a) and (b) are met and can provide that evidence to the PRA, if the PRA requests it; and
 4. (d) the firm has given the PRA advance notice in writing that it proposes to utilise an increase in deferred tax assets in accordance with this rule.
2. (2) The relevant matters for the purpose of (1)(b)(i) are:
 3. 1. (a) any legal or regulatory requirements on the time limits relating to the carry-forward of unused tax losses or the carry-forward of unused tax credits;
 2. (b) the magnitude of the loss referred to in 6.4(1) and its impact on the firm's current and future financial situation and on insurance product pricing, market profitability, insurance demand, reinsurance coverage and all other relevant macro-economic variables; and
 3. (c) the increased uncertainty in future profit following the loss referred to in 6.4(1), as well as the increasing degree of uncertainty relating to future taxable profit following that loss, as the projection horizon becomes longer.
4. (3) The relevant requirements for the purpose of (1)(b)(ii) are:
 5. 1. (a) a firm must not assume new business sales in excess of those projected for the purposes of the firm's business planning;
 2. (b) a firm must not assume new business sales after the end of the firm's business planning horizon and, for this purpose, a firm's business planning horizon must not exceed five years;
 3. (c) the rates of return on the firm's investments following the loss referred to in 6.4(1) must be assumed to be equal

to the implicit returns of the forward rates derived from the relevant risk-free interest rate term structure obtained after that loss, unless the firm is able to demonstrate that returns in excess of those implicit returns are likely;

4. (d) where a firm sets a projection horizon for profits from new business that is longer than its business planning horizon, it must:

1. (i) set a finite projection horizon;
 2. (ii) apply appropriate haircuts to the profits from new business projected beyond the business planning horizon;
 3. (iii) assume that such haircuts increase the further into the future the profits are projected; and
5. (e) a firm must not apply assumptions that are more favourable than those used for valuation and utilisation of deferred tax assets in accordance with Valuation 11.

7

Simplification in the Standard Formula

7.1

1. (1) A _firm_ may use a simplified calculation for a specific sub-module or risk module where the nature, scale and complexity of the risks it faces justifies it.
2. (2) A _firm_ must calibrate its simplified calculation in accordance with Solvency Capital Requirement ? General Provisions 3.3 to 3.4.

[Note: Art. 109 of the _Solvency II Directive_]

7.2

1. (1) For the purposes of 7.1(1), a firm must determine whether the simplified calculation is proportionate to the nature, scale and complexity of the risks by carrying out an assessment which must include all of the following:
 1. (a) an assessment of the nature, scale and complexity of the risks of the firm covered in the relevant module or sub-module;
 2. (b) an evaluation in qualitative or quantitative terms, as appropriate, of the error introduced in the results of the simplified calculation due to any deviation between the following:
 1. (i) the assumptions underlying the simplified calculation in relation to the risk;
 2. (ii) the results of the assessment referred to in (a).
2. (2) Where the error referred to in 1(b) would lead to a misstatement of the SCR that could influence the decision-making or judgement of the user of the information relating to the SCR, the use of a simplified calculation by a firm will not be proportionate to the nature, scale and complexity of the risks that the firm faces and the firm must not use that simplified calculation, unless it produces an SCR which exceeds the SCR; that results from the standard calculation.

7.3

A firm that is a captive insurer or captive reinsurer may use the simplified calculations set out in 7.4, 7.23, 7.25 and 7.27, where 7.2 is complied with and all of the following requirements are met:

1. (1) in relation to the insurance obligations of the firm, all insured persons and beneficiaries are legal entities of the group of which the firm is part;
2. (2) in relation to the reinsurance obligations of the firm, all insured persons and beneficiaries of the contracts of insurance underlying the reinsurance obligations are legal entities of the group of which the firm is part; and
3. (3) the insurance obligations and the contracts of insurance underlying the reinsurance obligations of the firm do not relate to any compulsory third-party liability insurance.

7.4

1. (1) Subject to 7.2 and 7.3, a firm that is a captive insurer or captive reinsurer may calculate the capital requirement for non-life premium and reserve risk in accordance with the following formula:

$$\sqrt{[SCR_{NL, prem, res}] = \sqrt{0.65 \cdot \sum_s NL^2_{\left(pr, s \right)} + 0.35 \cdot \left(\sum_s NL_{\left(pr, s \right)} \right)^2}}$$

where $_s__$ covers all segments set out in 3A3.

1. (2) For the purposes of (1), a firm must calculate the capital requirement for non-life premium and reserve risk of a particular segment $_s__$ set out in 3A3 in accordance with the following formula:

$$\sqrt{[NL_{pr, s}] = 0.6 \cdot \sqrt{V^2_{\left(prem, s \right)} + V_{\left(prem, s \right)} \cdot V_{\left(res, s \right)} + V^2_{\left(res, s \right)}}}$$

where:

1. (a) $_V(prem, s)__$ denotes the volume measure for premium risk of segment $_s__$ calculated in accordance with 3A2.3; and
2. (b) $_V(res, s)__$ denotes the volume measure for reserve risk of a segment calculated in accordance with 3A2.6.

7.5

For the purposes of 3A6.1(1), subject to 7.2, a firm may determine the insurance policies for which discontinuance would result in an increase in technical provisions without the risk margin on the basis of groups of policies, provided that the grouping complies with the requirements set out in Technical Provisions ? Further Requirements 20.1(2).

7.6

Subject to 7.2, a firm may calculate:

1. (1) the sum insured for windstorm risk referred to in 3A9.6(2) and 3A9.8 on the basis of groups of risk zones provided that:
 1. (a) each of the risk zones within a group must be situated within the same particular region set out in Annex V; and
 2. (b) where the sum insured for windstorm risk referred to in 3A9.6(2) is calculated on the basis of a group of risk zones, the risk weight for windstorm risk referred to in 3A9.6(1) must be the risk weight for windstorm risk in the risk zone within that group with the highest risk weight for windstorm risk set out in Annex X;
2. (2) the sum insured for earthquake risk referred to 3A11.3(2) and 3A11.5 on the basis of groups of risk zones, provided that:
 1. (a) each of the risk zones within a group must be situated within the same particular region set out in Annex VI; and
 2. (b) where the sum insured for earthquake risk referred to in 3A11.3(2) is calculated on the basis of a group of risk zones, the risk weight for earthquake risk referred to in 3A11.3(1) must be the risk weight for earthquake risk in the risk zone within that group with the highest risk weight for earthquake risk as set out in Annex X;
3. (3) the sum insured for flood risk referred to in 3A12.6(2) and 3A12.8 on the basis of groups of risk zones, provided that:
 1. (a) each of the risk zones within a group must be situated within the same particular region set out in Annex VII; and
 2. (b) where the sum insured for flood risk referred to in 3A12.6(2) is calculated on the basis of a group of risk zones, the risk weight for flood risk referred to in 3A12.6(1) must be the risk weight for flood risk in the risk zone within that group with the highest risk weight for flood risk as set out in Annex X;
4. (4) the sum insured for hail risk referred to in 3A13.6(2) and 3A13.8 on the basis of groups of risk zones, provided that:
 1. (a) each of the risk zones within a group must be situated within the same particular region set out in Annex VIII; and
 2. (b) where the sum insured for hail risk referred to in 3A13.6(2) is calculated on the basis of a group of risk zones, the

risk weight for hail risk referred to in 3A13.6(1) must be the risk weight for hail risk in the risk zone within that group with the highest risk weight for hail risk as set out in Annex X; and

5. (5) the weighted sum insured for subsidence risk referred to in 3A14.2 on the basis of groups of risk zones, provided that where the weighted sum insured referred to in 3A14.2 is calculated on the basis of a group of risk zones, the risk weight for subsidence risk referred to in 3A14.2(1) must be the risk weight for subsidence risk in the risk zone within that group with the highest risk weight for subsidence risk as set out in Annex X.

7.7

1. (1) Subject to 7.2, a firm may calculate the capital requirement for fire risk referred to in 3A21.1 in accordance with the following formula:

$$\text{SCR}_{\text{fire}} = \max(\text{SCR}_{\text{firei}}; \text{SCR}_{\text{firec}}; \text{SCR}_{\text{firer}})$$

where:

1. (a) $\text{SCR}_{\text{firei}}$ denotes the largest industrial fire risk concentration;
2. (b) $\text{SCR}_{\text{firec}}$ denotes the largest commercial fire risk concentration; and
3. (c) $\text{SCR}_{\text{firer}}$ denotes the largest residential fire risk concentration.

1. (2) A firm must calculate its largest industrial fire risk concentration in accordance with the following formula:

$$\text{SCR}_{\text{firei}} = \max(E_{1,i}; E_{2,i}; E_{3,i}; E_{4,i}; E_{5,i})$$

where $E_{k,i}$ denotes the total exposure within the perimeter of the k -th largest industrial fire risk exposure.

1. (3) A firm must calculate its largest commercial fire risk concentration in accordance with the following formula:

$$\text{SCR}_{\text{firec}} = \max(E_{1,c}; E_{2,c}; E_{3,c}; E_{4,c}; E_{5,c})$$

where $E_{k,c}$ denotes the total exposure within the perimeter of the k -th largest commercial fire risk exposure.

1. (4) A firm must calculate its largest residential fire risk concentration in accordance with the following formula:

$$\text{SCR}_{\text{firer}} = \max(E_{1,r}; E_{2,r}; E_{3,r}; E_{4,r}; E_{5,r}; \theta)$$

where:

1. (a) $E_{k,r}$ denotes the total exposure within the perimeter of the k -th largest residential fire risk exposure; and
2. (b) θ denotes the market share based residential fire risk exposure.

1. (5) For the purposes of (2), (3) and (4), the total exposure within the perimeter of the k -th largest industrial, commercial or residential fire risk exposure of a firm is the sum insured by the firm with respect to a set of buildings that meets all of the following requirements:

1. (a) in relation to each building, the firm has obligations in lines of business 7 and 19 which cover damage due to fire or explosion, including as a result of terrorist attacks; and
 2. (b) each building is partly or fully located within a radius of 200 metres around the industrial, commercial or residential building with the k -th highest sum insured after deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles.
3. For the purposes of determining the sum insured with respect to a building, a firm must take into account all reinsurance contracts and special purpose vehicles that would pay out in case of insurance claims related to that building, other than reinsurance contracts and special purpose vehicles that are subject to conditions not related to that building, which must not be taken into account.

2. (6) A firm must calculate the market share based residential fire risk exposure in accordance with the following

formula:

$$\theta = \{SI\}_{av} \cdot 500 \cdot \max(0.05; \{max\}_c \cdot (\{MarketShare\}_c))$$

where:

1. (a) $\{SI\}_{av}$ is the average sum insured by the firm with respect to residential property;
2. (b) $\{max\}_c$ denotes all countries where the firm has obligations in lines of business 7 and 19 covering residential property; and
3. (c) $\{MarketShare\}_c$ is the market share of the firm in country $\{c\}$ related to obligations in those lines of business covering residential property.

7.8

Subject to 7.2, a firm may calculate the capital requirement for life mortality risk in accordance with the following formula:

$$\{SCR\}_{mortality} = 0.15 \cdot q \cdot \sum_{k=1}^n \{CAR\}_k \cdot \frac{(1-q)^{k-1} \cdot (1+i_k)^{k-0.5}}{(1+i_k)^{k-0.5}}$$

where, with respect to insurance and reinsurance policies with a positive capital at risk:

1. (a) $\{CAR\}_k$ denotes the total capital at risk in year $\{k\}$, meaning the sum over all contracts of insurance of the higher of zero and the difference, in relation to each contracts of insurance between the following amounts:
 1. (i) the sum of:
 1. A. the amount that the firm would pay in year $\{k\}$ in the event of the death of the persons insured under the contracts of insurance after deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles; and
 2. B. the expected present value of amounts not covered in A. that the firm would pay after year $\{k\}$ in the event of the immediate death of the persons insured under the contracts of insurance after deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles; and
 2. (ii) the best estimate of the corresponding insurance and reinsurance obligations in year $\{k\}$ after deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles;
2. (b) $\{q\}$ denotes the expected average mortality rate over all the insured persons and over all future years weighted by the sum insured;
3. (c) $\{n\}$ denotes the modified duration in years of payments payable on death included in the best estimate; and
4. (d) $\{i_k\}$ denotes the annualised spot rate for maturity $\{k\}$ of the relevant risk-free interest rate term structure.

7.9

Subject to 7.2, a firm may calculate the capital requirement for life longevity risk in accordance with the following formula:

$$\{SCR\}_{longevity} = 0.2 \cdot q \cdot n \cdot 1.1^{(n-1)/2} \cdot \{BE\}_{long}$$

where, with respect to the policies referred to in 3B2.2:

1. (a) $\{q\}$ denotes the expected average mortality rate of the insured persons during the following 12 months weighted by the sum insured;
2. (b) $\{n\}$ denotes the modified duration in years of the payments to beneficiaries included in the best estimate; and
3. (c) $\{BE\}_{long}$ denotes the best estimate of the insurance and reinsurance obligations subject to longevity risk.

7.10

Subject to 7.2, a firm may calculate the capital requirement for life disability-morbidity risk in accordance with the following formula:

$$\begin{aligned} \text{SCR}_{\text{disability-morbidity}} = & 0.35 \cdot \text{CAR}_1 \cdot d_1 + 0.25 \cdot 1.1^{\left(\frac{n-3}{2}\right)} \cdot \left(\frac{n-1}{2}\right) \cdot \text{CAR}_2 \cdot d_2 + 0.2 \cdot 1.1^{\left(\frac{n-1}{2}\right)} \cdot n \cdot \text{BE}_{\text{dis}} \end{aligned}$$

where, with respect to insurance and reinsurance policies with a positive capital at risk:

1. (a) CAR_1 denotes the total capital at risk, meaning the sum over all contracts of insurance of the higher of zero and the difference between the following amounts:

1. (i) the sum of:

1. A. the amount that the firm would currently pay in the event of the death or disability of the persons insured under the contract of insurance after deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles; and

2. B. the expected present value of amounts not covered in A. that the firm would pay in the future in the event of the immediate death or disability of the persons insured under the contract of insurance after deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles; and

2. (ii) the best estimate of the corresponding insurance and reinsurance obligations after deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles;

2. (b) CAR_2 denotes the total capital at risk as defined in (a) after 12 months;

3. (c) d_1 denotes the expected average disability-morbidity rate during the following 12 months weighted by the sum insured;

4. (d) d_2 denotes the expected average disability-morbidity rate in the 12 months after the following 12 months weighted by the sum insured;

5. (e) n denotes the modified duration of the payments on disability-morbidity included in the best estimate;

6. (f) t denotes the expected termination rates during the following 12 months; and

7. (g) BE_{dis} denotes the best estimate of insurance and reinsurance obligations subject to disability-morbidity risk.

7.11

Subject to 7.2, a firm may calculate the capital requirement for life expense risk in accordance with the following formula:

$$\text{SCR}_{\text{expenses}} = 0.1 \cdot \text{EI} \cdot n + \text{EI} \cdot \left(\frac{1}{1+0.01} \right) \cdot \left(\frac{1}{1+i} \right)^{n-1} \cdot \left(\frac{1}{1+i} \right)^{\frac{n-1}{2}}$$

where:

1. (a) EI denotes the amount of expenses incurred in servicing long-term insurance or reinsurance obligations other than health insurance obligations and health reinsurance obligations during the last year;

2. (b) n denotes the modified duration in years of the cash-flows included in the best estimate of those obligations; and

3. (c) i denotes the weighted average inflation rate included in the calculation of the best estimate of those obligations, where the weights are based on the present value of expenses included in the calculation of the best estimate for servicing existing long-term insurance and reinsurance obligations.

7.12

1. (1) Subject to 7.2, a firm may calculate the capital requirement for the risk of a permanent increase in lapse rates in accordance with the following formula:

$$\text{Lapse}_{\text{up}} = 0.5 \cdot I_{\text{up}} \cdot n_{\text{up}} \cdot S_{\text{up}}$$

where:

1. (a) l_{up} denotes the higher of the average lapse rate of the policies with positive surrender strains and 67%;
2. (b) n_{up} denotes the average period in years over which the policies with positive surrender strains run off; and
3. (c) S_{up} denotes the sum of positive surrender strains referred to in (3).

1. (2) Subject to 7.2, a firm may calculate the capital requirement for the risk of a permanent decrease in lapse rates in accordance with the following formula:

$$\text{Lapse}_{down} = 0.5 \cdot l_{down} \cdot n_{down} \cdot S_{down}$$

where:

1. (a) l_{down} denotes the higher of the average lapse rate of the policies with negative surrender strains and 40%;
2. (b) n_{down} denotes the average period in years over which the policies with negative surrender strains run off; and
3. (c) S_{down} denotes the sum of negative surrender strains referred to in (3).

1. (3) The surrender strain of an insurance policy is the difference between the following:

1. (a) the amount currently payable by the firm on discontinuance by the policyholder, net of any amounts recoverable from policyholders or intermediaries; and
2. (b) the amount of technical provisions without the risk margin.

7.13

Subject to 7.2, a firm may calculate each of the following capital requirements on the basis of groups of policies, provided that the grouping complies with the requirements set out in Technical Provisions ? Further Requirements 20.1(2):

1. (1) the capital requirement for the risk of a permanent increase in lapse rates referred to in 3B6.2;
2. (2) the capital requirement for the risk of a permanent decrease in lapse rates referred to in 3B6.3; and
3. (3) the capital requirement for mass lapse risk referred to in 3B6.6.

7.14

Subject to 7.2, a firm may calculate the capital requirement for life-catastrophe risk in accordance with the following formula:

$$\text{SCR}_{life-catastrophe} = \sum_i 0.0015 \cdot \text{CAR}_i$$

1. where:

2. (a) the sum includes all policies with a positive capital at risk; and
3. (b) CAR_i denotes the capital at risk of the policy i , meaning the higher of zero and the difference between the following amounts:

1. (i) the sum of:

1. A. the amount that the firm would currently pay in the event of the death of the persons insured under the contract of insurance after deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles; and
2. B. the expected present value of amounts not covered in A. that the firm would pay in the future in the event of the immediate death of the persons insured under the contract of insurance after deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles; and

2. (ii) the best estimate of the corresponding insurance and reinsurance obligations after deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles.

7.15

For the purposes of 3C7.1(1), subject to 7.2, a firm may determine the insurance policies for which discontinuance would result in an increase in technical provisions without the risk margin on the basis of groups of policies, provided that the grouping complies with the requirements set out in Technical Provisions ? Further Requirements 20.1(2).

7.16

Subject to 7.2, a firm may calculate the capital requirement for health mortality risk in accordance with the following formula:

$$\mathbb{E}[\text{SCR}_{\text{health-mortality}}] = 0.15 \cdot q \cdot \sum_{k=1}^n \text{CAR}_k \cdot \frac{(1-q)^{k-1}}{(1+i_k)^{k-0.5}}$$

where, with respect to insurance and reinsurance policies with a positive capital at risk:

1. (a) CAR_k denotes the total capital at risk in year k , meaning the sum over all contracts of insurance of the higher of zero and the difference, in relation to each contract of insurance, between the following amounts:
 1. (i) the sum of:
 1. A. the amount that the firm would pay in year k in the event of the death of the persons insured under the contract of insurance after deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles; and
 2. B. the expected present value of amounts not covered in A. that the firm would pay after year k in the event of the immediate death of the persons insured under the contract of insurance after deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles; and
 2. (ii) the best estimate of the corresponding insurance and reinsurance obligations in year k after deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles;
2. (b) q denotes the expected average mortality rate over all insured persons and over all future years weighted by the sum insured;
3. (c) n denotes the modified duration in years of payments payable on death included in the best estimate; and
4. (d) i_k denotes the annualised spot rate for maturity k of the relevant risk-free interest rate term structure.

7.17

Subject to 7.2, a firm may calculate the capital requirement for health longevity risk in accordance with the following formula:

$$\mathbb{E}[\text{SCR}_{\text{health-longevity}}] = 0.2 \cdot q \cdot n \cdot 1.1^{\frac{(n-1)}{2}} \cdot \text{BE}_{\text{long}}$$

where, with respect to the policies referred to in 3C10.2:

1. (a) q denotes the expected average mortality rate of the insured persons during the following 12 months weighted by the sum insured;
2. (b) n denotes the modified duration in years of the payments to beneficiaries included in the best estimate; and
3. (c) BE_{long} denotes the best estimate of the obligations subject to health longevity risk.

7.18

Subject to 7.2, a firm may calculate the capital requirement for medical expense disability-morbidity risk in accordance with the following formula:

$$\mathbb{E}[\text{SCR}_{\text{medical expense}}] = 0.05 \cdot \text{MP} \cdot \left(\frac{1}{1+0.01} \right)^n \cdot \left(\frac{1}{1+0.01} \right)^{n-1} \cdot \left(\frac{1}{1+i} \right)^{n-1}$$

where:

1. (a) MP denotes the amount of medical payments on medical expense insurance obligations or medical expense reinsurance obligations during the last year;
2. (b) n denotes the modified duration in years of the cash-flows included in the best estimate of those obligations; and
3. (c) i denotes the average rate of inflation on medical payments included in the calculation of the best estimate of those obligations, where the weights are based on the present
4. value of medical payments included in the calculation of the best estimate of those obligations.

7.19

Subject to 7.2, a firm may calculate the capital requirement for income protection disability-morbidity risk in accordance with the following formula:

$$\text{SCR}_{\text{income-protection-disability-morbidity}} = 0.35 \cdot \text{CAR}_1 \cdot d_1 + 0.25 \cdot \{1.1\}^{\left(\frac{n-3}{2}\right)} \cdot \left\{ \frac{n-1}{n} \right\} \cdot \text{CAR}_2 \cdot d_2 + 0.2 \cdot \{1.1\}^{\left(\frac{n-1}{2}\right)} \cdot \left\{ \frac{n}{n} \right\} \cdot \text{BE}_{\text{dis}}$$

where, with respect to insurance and reinsurance policies with a positive capital at risk:

1. (a) CAR_1 denotes the total capital at risk, meaning the sum over all contracts of insurance of the higher of zero and the difference between the following amounts:
 1. (i) the sum of:
 1. A. the amount that the firm would currently pay in the event of the death or disability of the persons insured under the contract of insurance after deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles; and
 2. B. the expected present value of amounts not covered in A. that the firm would pay in the future in the event of the immediate death or disability of the persons insured under the contract of insurance after deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles;
 2. (ii) the best estimate of the corresponding insurance and reinsurance obligations after deduction of the amounts recoverable from reinsurance contracts and special purpose vehicles;
2. (b) CAR_2 denotes the total capital at risk as defined in (a) after 12 months;
3. (c) d_1 denotes the expected average disability-morbidity rate during the following 12 months weighted by the sum insured;
4. (d) d_2 denotes the expected average disability-morbidity rate in the 12 months after the following 12 months weighted by the sum insured;
5. (e) n denotes the modified duration of the payments on disability-morbidity included in the best estimate;
6. (f) t denotes the expected termination rates during the following 12 months; and
7. (g) BE_{dis} denotes the best estimate of obligations subject to disability-morbidity risk.

7.20

Subject to 7.2, a firm may calculate the capital requirement for health expense risk in accordance with the following formula:

$$\text{SCR}_{\text{health-expense}} = 0.1 \cdot \left(\text{EI} \cdot \left(\frac{1}{1+0.01} \right)^n + \text{EI} \cdot \left(\frac{1}{1+0.01} \right)^{n-1} \cdot \left(\frac{1}{1+0.01} \right)^{\left(\frac{n-1}{2} \right)} \cdot \left(\frac{1}{1+0.01} \right)^{\left(\frac{n-1}{2} \right)} \right)$$

where:

1. (a) EI denotes the amount of expenses incurred in servicing health insurance obligations and health reinsurance

obligations during the last year;

2. (b) $_n$ denotes the modified duration in years of the cash-flows included in the best estimate of those obligations; and
3. (c) $_i$ denotes the weighted average inflation rate included in the calculation of the best estimate of those obligations, weighted by the present value of expenses included in the calculation of the best estimate for servicing existing health insurance obligations and health reinsurance obligations.

7.21

1. (1) Subject to 7.2, a firm may calculate the capital requirement for the risk of a permanent increase in lapse rates referred to in 3C16.1(1) in accordance with the following formula:

$$\mathbb{E}[\text{Lapse}_{\text{up}}] = 0.5 \cdot I_{\text{up}} \cdot n_{\text{up}} \cdot S_{\text{up}}$$

1. where:

1. (a) $_I_{\text{up}}$ denotes the higher of the average lapse rate of the policies with positive surrender strains and 83%;
 2. (b) $_n_{\text{up}}$ denotes the average period in years over which the policies with positive surrender strains run off; and
 3. (c) $_S_{\text{up}}$ denotes the sum of positive surrender strains referred to in (3).
2. (2) Subject to 7.2, a firm may calculate the capital requirement for the risk of a permanent decrease in lapse rates referred to in 3C16.1(2) in accordance with the following formula:

$$\mathbb{E}[\text{Lapse}_{\text{down}}] = 0.5 \cdot I_{\text{down}} \cdot n_{\text{down}} \cdot S_{\text{down}}$$

1. where:

1. (a) $_I_{\text{down}}$ denotes the average lapse rate of the policies with negative surrender strains;
 2. (b) $_n_{\text{down}}$ denotes the average period in years over which the policies with negative surrender strains run off; and
 3. (c) $_S_{\text{down}}$ denotes the sum of negative surrender strains referred to in (3).
2. (3) The surrender strain of an insurance policy is the difference between the following:
1. (a) the amount currently payable by the firm on discontinuance by the policyholder, net of any amounts recoverable from policyholders or intermediaries; and
 2. (b) the amount of technical provisions without the risk margin.

7.22

Subject to 7.2, a firm may calculate each of the following capital requirements on the basis of groups of policies, provided that the grouping complies with the requirements set out in Technical Provisions ? Further Requirements 20.1(2):

1. (1) the capital requirement for the risk of a permanent increase in SLT health lapse rates referred to in 3C16.2;
2. (2) the capital requirement for the risk of a permanent decrease in SLT health lapse rates referred to in 3C16.3; and
3. (3) the capital requirement for SLT health mass lapse risk referred to in 3C16.6.

7.23

1. (1) Subject to 7.2 and 7.3, a firm that is a captive insurer or captive reinsurer may calculate the capital requirement for interest-rate risk referred to in 3D4 as follows:

1. (a) the sum, for each currency, of the capital requirements for the risk of an increase in the term structure of interest rates as set out in (2); and
 2. (b) the sum, for each currency, of the capital requirements for the risk of a decrease in the term structure of interest rates as set out in (3).
2. (2) For the purposes of (1)(a), a firm must calculate the capital requirement for the risk of an increase in the term

structure of interest rates for a given currency in accordance with the following formula:

$$\{IR\}_{up} = \sum_i \{MVAL\}_i \cdot \{dur\}_i \cdot \{rate\}_i \cdot \{stress\}_{(i,up)} - \sum_{lob} \{BE\}_{lob} \cdot \{dur\}_{lob} \cdot \{rate\}_{lob} \cdot \{stress\}_{(lob,up)}$$

where:

1. (a) the first sum covers all maturity intervals $_i$ set out in (4);
2. (b) $_i MVAL$ denotes the value in accordance with Valuation 2.1 to 2.2 of assets less liabilities other than technical provisions for maturity interval $_i$;
3. (c) $_i dur$ denotes the simplified duration of maturity interval $_i$;
4. (d) $_i rate$ denotes the relevant risk-free rate for the simplified duration of maturity interval $_i$;
5. (e) $_{(i,up)} stress$ denotes the relative upward stress of the interest rate for simplified duration of maturity interval $_i$;
6. (f) the second sum covers all lines of business;
7. (g) $_{lob} BE$ denotes the best estimate for line of business $_{lob}$;
8. (h) $_{lob} dur$ denotes the modified duration of the best estimate in line of business $_{lob}$;
9. (i) $_{lob} rate$ denotes the relevant risk-free rate for modified duration in line of business $_{lob}$; and
10. (j) $_{(lob,up)} stress$ denotes the relative upward stress of the interest rate for the modified duration $_{lob} dur$.

1. (3) For the purposes of (1)(b), a firm must calculate the capital requirement for the risk of a decrease in the term structure of interest rates for a given currency in accordance with the following formula:

$$\{IR\}_{down} = \sum_i \{MVAL\}_i \cdot \{dur\}_i \cdot \{rate\}_i \cdot \{stress\}_{(i,down)} - \sum_{lob} \{BE\}_{lob} \cdot \{dur\}_{lob} \cdot \{rate\}_{lob} \cdot \{stress\}_{(lob,down)}$$

where:

1. (a) the first sum covers all maturity intervals $_i$ set out in (4);
2. (b) $_i MVAL$ denotes the value in accordance with Valuation 2.1 to 2.2 of assets less liabilities other than technical provisions for maturity interval $_i$;
3. (c) $_i dur$ denotes the simplified duration of maturity interval $_i$;
4. (d) $_i rate$ denotes the relevant risk-free rate for the simplified duration of maturity interval $_i$;
5. (e) $_{(i,down)} stress$ denotes the relative downward stress of the interest rate for simplified duration of maturity interval $_i$;
6. (f) the second sum covers all lines of business;
7. (g) $_{lob} BE$ denotes the best estimate for line of business $_{lob}$;
8. (h) $_{lob} dur$ denotes the modified duration of the best estimate in line of business $_{lob}$;
9. (i) $_{lob} rate$ denotes the relevant risk-free rate for modified duration in line of business $_{lob}$; and (j) $_{(lob,down)} stress$ denotes the relative downward stress of the interest rate for modified duration $_{lob} dur$.

1. (4) The maturity intervals $_i$ and the simplified duration $_i dur$ referred to in (2)(a), 2(c), (3)(a) and 3(c) must be as follows:

1. (a) up to the maturity of one year, the simplified duration must be 0.5 years;
2. (b) between maturities of one and three years, the simplified duration must be two years;
3. (c) between maturities of three and five years, the simplified duration must be four years;
4. (d) between maturities of five and 10 years, the simplified duration must be seven years; and
5. (e) from the maturity of 10 years onwards, the simplified duration must be 12 years.

1. (1) Subject to 7.2, a firm may calculate the capital requirement for spread risk referred to in 3D17 in accordance with the following formula:

$$\frac{1}{\text{SCR}_{\text{bonds}}} = \frac{\text{MV}_{\text{bonds}}}{\text{Liab}_{\text{ul}}} \cdot \left(\sum_i \left(\frac{\text{MV}_i}{\text{MV}_{\text{bonds}}} \right)^{\text{bonds}} \cdot \text{stress}_i + \left(\frac{\text{MV}_{\text{norating}}}{\text{MV}_{\text{bonds}}} \right)^{\text{bonds}} \cdot \min \left(\frac{\text{dur}_i}{\text{dur}_{\text{norating}}} \cdot 0.03; 1 \right) \right) + \text{Liab}_{\text{ul}}$$

where:

1. (a) $\text{SCR}_{\text{bonds}}$ denotes the capital requirement for spread risk on bonds and loans;
2. (b) MV_{bonds} denotes the value in accordance with Valuation 2.1 to 2.2 of the assets subject to capital requirements for spread risk on bonds and loans;
3. (c) MV_i denotes the proportion of the portfolio of the assets subject to a capital requirement for spread risk on bonds and loans with credit quality step i , where a credit assessment by a nominated external credit assessment institution is available for those assets;
4. (d) $\text{MV}_{\text{norating}}$ denotes the proportion of the portfolio of the assets subject to a capital requirement for spread risk on bonds and loans for which no credit assessment by a nominated external credit assessment institution is available;
5. (e) dur_i and $\text{dur}_{\text{norating}}$ denote the modified duration denominated in years of the assets subject to a capital requirement for spread risk on bonds and loans where no credit assessment by a nominated external credit assessment institution is available;
6. (f) stress_i denotes a function of the credit quality step i and of the modified duration denominated in years of the assets subject to a capital requirement for spread risk on bonds and loans with credit quality step i , set out in (2); and
7. (g) Liab_{ul} denotes the increase in the technical provisions without risk margin for policies where the policyholders bear the investment risk with embedded options and guarantees that would result from an instantaneous decrease in the value of the assets subject to the capital requirement for spread risk on bonds of:

$$\frac{\text{MV}_{\text{bonds}}}{\text{Liab}_{\text{ul}}} \cdot \left(\sum_i \left(\frac{\text{MV}_i}{\text{MV}_{\text{bonds}}} \right)^{\text{bonds}} \cdot \text{stress}_i + \left(\frac{\text{MV}_{\text{norating}}}{\text{MV}_{\text{bonds}}} \right)^{\text{bonds}} \cdot \min \left(\frac{\text{dur}_i}{\text{dur}_{\text{norating}}} \cdot 0.03; 1 \right) \right)$$

1. (2) stress_i referred to in (1)(f), for each credit quality step i , must be equal to: $\text{dur}_i \cdot b_i$ where dur_i is the modified duration denominated in years of the assets subject to a capital requirement for spread risk on bonds and loans with credit quality step i , and b_i is determined in accordance with the following table:

Credit quality step i | 0 | 1 | 2 | 3 | 4 | 5 | 6

---|---|---|---|---|---|---

b_i | 0.9% | 1.1% | 1.4% | 2.5% | 4.5% | 7.5% | 7.5%

1. (3) $\text{dur}_{\text{norating}}$ referred to in (1)(e) and dur_i referred to in (2) must not be lower than one year.

7.25

Subject to 7.2 and 7.3, a firm that is a captive insurer or captive reinsurer may base the calculation of the capital requirement for spread risk referred to in 3D17 on the assumption that all assets are assigned to credit quality step 3.

7.26

Subject to 7.2, a firm may assign a bond other than those to be included in the calculations under paragraphs 3D24.2 to 3D24.21 a risk factor stress_i equivalent to that for credit quality step 3 for the purposes of 3D17.3 and assign the bond to credit quality step 3 for the purpose of calculating the weighted average credit quality step in accordance with 3D26.4, provided that all of the following requirements are met:

1. (1) credit assessments from a nominated external credit assessment institution are available for at least 80% of the

total value of the bonds other than those to be included in the calculations under 3D24.2 to 3D24.21;

2. (2) a credit assessment by a nominated external credit assessment institution is not available for the bond in question;
3. (3) the bond in question provides a fixed redemption payment on or before the date of maturity, in addition to regular fixed or floating rate interest payments;
4. (4) the bond in question is not a structured note or collateralised security as referred to in the CIC table set out in the Section IR.06.02 instructions referred to in Reporting 10; and
5. (5) the bond in question does not cover liabilities that provide long-term insurance obligations with profit participation, nor does it cover unit-linked liabilities or index-linked liabilities, nor liabilities where a matching adjustment is applied.

7.27

Subject to 7.2 and 7.3, a firm that is a captive insurer or captive reinsurer may use all of the following assumptions for the calculation of the capital requirement for concentration risk:

1. (1) Intra-group asset pooling arrangements of the firm may be exempted from the calculation base referred to in 3D28.2 to the extent that there exist legally enforceable contractual terms which ensure that the liabilities of the firm will be offset by the intra-group exposures it holds against other entities of the group.
2. (2) The relative excess exposure threshold referred to in 3D28.1(c) must be equal to 15% for the following single name exposures:
 1. (a) exposures to credit institutions that do not belong to the same group and that have been assigned to the credit quality step 2; and
 2. (b) exposures to entities of the group that manage the cash of the firm that have been assigned to the credit quality step 2.

7.28

1. (1) Subject to 7.2 and where the best estimate of amounts recoverable from a reinsurance arrangement or securitisation and the corresponding debtors is not negative, a firm may calculate the risk-mitigating effect on underwriting risk of that reinsurance arrangement or securitisation referred to in 3E9 in accordance with the following formula:

$$\frac{\text{RM}_{re,all}}{\text{Recoverables}_i} \cdot \frac{\text{Recoverables}_i}{\text{Recoverables}_{all}}$$

where:

1. (a) $\text{RM}_{re,all}$ denotes the risk-mitigating effect on underwriting risk of the reinsurance arrangements and securitisations for all counterparties calculated in accordance with (2); and
 2. (b) Recoverables_i denotes the best estimate of amounts recoverable from the reinsurance arrangement or securitisation and the corresponding debtors for counterparty i and $\text{Recoverables}_{all}$ denotes the best estimate of amounts recoverable from the reinsurance arrangements and securitisations and the corresponding debtors for all counterparties.
1. (2) A firm may calculate the risk-mitigating effect on underwriting risk of the reinsurance arrangements and securitisations for all counterparties referred to in (1) as the difference between the following capital requirements:
 1. (a) the hypothetical capital requirement for underwriting risk of the firm if none of the reinsurance arrangements and securitisations exist; and
 2. (b) the capital requirement for underwriting risk of the firm.

7.29

Subject to 7.2 and where the best estimate of amounts recoverable from a proportional reinsurance arrangement and the corresponding debtors for a counterparty i is not negative, a firm may calculate the risk-mitigating effect on

underwriting risk j of the proportional reinsurance arrangement for counterparty i referred to 3E9 in accordance with the following formula:

$$\frac{\text{Recoverables}_i}{\text{BE-Recoverables}_{\text{all}}} \cdot \text{SCR}_j$$

where:

1. (1) BE_j denotes the best estimate of obligations gross of the amounts recoverable;
2. (2) Recoverables_i denotes the best estimate of amounts recoverable from the proportional reinsurance arrangement and the corresponding debtors for counterparty i ;
3. (3) $\text{Recoverables}_{\text{all}}$ denotes the best estimate of amounts recoverable from the proportional reinsurance arrangements and the corresponding debtors for all counterparties; and
4. (4) SCR_j denotes the capital requirement for underwriting risk j of the firm.

7.30

Subject to 7.2, a firm may use the following simplified calculations for the purposes of 3E6, 3E7 and 3E8:

1. (1) The best estimate referred to in 3E7.1(d) may be calculated in accordance with the following formula:

$$\text{BE}_C = \frac{P_C}{P_U} \cdot \text{BE}_U$$

where:

1. (a) BE_U denotes the best estimate of the liability ceded to the pooling arrangement by the firm, net of any amounts reinsured with counterparties external to the pooling arrangement;
2. (b) P_C denotes the counterparty member's share of the risk according to the terms of the pooling arrangement; and
3. (c) P_U denotes the firm's share of the risk according to the terms of the pooling arrangement.

1. (2) The best estimate referred to in 3E8.1(c) may be calculated in accordance with the following formula:

$$\text{BE}_{CE} = \frac{1}{P_U} \cdot \text{BE}_{CEP}$$

where:

1. (a) BE_{CEP} denotes the best estimate of the liability ceded to the external counterparty by the pooling arrangement, in relation to risk ceded to the pooling arrangement by the firm; and
2. (b) P_U denotes the firm's share of the risk according to the terms of the pooling arrangement.
3. (3) A firm may calculate the risk-mitigating effect referred to in 3E8.1(d) in accordance with the following formula:

$$\Delta \text{RM}_{CE} = \frac{\text{BE}_{CE}}{\sum \text{BE}_{CE}} \cdot \Delta \text{RM}_{CEP}$$

where:

1. (a) BE_{CE} denotes the best estimate of the liability ceded to the external counterparty by the pooling arrangement as a whole; and
2. (b) ΔRM_{CEP} denotes the contribution of all external counterparties to the risk-mitigating effect of the pooling arrangement on the underwriting risk of the firm.

1. (4) The counterparty pool members and the counterparties external to the pooling arrangement may be grouped according to the credit assessment by a nominated external credit assessment institution, provided there are separate groupings for pool exposure of type A, pool exposure of type B and pool exposure of type C.

7.31

Subject to 7.2, a firm may calculate the loss-given-default set out in 3E4, including the risk-mitigating effect on underwriting risk and market risk and the risk-adjusted value of collateral provided by way of a collateral arrangement, for a group of single name exposures provided that the group of single name exposures are assigned the highest probability of default assigned to single name exposures included in the group in accordance with 3E12.

7.32

Subject to 7.2, a firm may calculate the risk-mitigating effect on underwriting risk and market risk of a reinsurance arrangement, securitisation or derivative referred to in 3E9 as the difference between the following capital requirements:

1. (1) the sum of the hypothetical capital requirement for the sub-modules of the underwriting risk and market risk modules of the firm, calculated in accordance with this Part but as if the reinsurance arrangement, securitisation or derivative did not exist; and
2. (2) the sum of the capital requirements for the sub-modules of the underwriting risk and market risk modules of the firm.

7.33

For the purposes of 3E9, subject to 7.2 and where the reinsurance arrangement, securitisation or derivative covers obligations from only one of the segments (segment s) set out in 3A3 or, as applicable, 3C4, a firm may calculate the risk-mitigating effect of that reinsurance arrangement, securitisation or derivative on its underwriting risk in accordance with the following formula:

$$\sqrt{\left(\text{SCR}_{\text{CAT}}^{\text{hyp}} - \text{SCR}_{\text{CAT}}^{\text{without}}\right)^2 + \left(3 \cdot \sigma_s \cdot \left(P_s^{\text{hyp}} - P_s^{\text{without}} + \text{Recoverables}\right)\right)^2 + 1.5 \cdot \sigma_s \cdot \left(P_s^{\text{hyp}} - P_s^{\text{without}} + \text{Recoverables}\right) \cdot \left(\text{SCR}_{\text{CAT}}^{\text{hyp}} - \text{SCR}_{\text{CAT}}^{\text{without}}\right)}$$

1. where:

2. (1) $\text{SCR}_{\text{CAT}}^{\text{hyp}}$ denotes the hypothetical capital requirement for the non-life catastrophe risk sub-module referred to in 3A7.2 or, as applicable, the hypothetical capital requirement for the health catastrophe risk sub-module referred to in 3C17, that would apply if the reinsurance arrangement, securitisation or derivative did not exist;
3. (2) $\text{SCR}_{\text{CAT}}^{\text{without}}$ denotes the capital requirement for the non-life catastrophe risk sub-module referred to in 3A7.2 or, as applicable, the capital requirement for the health catastrophe risk sub-module referred to in 3C17;
4. (3) σ_s denotes the standard deviation for non-life premium risk of segment s determined in accordance with 3A4.3 and 3A4.4 or, as applicable, the standard deviation for the NSLT health premium risk of segment s determined in accordance with 3C5.3;
5. (4) P_s^{hyp} denotes the hypothetical volume measure for premium risk of segment s determined in accordance with 3A2.3 or 3A2.4 or, as applicable, 3C3.3 or 3C3.4 that would apply if the reinsurance arrangement, securitisation or derivative did not exist;
6. (5) P_s^{without} denotes the volume measure for premium risk of segment s determined in accordance with 3A2.3 or 3A2.4 or, as applicable, 3C3.3 or 3C3.4; and
7. (6) Recoverables denotes the best estimate of amounts recoverable from the reinsurance arrangement, securitisation or derivative and the corresponding debtors.

7.34

1. (1) Subject to 7.2, and where the counterparty requirement and the third party requirement referred to in 3E10.1 are both met, a firm may, for the purposes of 3E10, calculate the risk-adjusted value of a collateral provided by way of a collateral arrangement under which collateral is provided by way of security, as 85% of the value of the assets held as collateral, valued in accordance with Valuation 2.1 to 2.2.

2. (2) Subject to 7.2 and 3G8, and where the counterparty requirement referred to in 3E10.1 is met and the third party requirement referred to in 3E10.1 is not met, a firm may, for the purposes of 3E10, calculate the risk-adjusted value of a collateral provided by way of a collateral arrangement under which collateral is provided by way of security, as 75% of the value of the assets held as collateral, valued in accordance with Valuation 2.1 to 2.2.

7.35

Subject to 7.2, a firm may calculate the loss-given-default on a reinsurance arrangement or insurance securitisation referred to in 3E4.3 in accordance with the following formula:

$$\text{LGD} = \max\left(90\% \cdot (\text{Recoverables} + 50\% \cdot \text{RM}_{\text{re}}) - F \cdot \text{Collateral}; 0\right)$$

1. where:

2. (1) *Recoverables* denotes the best estimate of amounts recoverable from the reinsurance arrangement or insurance securitisation and the corresponding debtors;
3. (2) *RM re* denotes the risk-mitigating effect on underwriting risk of the reinsurance arrangement or securitisation;
4. (3) *Collateral* denotes the risk-adjusted value of collateral provided by way of a collateral arrangement in relation to the reinsurance arrangement or securitisation; and
5. (4) *F* denotes a factor to take into account the economic effect of the collateral arrangement in relation to the reinsurance arrangement or securitisation in case of any credit event related to the counterparty, determined in accordance with 3E10.7.

7.36

Subject to 7.2 and where the standard deviation of the loss distribution of type 1 exposures, as determined in accordance with 3E13.4 is lower than or equal to 20% of the total loss-given default on all type 1 exposures referred to in 3.13 to 3.19, a firm may calculate the capital requirement for counterparty default risk referred to in 3E13.1 in accordance with the following formula:

$$\text{SCR}_{\text{def},1} = 5 \cdot \sigma$$

where σ denotes the standard deviation of the loss distribution of type 1 exposures as determined in accordance with 3E13.4.

8

Lloyd's

8.1

This Chapter applies to the *Society* in relation to the use of the *standard formula* for the purpose of Solvency Capital Requirement ? General Provisions 3.1.

8.2

The *Society* must aggregate the results of each notional *SCR* referred to in Solvency Capital Requirement ? General Provisions 8.4 together with the *central requirement*, in order to obtain the *SCR* for Lloyd's.

9

Ring-Fenced Funds and Matching Adjustment Portfolios

9.1

A firm must calculate the adjustment in respect of any ring-fenced fund or matching adjustment portfolio referred to in 2.2 as follows:

1. (1) the firm must calculate a notional SCR for each ring-fenced fund and each matching adjustment portfolio, as well as for the remaining part of the firm, in the same manner as if each of those ring-fenced funds, each of those matching adjustment portfolios and the remaining part of the firm were separate firms;
2. (2) the firm must calculate its SCR as the sum of the notional SCRs for each of its ring-fenced fund, each of its matching adjustment portfolios and for the remaining part of the firm;
3. (3) subject to (4), where the calculation of the capital requirement for a risk module or sub-module of the basic SCR is based on the impact of a scenario on the firm's basic own funds, the firm must calculate the impact of the scenario on its basic own funds at the level of each ring-fenced fund, each matching adjustment portfolio and the remaining part of the firm;
4. (4) the basic own funds at the level of each ring-fenced fund or each matching adjustment portfolio must, for the purposes of (3), only include restricted own funds;
5. (5) where profit participation arrangements exist in respect of any insurance and reinsurance obligations within a ring-fenced fund, the firm must apply the following approach:
 1. (a) where the calculation referred to in (3) would result in an increase in the basic own funds at the level of the ring-fenced fund, the estimated increase in those basic own funds must, in order to reflect the existence of those profit participation arrangements in the ring-fenced fund, be adjusted by an amount equal to the increase in technical provisions resulting from the increase in future discretionary benefits that the firm would expect to pay to policyholders in that scenario;
 2. (b) where the calculation referred to in (3) would result in a decrease in the basic own funds at the level of the ring-fenced fund, the estimated decrease in those basic own funds for the calculation of the net basic SCR as referred to in 6.3(2), must, subject to (c), be adjusted by an amount equal to the reduction in future discretionary benefits that the firm would expect to pay to policyholders in that scenario;
 3. (c) the amount of the adjustment referred to in (b) must not exceed the amount of future discretionary benefits that are included in the firm's technical provisions in respect of that ring-fenced fund.
6. (6) notwithstanding (1), the firm must calculate the notional SCR for each ring-fenced fund and each matching adjustment portfolio using the scenario-based calculations under which the basic own funds of the firm as a whole are most negatively affected.
7. (7) for the purposes of determining the scenario under which basic own funds are most negatively affected for the firm as a whole, the firm must:
 1. (a) calculate the sum of the results of the impacts of the scenarios on the basic own funds at the level of each ring-fenced fund and each matching adjustment portfolio, in accordance with (3) and (5); and
 2. (b) add sums at the level of each ring-fenced fund and each matching adjustment portfolio to one another and to the results of the impact of the scenarios on the basic own funds in the remaining part of the firm.
8. (8) the firm must determine the notional SCR for each ring-fenced fund and each matching adjustment portfolio by aggregating the capital requirements for each sub-module and risk module of the basic SCR.
9. (9) notwithstanding 3.4, the firm must not allow for diversification effects between any, or a combination of, the following:
 1. (a) its ring-fenced fund;
 2. (b) its matching adjustment portfolios; or
 3. (c) the remaining part of the firm.